

Friends in Higher Places: Social Fit and University Choice*

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Abstract

Low-SES students apply to elite universities at lower rates than high-SES students with similar grades. Combining UK administrative data with a field experiment, we examine how social exposure to elite universities shapes this disparity. After a “breakthrough” event, when a high school first sends a student to an elite university, applications to that university rise. To isolate mechanisms, we randomize social exposure to elite university students through videos and one-on-one mentorship. The intervention increases applications to the treatment university by 17 percentage points among low-SES students, with no effect among high-SES students. Baseline surveys show that low-SES students are pessimistic about their chances of fitting in at elite universities, but not their chances of receiving an offer. Student beliefs about social fit update positively, with no change in other beliefs. Demographic overlap with the elite university student amplifies treatment effects, suggesting that social exposure alleviates information frictions about whether *someone like them* would fit in. (JEL D71, D91, I23, I24, I26, Z13)

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I. Introduction

Elite universities are vehicles for social mobility (Chetty et al. 2020; Van Der Erve, Drayton, and Britton 2021). Yet socioeconomically disadvantaged students are substantially less likely to attend elite universities in the UK (Campbell et al. 2022), US (Chetty, Deming, and Friedman 2026), Norway (Cattan, Salvanes, and Tominey 2025), and Chile (Barrios-Fernández 2022), even conditional on high-school grades. In the UK, the setting of this paper, socioeconomic gaps in elite university access persist despite uniform tuition fees across universities and income-based affirmative action. The ubiquity of these disparities points to causes beyond any single university system’s design.

A natural candidate for why academically qualified low-SES students apply to elite universities at lower rates is differences in *social exposure*. Low-SES students are less likely to know someone who attends an elite university, a channel that has been shown to matter in university enrollment decisions (Altmejd et al. 2021; Barrios-Fernández 2022; Bechichi and Kenedi 2024). Students may respond to the university choices of family members or schoolmates for various reasons. Through social contact, students may learn about aspects of the university experience that are highly specific to a university-person pair, such as social fit. Social exposure may also ease general information frictions about admissions criteria or labor market returns. Yet another possibility is that students want to be co-located with friends and family. These mechanisms have vastly different policy implications, yet we know relatively little about *why* students respond to their peers’ university choices and which information frictions, if any, social exposure alleviates.

This paper examines how social exposure shapes elite university applications. We conduct two complementary studies using UK administrative data and a high-stakes field experiment with 640 university applicants across 21 high schools. In the first, we study the effects of quasi-random variation in exposure to elite universities on university and labor market outcomes. In the second, we randomize social exposure through videos and one-on-one mentorship with university students. Our field experiment allows us to identify specific information frictions and test whether scalable policy interventions can address socioeconomic disparities in elite university applications.

We begin by documenting socioeconomic gaps in both applications and social exposure to elite universities. Low-income students are less likely to attend one of the 10 most academically selective UK universities and around 40% of this enrollment gap can be attributed to disparities in applications. At the same time, low-income students have substantially less social exposure to top universities through their high schools. Low-income students are 5–10 percentage points more likely to attend high schools where *no*

student has attended any of the top 10 universities in the preceding three years.

We develop a simple framework to illustrate how systematic lack of exposure to elite universities can generate persistent pessimistic beliefs and uncertainty about elite universities. Students learn from past cohorts of schoolmates about the payoffs at an elite university, but learn the exact payoff only if a past schoolmate attended the elite university. Under an assumption of naïve social learning, students infer that non-attendees had low payoffs and thus revise their expected payoffs downward if their older schoolmates did not attend. This can lead to herding patterns, where students from schools with no applications to elite universities are persistently less likely to apply. However, a student with an idiosyncratically positive expected payoff may still choose to attend the elite university. This first student to attend the elite university can raise applications among subsequent cohorts at that school.

Next, we show that application choices respond to quasi-random exposure to elite universities using UK administrative data. Specifically, we study *breakthroughs* to a university—cases where a student attends a particular university for the first time after several years in which no student from their school attended. In an event study framework, we find that breakthroughs persistently encourage applications to the specific university where the breakthrough occurs, raising application rates by around 30–50%. Breakthroughs have no effect on applications to other similarly ranked universities; breakthroughs to Cambridge increase applications to Cambridge among subsequent cohorts with no impact on applications to Oxford and vice versa. This suggests that low-SES students may face information frictions about *match-specific* payoffs for a particular university, rather than components that are transferable across universities or across all students. We also find that breakthrough effects are strongest among students who are demographically *similar* to the breakthrough student.

Students shifted into elite universities graduate successfully and see substantial labor market returns, suggesting that these students are well-matched to these universities. Enrollment increases by 60–85% of the increase in applications and almost all of these students graduate successfully and on time. Using a matched-control design, we show that students who apply to an elite university following a breakthrough earn around £4,000 (US\$5,120) more per year than matched controls by age 27. This effect size is comparable to benchmarks of the overall returns to elite UK universities from our own data and other research (Britton et al. 2022), indicating that the students who respond to breakthroughs successfully capture the large potential earnings gains from attending an elite university.

Why did these students forgo the large benefits to attending an elite university *prior* to the breakthrough, and what changed *after* the breakthrough? One potential explanation

for the breakthrough effects is that students feel more comfortable attending a university at which they *already have a friend*. Another possibility is that students gained information about universities following a breakthrough. Our findings in the administrative data can neither distinguish between these two channels nor isolate the specific aspects of universities about which students lack information. For example, low-SES students may be systematically misinformed about admissions requirements or the returns to elite universities. Alternatively, larger effects among demographically similar students may suggest that students learn about a demographic-group specific aspect of the university experience, such as social fit. Identifying which beliefs, if any, are shifted in response to social exposure is essential for designing policies to address socioeconomic disparities in higher education.

To distinguish between potential mechanisms for the breakthrough effects, we conduct a high-stakes field experiment that randomizes social exposure to elite universities. We vary the demographic overlap and provide both passive exposure (video clips) and relational exposure (one-on-one mentorship). Our experimental sample consists of 640 high school students in their final year who expressed some interest in applying to universities in January 2026. All participants attend an in-school workshop that provides information about the application process, admissions requirements, and average postgraduate earnings at various universities. Participants randomized into treatment additionally watch two video clips of university students discussing their experiences. Treated participants are subsequently randomized into one of two treatment subarms for one-on-one mentorship with university students: T1 connects participants to demographically *dissimilar* mentors and T2 connects participants to demographically *similar* mentors.

The design of our experiment allows us to measure whether applications respond to social exposure, how this response varies with demographic similarity, and which beliefs shift in response to exposure. We ask participant to report their perceived likelihood that they would (a) get an offer from, (b) make friends and fit in at, or (c) graduate successfully from specific universities. By eliciting beliefs across baseline, midline, and endline surveys, we document how each belief evolves over the course of the experiment. We collect realized application choices directly from participating schools at the end of the experiment. The active control arm ensures that any estimated treatment effects reflect the impact of social exposure over and above that of acquiring admissions and earnings information.

Baseline surveys indicate that low-SES students are pessimistic about their probability of fitting in and making friends at elite universities, but not about their chances of receiving an offer or graduating successfully. Both low-income and first-generation students report perceived probabilities of social fit that are 6-8 percentage points lower than high-SES students with similar coursework and test score. Furthermore, we find that low-SES

students hold beliefs about admissions requirements at elite universities that are correctly calibrated. Our findings suggest that a baseline lack of exposure can generate pessimism about match-specific aspects of elite universities without distorting beliefs about more general aspects such as admissions requirements.

Experimentally induced social exposure substantially shifts participants' application choices. Pooling across treatments, assignment to treatment increases the probability of applying to one of the universities that the mentor or video speaker attended by 14 percentage points ($p = 0.004$). Treatment effects in our experiment are concentrated among low-SES students. We find a 17 percentage point ($p = 0.007$) increase in the probability of applying to the treatment university for low-SES students, while the effect is substantially smaller and insignificant for high-SES students ($b = 0.089, p = 0.258$).

The effects of social exposure operate through the provision of new information about elite universities, rather than students seeking out proximity to their friends. While video clips provide participants with information, they generally do not result in a social connection to a specific university as students cannot contact the video speakers. One-on-one calls with mentors, on the other hand, can plausibly generate a friendship between mentors and participants. Because mentors' universities are generally not the same as those in the videos, we can separately estimate exposure effects from videos and mentors. If breakthrough effects are driven by students following their friends rather than information frictions, we would expect videos to have no effect on students' application choices. Instead, we find that videos and mentors are equally effective at inducing applications.

Consistent with our breakthrough results in the administrative data, we find that demographic similarity amplifies treatment effects. Mentor treatment effects are driven entirely by participants randomized into T2, which assigned participants to demographically similar mentors. A similar mentor raises the probability of applying to the mentor's university by 21 percentage points ($p = 0.001$), while a dissimilar mentor has no effect. Video treatment effects are also stronger among participants who are demographically similar to the video speakers. Our experimental results suggest that the heterogeneity in breakthrough effects by similarity arises because interactions with more similar peers have larger effects on beliefs, rather than because similar peers are more likely to interact.

Social exposure induces a positive shift in belief updates about the probability of social fit but has no effect on beliefs about admissions or graduation. Low-SES students in the treatment arm have a more optimistic distribution of belief updates about the probability of fitting in compared to low-SES students in the control arm ($p = 0.024$). We find no effects on belief updates about admissions probability ($p = 0.732$) or successful graduation ($p = 0.386$). These results are consistent with our finding that there are no socioeconomic

discrepancies in these beliefs at baseline. Among high-SES students, the distribution of belief updates about the probability of fitting in is statistically indistinguishable between treatment and control arms ($p = 0.944$). Social exposure also has no effects on high-SES students' belief updates about the likelihood of admission and graduation. These findings shed light on why breakthrough effects have no spillover effects on other elite universities. While Oxford and Cambridge have similar admission requirements, they may be seen to differ in social atmosphere.

During conversations with mentors, participants are more likely to ask about fitting in at university than to discuss careers, teaching quality, or advice about choosing a major. We surveyed mentors to collect information about the topics discussed during the calls. Mentors reported that the two most commonly discussed topics, which came up in 70% of the conversations, were “student life” and “fitting in”. In 57% of the calls, mentors indicated that “whether they would fit in and enjoy their time” was the single most important factor. While descriptive in nature, these results align with our previous findings that the effects of exposure operate through changes in perceptions of the social environment.

Taken together, our paper shows that lack of exposure to elite universities—and the resulting negative perceptions about social fit—discourages applications from low-SES students who would see substantial labor market benefits from attending these universities. Low-SES students' application choices are far from immutable and are highly responsive to encountering an individual who attends an elite university. These shifts in application choices operate through learning about aspects of the university experience that are specific to a particular university and demographic group, such as *whether someone like them would fit in*. When perceptions about the social environment at elite universities deter applications, policies aimed at lowering admissions thresholds for disadvantaged groups may be insufficient to close socioeconomic gaps at elite universities. Our findings suggest that interventions offering social connectedness at scale can be an effective way to address disparities resulting from socioeconomically stratified school networks. Back-of-the-envelope calculations suggest that equalizing exposure to elite universities can close up to 30% of the socioeconomic gap in applications.

We make three primary contributions in this paper. First, we build on the extensive literature showing that university enrollment responds to the destinations of family members (Altmejd et al. 2021; Barrios-Fernández 2022), neighbors (Avdeev et al. 2024), and schoolmates (Bechichi and Kenedi 2024; Cattán, Salvanes, and Tominey 2025). Most closely related to our paper, Bechichi and Kenedi (2024) use a fuzzy regression discontinuity design in France to show that an older student enrolling increases the probability of applications from that school to the same college-major pair by 19%. Cattán, Salvanes, and

Tominey (2025) show that idiosyncratic variation in classmates with elite parents affects enrollment at elite universities and earnings outcomes. We extend this literature by providing novel experimental evidence on the underlying mechanisms behind these peer effects. In particular, we demonstrate that learning about whether someone like them would fit in at a given elite university is a key mechanism.

Second, we bridge the literature establishing peer effects in university choice with the undermatching literature and the work on returns to elite universities (Dale and Krueger 2002; Zimmerman 2014; Britton et al. 2022; Chetty, Deming, and Friedman 2026; Barrios-Fernández, Neilson, and Zimmerman 2026). A large body of work has established undermatching, where low-SES students enroll in lower ranked universities compared to high-SES students with similar test scores (Hoxby and Avery 2012; Hoxby and Turner 2013, 2015; Black, Cortes, and Lincove 2015; Sanders, Chande, and Selley 2017; Sanders et al. 2018; Dillon and Smith 2020; Andrews, Imberman, and Lovenheim 2020; Campbell et al. 2022; Cohodes, Ho, and Robles 2022; Wyness 2023). Past literature has discussed a wide range of potential factors that drive undermatching, including information frictions about financial aid (Dynarski et al. 2021); choice architecture (Pallais 2015); inaccurate beliefs about admissions chances or academic ability (Dizon-Ross 2019; Larroucau et al. 2025); and access to school counselors (Mulhern 2023; Barr and Castleman 2025). The sociology literature provides qualitative evidence on perceptions of social fit at elite universities (see for instance Jack 2020). Yet there is limited evidence quantifying how beliefs about social fit affect socioeconomic disparities in elite university applications. We present evidence for an overlooked mechanism, *social fit and belonging at elite universities*, while ruling out information frictions about admissions chances and labor market returns in our context.

Finally, our paper also contributes to the literature on social learning beyond the economics of education. Banerjee (1992) characterizes herding equilibria under rational social learning, and Eyster and Rabin (2010) characterize herding under bounded rationality. Malmendier, Oprea, and Veldkamp (2026) provide theoretical and empirical evidence on when learning is amplified by demographic similarity. Chetty et al. (2022) show that social interactions between low-SES and high-SES individuals are one of the strongest predictors of income mobility. We provide empirical evidence for a context where information frictions result in socially inefficient herding: low-SES students forgo substantial labor market benefits by not applying to elite universities. We further demonstrate that this can be corrected by experimentally providing social exposure to demographically similar individuals at elite universities.

II. Setting and data

A. UK Educational System

The UK educational system has several features that make it uniquely advantageous to study social exposure as a barrier to university access. Further details on the UK educational system can be found in the Online Appendix [OA2](#).

First, university applications and admissions are centralized and largely based on performance on a series of standardized exams embedded in the core school curriculum. Students in England and Wales take compulsory General Certificate of Secondary Education (GCSE) exams at age 16.¹ Students spend the last two years of secondary education completing academic qualifications (A-levels) or vocational qualifications. Upon application, students submit a short personal statement, GCSE grades, and predicted grades in each of their A-Level or vocational subjects.² All applications go through the Universities and Colleges Applications Service (UCAS) platform. Universities independently decide whether to make an offer to an applicant or reject them outright, and final university placements are decided based in part on students' final grades.

Second, financial frictions are less relevant to university choice than they are in countries such as the US, as university tuition is uniform across universities. Tuition for domestic students was capped at £3,000 from 2006–2011 and increased to £9,000 in 2012, with irregular increases thereafter generally below the rate of inflation. Essentially all universities charge tuition fees exactly at the cap, meaning that there is no variation in tuition between universities.³ Financial aid is also widely available and standardized, with government loans that are repaid via automatic deduction from paychecks based on income bracket.

Third, the application set is a high-stakes decision, as students can apply to *at most* five university-major pairs, called “courses” in a given cycle.⁴ This cap on the number of applications raises the likelihood that academically qualified students omit elite universities from their application portfolio.

Finally, school-to-university pipelines are quite dense in the UK, which restricts students'

1. We use GCSE grades in the ‘core’ subjects of Math, English, and Science, converted to a percentile, throughout the administrative data analysis as a measure of students’ academic ability.

2. Students typically apply to university and receive decisions before they complete their A-Level exams or vocational qualifications, so a standard component of the application system is that teachers assign their students predicted grades in each of these subjects that are reported to universities as part of students’ application. These grades are noisy and generally upward-biased (Murphy and Wyness 2020). See (Tadjfar and Vira 2025) for more details on how predicted grades and test scores differentially impact university outcomes.

3. Scottish universities have no tuition associated for Scottish students, but students from England who attend these universities pay the same tuition fees that they would pay at English universities.

4. This cap typically binds, with 80% of students in our administrative data exhausting the five courses.

social exposure to elite universities. The Universities of Oxford and Cambridge, commonly denoted by the portmanteau “Oxbridge”, enroll as many students from just eight elite high schools as they do the remaining 3,000 (Sutton Trust 2025). Elite universities in the UK have also been reported to create socially isolating environments for socioeconomically disadvantaged students.⁵

B. Administrative Data

We draw on four UK administrative datasets using the Longitudinal Education Outcomes database developed by the Department for Education (Office for National Statistics 2023). The National Pupil Database (NPD) provides data on students attending English schools, including demographics, test scores, school type, and subjects taken. The Universities and Colleges Applications Service (UCAS) provides data on applications to undergraduate universities, offers, student responses to offers, and final offers accepted.⁶ For university graduation and performance outcomes, we link these with data from the Higher Education Statistics Agency (HESA), which collates annual university-level data on student enrollment, graduation, degree class, course studied, and various other details about outcomes at university. For long-run labor market outcomes, we link data from HM Revenue and Customs (HMRC). HMRC provides data on employment spells, pseudonymized employer ID, and annual earnings, drawn from tax records.

We focus on the sample of students completing secondary education and applying to university between 2007 and 2021, which is the timespan for which we observe all four of these datasets. Within these cohorts, we apply the following sample restrictions. We restrict to students who apply to university as part of the UCAS “main scheme” and are aged 18 as of 31st August of the year of their application cycle, and exclude students who do not apply to *any* university.⁷ Online Appendix OA1 provides additional details on variable definitions and sample restrictions.

5. See for example Bhopal and Myers (2023), writing in the Guardian newspaper.

6. The UCAS dataset available in the LEO database has pseudonymised institutional identifiers, which prevents identification of particular schools or universities by name. For analysis that requires this, we draw on standalone data provided directly by UCAS that is not linkable to labor market outcomes or the NPD.

7. Specifically, we exclude students who do not apply to at least one university that can be linked with HESA data. UCAS courses that do not link with HESA data are generally more specialised courses, such as arts academies or music conservatoires, rather than traditional university courses.

C. Survey Data and Experimental Sample

We conducted a series of surveys with one cohort of UK university applicants applying in January 2026 as part of our field experiment (see Section VI.A for details on the experimental design). We recruited 21 high schools across England and Wales to participate in our study, yielding a geographically spread out and nationally representative sample.⁸ Table 1 presents summary statistics for students participating in our experiment, as well as comparable statistics drawn from the administrative data. Compared to the national administrative data, we find there are substantial differences in ethnicity (56% of students in the RCT sample are white, compared with 80% in the general population) and gender (45% of students in the RCT are female, compared to 50% in the general population and 56% among university applicants). In terms of disadvantage, RCT participants are comparable to the overall population of age-18 school leavers and somewhat more disadvantaged than the typical university applicant.

III. Socioeconomic Gaps in Elite University Enrollment

We begin our empirical analysis by documenting socioeconomic gaps in the probability of enrolling at one of the ten most academically selective universities in the UK, henceforth referred to as *elite universities*.⁹ Figure 1 reports the differences the likelihood of enrolling, applying to, and being exposed to one of the elite universities, between low-income and high-income students. We focus on students above the 90th percentile of age-16 standardized test scores (GCSE grades) since application and admission rates at elite universities are very low below this level.

The highest-achieving students from low-income backgrounds are up to 12 percentage points less likely to enroll at an elite university compared to high-income students with the same grades. Crucially, this disparity largely begins at the application stage, as low-income students are up to 12 percentage points less likely to *apply* to an elite university. If low-income students applied to elite universities at the same rate as high-income students, holding offer rates and enrollment conditional on receiving an offer fixed, enrollment

8. Schools were recruited to participate in the programme through various channels, including school networks from our partner organisation, WISE/IET, as well as direct outreach to schools. WISE (Women Into Science and Engineering) is a UK social enterprise that aims to support women’s access to STEM university courses and careers, but has a network of coeducational schools that we worked with to recruit students. We collected data through Qualtrics surveys administered to students at home and in school, as well as data on student applications and enrollment directly from participating schools.

9. Universities are ranked using the mean A-level grades of students enrolled at each university as a measure of the university’s selectivity.

disparities for the highest-ability students would fall by 40-50%.¹⁰ At the same time, low-income students are substantially less likely to encounter individuals who have attended elite universities, either through family members or schoolmates. High-achieving low-income students are 5-10 percentage points less likely to attend a school where a student has attended an elite university sometime in the last three years.

IV. A Simple Framework of Social Exposure and University Choice

We provide a simple framework to illustrate how systematic lack of exposure to elite universities can deter applications among future cohorts within a school.

A. Setup

Time is discrete and indexed by cohorts $t = 1, 2, \dots$; for simplicity we assume there is one student per cohort choosing between an elite university and a local default. The payoff to student t from attending an elite university is

$$(1) \quad v_t = \theta + \gamma + e_t,$$

where θ is a publicly known common return, $\gamma \sim N(0, \sigma_\gamma^2)$ is an unobservable *community-level match component*—how students from this community fare at the university—and $e_t \sim N(0, \sigma_e^2)$, i.i.d. across cohorts, is an *individual component* that student t observes privately before choosing. The student’s outside option is a known local default with value $\bar{u}$, so students choose by comparing the unknown components of payoff $\gamma + e_t$ to $\tau \equiv \bar{u} - \theta$.

Students have CARA expected utility preferences with risk aversion λ . Given a posterior distribution of $\gamma \sim N(\mu_t, \sigma_t^2)$, a student attends the elite university if and only if

$$(2) \quad e_t \geq \kappa_t \equiv \tau - \mu_t + \frac{\lambda}{2} \sigma_t^2.$$

B. Signals From Peer Choices

Student t observes the previous cohort’s action $a_{t-1} \in \{0, 1\}$. If the predecessor attended, her realized payoff is also observed, revealing $x = \gamma + e_{t-1}$. If she did not attend, nothing

10. The probability of enrolling at a given university through the main UCAS scheme can be decomposed as $P(\text{enroll}) = P(\text{apply}) \times P(\text{offer} \mid \text{apply}) \times P(\text{accept} \mid \text{offer}) \times P(\text{enroll} \mid \text{accept})$. Holding the last three terms at their low-income values and replacing $P(\text{apply})$ with the high-income application rate gives the counterfactual enrollment rates; taking logs gives an additive decomposition in which application gaps account for roughly 40–50% of the enrollment gap.

further is observed. Each cohort updates from the common prior using a simple theory of the predecessor's behavior: it treats the predecessor as having known her own total match $\gamma + e_{t-1}$ and attended if and only if it exceeded τ . Cohorts model their predecessors as making decisions under certainty, rather than as uncertain social learners themselves.¹¹ Let $\sigma^2 \equiv \sigma_\gamma^2 + \sigma_e^2$, $r \equiv \sigma_\gamma^2 / \sigma^2$, $\beta \equiv \tau / \sigma$, and let $h(\beta) \equiv \phi(\beta) / \Phi(\beta)$ denote the inverse Mills ratio.

Observing non-attendance. A student who observes $a_{t-1} = 0$ infers (possibly incorrectly) that $\{\gamma + e_{t-1} < \tau\}$. The event truncates beliefs about $\gamma + e_{t-1}$; projecting onto γ then yields¹²

$$(3) \quad \mu^- = -r\sigma h(\beta) < 0, \quad (\sigma^-)^2 = \sigma_\gamma^2 [1 - r(\beta h(\beta) + h(\beta)^2)].$$

so non-attendance is an unambiguously negative signal. As the payoff is not observed directly, the variance reduction is strictly smaller than that from observing an attendee (below); under risk aversion, this is an additional force that leads non-attendance to discourage future applications.

Observing attendance. The realized payoff reveals $x = \gamma + e_{t-1}$ exactly. The observer cannot decompose the sum, and optimally attributes fraction r of it to the community:

$$(4) \quad \mu^+ = rx, \quad (\sigma^+)^2 = \sigma_\gamma^2 (1 - r) < (\sigma^-)^2.$$

Attendance may move the mean in either direction depending on the realized outcome, but always delivers the larger variance reduction.¹³

11. This naïve interpretation of predecessors' actions is similar to Eyster and Rabin (2010). Without this assumption, a history of non-attendance would carry no information about γ if social learners were themselves uncertain, as students would understand that past cohorts in communities with a history of non-attendance would not have any more information than they do about the community-level payoff γ .

12. The posterior on γ itself is not truncated — any value of γ remains possible if the predecessor's individual draw was sufficiently low — but is smoothly shifted left, with density proportional to $\phi(\gamma/\sigma_\gamma) \Phi((\tau - \gamma)/\sigma_e)$. The moments in (3) are exact.

13. We can relax the assumption that sharing a community is a binary by introducing imperfect correlation between community payoffs: assume γ_{t-1} and γ_t are jointly normal with variance σ_γ^2 and covariance $\rho \in [0, 1]$. A factor structure generates ρ from primitives: let $\gamma_i = s_i'g$, where g collects match components shared at different levels of social aggregation (school, neighbourhood, class background, gender, ethnicity), $g \sim N(0, \Sigma)$, and loadings are normalised so that $s_i' \Sigma s_i = \sigma_\gamma^2$ for all i . Then $\rho_{ij} = s_i' \Sigma s_j / \sigma_\gamma^2$: similarity is overlap in group membership. Weighting others' experience by the covariance between their circumstances and one's own parallels the technology-learning weights in Conley and Udry (2010). For $\rho < 1$ the observer extracts information about γ_i from $\gamma_j + e_j$. Then $\mu^- = -\rho r \sigma h(\beta)$, $(\sigma^-)^2 = \sigma_\gamma^2 [1 - \rho^2 r(\beta h(\beta) + h(\beta)^2)]$, $\mu^+ = \rho r x$, $(\sigma^+)^2 = \sigma_\gamma^2 (1 - \rho^2 r)$ with $(\sigma^+)^2 < (\sigma^-)^2$ for every $\rho > 0$.

C. Dynamics: Persistent Non-Attendance and Breakthrough Students

Because each cohort starts from the prior and observes only the adjacent cohort, behavior follows a two-regime Markov process with cutoffs

$$\kappa^- \equiv \tau + \frac{\lambda}{2}\sigma_\gamma^2 + \lambda^{-1} \log \left[\frac{\Phi(\beta + \lambda r \sigma)}{\Phi(\beta)} \right], \quad \kappa^+(x) \equiv \tau - \mu^+ + \frac{\lambda}{2}(\sigma^+)^2,$$

implied by (2)–(4). Let $\kappa_0 \equiv \tau + \frac{\lambda}{2}\sigma_\gamma^2$ denote the cutoff under prior beliefs.

Proposition 1 (Persistence of non-attendance). $\kappa^- > \kappa_0$: *following a non-attendance, the probability of a further non-attendance strictly exceeds the probability of non-attendance under prior beliefs.*

Non-attendance in period $t - 1$ thus increases non-attendance in period t , and can originate in a single unlucky individual draw: an idiosyncratic reason not to attend is misread by the next cohort as evidence about the community match, raising its own bar to attend, and so on. Conditional on true γ , beliefs in the non-attendance regime are systematically too pessimistic (as long as true γ is weakly positive) and communities with the same γ make persistently different choices based on early idiosyncratic history.

Proposition 2 (Breakthrough). *Upon attendance with realized outcome $x = \gamma + e_{t-1}$, the next cohort's beliefs jump to (4): the mean is re-anchored to $\mu^+ = rx$, a quantity containing the true γ , and the variance falls discretely, $(\sigma^+)^2 < (\sigma^-)^2$ and following any breakthrough with outcome above $\bar{x} \equiv -\sigma h(\beta) - \frac{\lambda}{2r}[(\sigma^-)^2 - (\sigma^+)^2] < 0$, the probability of attendance strictly exceeds its level in the non-attendance regime.*¹⁴

We refer to a student who exits the non-attendance regime as a *breakthrough student*; these individuals have sufficiently high idiosyncratic draws to overcome their community's pessimism.

V. Breakthroughs to Elite Universities

A. Empirical Specification

To understand whether and how social exposure to elite universities influence application choices, we use quasi-random variation in exposure generated by university-specific “breakthrough” events schools. Consider two schools, School A and School B, that have

14. The non-attendance regime is persistent but not absorbing. In each cohort of the regime, attendance occurs with probability $1 - \Phi(\kappa^-/\sigma_e) \in (0, 1)$, so the regime is exited in finite time almost surely.

both had no student attend University X for several years. If a student from School A attends University X, we refer to this as a breakthrough event to University X at School A, and refer to the student who first attends as a breakthrough student. This is a discrete change in the exposure to University X for future cohorts at School A. We examine how university applications at school A change relative to application patterns at school B following this breakthrough.

We pool our analysis in an event study design to exploit this variation across the large number of schools and universities in UK administrative data. To implement the strategy outlined above, we identify, for each university, the schools where no students enrolled at the university between 2007 and 2009, the first three years of our data. We then define the first year a student attends that university from that school as the breakthrough year for that school. If no student attends the university during the period covered in our data – from 2010 to 2021 – we assign that school to the control group for that university. By construction, no students from any school in the sample are enrolled at that university before their school’s event year, and no students from control schools are enrolled at that university at any point in the sample. Our estimating equation is then:

$$(5) \quad Y_{ist} = \alpha_s + \gamma_t + \delta X_{ist} + \sum_{\tau \neq -1} \beta_{\tau} \mathbb{I}(t - T_s = \tau) D_s + \varepsilon_{ist}$$

where X_{ist} is a vector of individual-specific covariates, T_s is the year in which school s had a breakthrough and D_s is the treatment indicator. Our primary specification omits individual-level covariates to show trends in applications at the affected universities transparently, but in robustness checks included in appendix figures, we include in X_{ist} students’ GCSE percentile (see Appendix OA1) and indicators for the number of A-levels they took, as well as the number of A-levels in facilitating subjects.¹⁵ We exclude the breakthrough student from the sample (or if there are multiple breakthroughs to the same university in the same year, we drop one of the breakthrough students), in order to isolate the effect of a breakthrough on the rest of the breakthrough student’s cohort in period $\tau = 0$. To allow for heterogeneous treatment effects by treatment year, we use the Sun and Abraham (2021) estimator for event studies.

To provide a more aggregated picture of the effects of breakthroughs, we stack breakthroughs across 10 elite universities. We construct a dataset for each university and stack them, indexing data from each by the breakthrough university u , and then run the following

15. These are subjects that the Russell Group of universities identify as most strongly supporting applications to elite universities.

stacked event study regression¹⁶:

$$(6) \quad Y_{istu} = \alpha_{su} + \gamma_{tu} + \delta_u X_{istu} + \sum_{\tau \neq -1} \beta_{\tau} \mathbb{I}(t - T_{su} = \tau) D_{su} + \varepsilon_{istu}$$

We also report results for certain outcomes using the analogous difference-in-differences specification of these effects, which pools effects across the post-treatment periods:

$$(7) \quad Y_{istu} = \alpha_{su} + \gamma_{tu} + \delta_u X_{istu} + \beta \mathbb{I}(t - T_{su} \geq 0) D_{su} + \varepsilon_{istu}$$

B. Shifts in Applications of Future Cohorts

First, we present an illustrative example using breakthroughs to the universities of Oxford and Cambridge. We use these two universities as an example of the university-level event study design for two reasons: first, they are generally considered to be particularly elite universities; second, they are generally seen as *equally* selective and elite, to the extent that they are often just referred to by the portmanteau ‘Oxbridge’. Figure 2a plots coefficients on the relative time indicators from (5), where the treated schools are those that experience a breakthrough to Cambridge and the outcomes are application to Oxford and Cambridge as indicated; Figure 2b does the same, but for schools that experience a breakthrough to Oxford. We see an increase in applications to Cambridge in panel (a) at the year of the breakthrough to Cambridge, rising by 0.6–0.8 percentage points, while the application rate to Oxford does not significantly increase. Similarly, in panel (b), there is an increase of 0.4–0.8pp in applications to Oxford following the breakthrough to Oxford while the application rate to Cambridge stays largely constant. The effects persist at least four years after the breakthrough, which may reflect the creation of a pipeline, in which students apply to and attend the university at higher rates in the years after the breakthrough.

This pattern illustrates our key finding, which generalizes across universities: following a breakthrough to a particular university at a school, applications to that university persistently increase, but applications to comparable, similarly-ranked universities do not. This rules out any explanation that attributes the increase in applications following a breakthrough to a common shock that would affect applications to a range of universities.

16. A given student may appear multiple times in the stacked dataset if their school sees breakthroughs to, or is in the control group for, multiple universities. We cluster standard errors at the school level (rather than the school-by-breakthrough-university level), which accounts for correlation within a school across the breakthrough university samples (Wing, Freedman, and Hollingsworth 2024).

Ex ante, many plausible explanations for the effect of breakthroughs to Cambridge on applications to Cambridge would center on information about these procedures that would also be informative about and encourage applications to Oxford. Yet instead we find effects concentrated at Cambridge, and vice versa for breakthroughs to Oxford; any mechanism that explains the effects of breakthroughs must therefore be highly *university-specific*.

If applications to breakthrough universities increase following a breakthrough, this must draw applications away from other universities, given that the cap of five applications is typically binding in the UK. To understand where applications are drawn from, we construct a set of mutually exclusive and exhaustive outcomes based on students' portfolios of five applications. These are portfolios that include the breakthrough university; portfolios that exclude the breakthrough university, but include a different university in the same 5-university selectivity tier; portfolios that include no universities from the breakthrough tier but include at least one university from a higher tier; and portfolios that include only applications from lower tiers than the breakthrough university.

Figure 2c reports the results of this analysis, pooling across the top 10 universities. First, Figure 2c shows that the breakthrough effects to Cambridge and Oxford generalize to a broader set of elite universities. We see substantial and persistent increase in applications of 0.4–0.6 percentage points pooled across these universities. We also observe that there is no change in the probability of applications to higher-ranked universities or universities of the same rank, consistent with the result that applications to Oxford do not change in response to a breakthrough to Cambridge and vice versa. Instead, the applications to the breakthrough draw applications away from lower-ranked universities. Breakthroughs to elite universities increase the ambition of future students' application portfolios. Students exposed to a breakthrough event at their school apply to the breakthrough university when they would otherwise have applied to lower-ranked universities.

To benchmark the size of these effects, we quantify the effect that equalizing exposure across income groups would have on application rates given the observed effect sizes using a back-of-the-envelope exercise. We find that around 30% of the overall socioeconomic disparity in applications can be explained by differences in exposure, with the largest effects among students with the highest GCSE grades. Details of the calculation are provided in Online Appendix Section OA4.

Finally, we assess the extent to which an increase in applications goes on to increase enrollment at and graduation from the university in question. Figure 2d reports event study coefficients for applications, receiving an admission offer, enrolling, and graduating successfully from the breakthrough university. Enrollment increases by 0.3–0.4 percentage points and constitutes 60–85% of the increase in applications. Furthermore, among those

induced to enrol by the breakthrough, almost all students graduate successfully, as the coefficients in the enrolment and graduation event studies are nearly identical.

C. Early-career Earnings Outcomes

Virtually all of the students who enroll at an elite university following a breakthrough go on to graduate successfully. These students may nevertheless fail to capture labor market gains from attending an elite university (for instance, see Michelman, Price, and Zimmerman 2022). To address these concerns, we use a matched controls design to directly examine earnings gains several years out of university.¹⁷

We identify students who apply to the breakthrough university following a breakthrough at their school and compare them with untreated students who are matched on a wide set of covariates.¹⁸ We conduct two different matching procedures. In the first, induced applicants are matched with students from untreated schools applying to university for the same major in the same year; in the second, induced applicants are matched with students from the same school in the pre-breakthrough cohorts. Within each of these sets, we match exactly on students' quintile of neighborhood income, ventile of standardized test scores (GCSE grades), and an indicator for taking at least three A-Levels. Finally, we select one nearest neighbor from within the exactly matched set for each student, based on the Mahalanobis distance over gender, ethnicity, region, and standardized test scores (GCSE grades). Induced applicants who have no available exact matches on the relevant variables are discarded from the estimation.

With these matched control and treatment observations, we estimate the returns to applying to an elite university by regressing earnings at ages 22, 25, and 27 on matched-pair fixed effects and an indicator for applying. We report estimates in Figure 3. By age 27, induced applicants have earnings £4,414 higher than matched controls from control group schools, and £4,003 higher than matched controls from pre-treatment years at their own school (~ 15% increase relative to mean earnings). These are comparable to other estimates; Britton et al. (2022) estimate returns to elite UK universities by controlling for observable test scores and other characteristics and find that age-30 earnings are 5–35%

17. Given the relatively small magnitude of breakthrough effects in absolute terms, event studies using earnings as the outcome are underpowered. Similarly, a 2SLS regression instrumenting for applying to the breakthrough university with the post-breakthrough indicator yields a 95% confidence interval of around (-£12000, +£92000) for effects on age-27 earnings, which includes implausibly high positive and negative effects.

18. We refer to these students as 'induced applicants', though note that this set of students includes both compliers (who apply to the breakthrough university only because of the breakthrough at their school) and always-takers (who would have applied even without the breakthrough). Note that this includes all induced applicants, so we are not restricting to students who successfully place at the breakthrough university.

higher for students who attend elite universities.¹⁹ As a robustness exercise, we estimate effect sizes for various universities using an empirical strategy similar to Dale and Krueger (2002) controlling for application portfolios and find that the average graduate of a top 10 university earns £4,600 more than a student attending the median university (we present estimates in Appendix Figure OA5c). A simple back-of-the-envelope calculation yields a discounted lifetime earnings effect of £92,362, which substantially exceeds tuition fees.²⁰

D. Heterogeneity by Demographic Similarity and Social Proximity

How do breakthrough effects vary by extent of exposure and demographic similarity with the breakthrough student? We first consider demographic overlap in low-income status, free school meal (FSM) eligibility, gender, and ethnicity separately. Next, we examine heterogeneity by exposure using two proxies for exposure intensity: an indicator for attending same school as the breakthrough student prior to age 16 and the number of overlapping subjects with the breakthrough student.²¹ For each characteristic, we estimate a difference-in-differences version of the breakthrough design and interact the main difference-in-differences indicator with a binary indicator for sharing the specified characteristic with the breakthrough student. Figure 4 reports estimates of the main effect and the interaction effect for each dimension for breakthroughs elite universities.

We find that both demographic overlap and extent of exposure matter for breakthrough effect sizes. Demographic overlap raises the effect of a breakthrough by around 0.1–0.25 percentage points (around 30–50% of the total effect), and stronger exposure raises the effect of a breakthrough by 0.5–0.75 percentage points (64–80% of the total effect). These patterns are consistent with the transmission of information about some aspect of the university experience or application process through conversations with peers. Our results on demographic similarity may be capturing the fact that there is demographic homophily in the likelihood of friendship formation. Alternatively, larger effect sizes among demographically similar peers may reflect learning about group-specific university payoffs.

19. Specifically, Britton et al. (2022) study the returns to Russell Group universities.

20. We take the point estimates of earnings premia using each approach at ages 22, 25, and 27, and linearly interpolate earnings premia between these ages. After age 27, we assume there is no real growth in the earnings premium for induced applicants, so this premium remains constant until retirement at age 68; this is a conservative assumption given the substantial increase in the earnings premium between even ages 25 and 27. We discount the earnings premium at 3% per year back to age 22, following Angrist, Autor, and Pallais (2022).

21. We focus on school at age 16 because it is common—but not ubiquitous—for students in the UK to change school after completing their GCSEs at age 16, and attend a different school for the final two years of their education.

E. Validity

Common School-level Shocks.— One potential confounder that threatens our interpretation of the breakthrough effects is the presence of common school-level shocks. For instance, schools may receive sudden boosts in funding, improvement in student quality, or receive an influx of better teachers or guidance counselors. In Figure 2, breakthroughs induce an increase in applications to the specific university that sees a breakthrough, but not to other equally selective universities. This pattern rules out common shocks such as a sudden increase in students’ ability or academic performance, as such shocks should lead to a broader increase in the selectivity of applications. Furthermore, Table OA1 shows that our results are robust to controlling for various measures of academic ability.²²

University-specific School-level Shocks.— Another more challenging confounder is a *university-specific* common shock affecting all students at the school. For example, a breakthrough may coincide with the year when a university does an outreach event at that high school. Outreach activities are generally targeted within the local area of the university.²³ If breakthrough effects were driven by outreach, the effects would be concentrated in the geographical area of the university. Table OA2 indicates that this is not the case. Yet another possibility is that breakthroughs are caused by the arrival of a new teacher who has a connection to a specific university and encourages applications to this university. Figure 4 shows that there is a statistically significant effect of breakthroughs even on students who had no subjects in common with the breakthrough student and who were thus unlikely to interact with the same teacher. Ultimately, however, fundamental limitations of the administrative data mean that it is challenging to entirely rule out alternative mechanisms. Our field experiment addresses this limitation by inducing precisely defined exposure to past university attendees and measuring effects on beliefs and application choices.

VI. Evidence from the Field

Our field experiment is designed to isolate potential mechanisms such as changes in beliefs about admissions chances, social fit and sense of belonging, and returns to university

22. We additionally report coefficients from a matched event study design. In the matched specification, we conduct 1:1 matching of treatment and control schools based on students’ university application behaviour and ability in the pre-breakthrough window, 2007–2009, and include match-pair by year fixed effects in the specification, so effects are identified from changes in outcomes for treated schools relative to a matched control school. This reduces pre-treatment disparities in outcomes between treatment and control schools.

23. University outreach to schools is often coordinated through the ‘Uni Connect’ Programme, which is organised around 29 regional hubs across England. This programme connects universities and schools within geographical regions of the UK but does less to coordinate outreach across regions. See Burtonshaw et al. 2024 for an evaluation of Uni Connect.

and rule out potential confounds from our findings in the administrative data.

A. Experimental Design

We recruited 21 high schools in the UK to participate in our field experiment, which randomizes exposure to current university students at academically selective universities. Panel A in Figure 5 overlays our participating schools on a map of England shaded based on the local probability of not being exposed to a top 10 university.²⁴ The experiment consisted of a baseline survey (Jan–Jun 2025), an in-person workshop led by a school alumnus or current university student and a midline survey (Apr–Jun 2025), an endline survey (Oct 2025), and realized university applications data collected directly from schools (Mar–Oct 2026).²⁵ Participants randomized into one of the treatment arms additionally watch video clips from current university students and receive one-on-one mentorship through calls and emails with university students.²⁶ Panel B in Figure 5 provides an overview of the experimental design. Randomization was at the individual-level and participants were assigned to treatments at the beginning of the baseline survey. Table OA6 presents balance checks among our primary sample of students who completed both the baseline and midline surveys, pooling across the treatment arms; all covariates that we test are balanced across treatment and control groups. Our main analysis sample consists of 640 participants who were in their final year of study (Year 13), expressed some interest in applying to universities in January 2026, and took both the baseline survey and midline survey.²⁷ Appendix OA7.A provides additional details on the experiment.

Active Control: In-person Workshops— We invited all participants, regardless of treatment assignment, to attend a one-hour in-person workshop delivered by a current university student held at their school. This workshop was designed to ensure that participants had a baseline level of information about the university application process and provided statistics on average earnings of graduates from different universities and typical admissions requirements at various universities.²⁸ In principle, this helps rule out that any treatment

24. We have one participating school in Wales which is omitted from this map, since our LEO data on university access only covers England.

25. As of September 9, 2026, we have collected final outcomes for 473 of the 640 students. Some schools are still finalizing the data collection process.

26. Note that in our pre-registration, we also described a treatment offering subsidised in-person visits to universities. For various logistical reasons, this arm was substantially underpowered and we do not report results from this arm.

27. Students who indicated they will not apply to university terminated the baseline survey early and were not part of our study.

28. In addition to calibrating student beliefs by providing statistics about earnings and grade requirements

effects estimated from the experiment are explained by changes in beliefs about the average returns to university or admissions criteria as this information was provided to all participants. The workshop also discussed the application process and timeline; student loans, cost of living and student finance; and how to write personal statements. The full set of workshop slides can be found in Online Appendix [OA9](#).

Treatment: Videos from University Students.—This treatment is designed to replicate, at scale, the information that a student could learn about a particular university from a conversation with a schoolmate who went to that university. At the end of the baseline survey, participants randomized into treatment watch two video clips recorded by a subset of our mentors. We group participants into one of three academic tiers based on their grades and assign them videos within their academic tier. This ensures that treated participants watch videos from universities that are aspirational but not academically out of reach given their grades and coursework. Table [OA7](#) presents the full set of video universities and the grades corresponding to each tier. Each video is 3-4 minutes long and discusses the speaker’s university application process and experience at their particular university.²⁹

Treatment: Demographically Matched vs. Unmatched Mentors.—This treatment arm is designed to test whether participants respond differently to information from a demographically *dissimilar* university student compared to a demographically *similar* university student. Students randomized into the treatment arm T at baseline are further randomized, with equal probability at the individual level, into one of T1 and T2. Students in T1 are assigned up to two mentors that are demographically dissimilar, i.e., mentors who overlap in *none* of the following characteristics: gender, region of hometown, and ethnicity. Students in T2 are assigned up to two mentors that are demographically similar, i.e., mentors

for admissions, the workshops also served several logistical functions. Having a component of the program available to all students rather than just the treatment group was important for school recruitment, as schools would have been less enthusiastic about a program in which a large fraction of their students would get no benefit. Collect midline survey responses in person during the workshop and communicating mentor assignments also helped reduce attrition.

29. Specifically, mentors were told to answer the following: “Say your first name, secondary school attended, and city you’re from”, “Which A-level / BTEC subjects did you take? If comfortable sharing, predicted and achieved grades in A-levels / BTECs and how those affected your choices”, “Subjects you applied to, universities you applied to, where you got offers from, where you chose as your firm / insurance choice and why?”, “Who you talked about the university application process with / resources you consulted”, “How you approached your personal statement”, “Anything that surprised you about the process?”, “Do you enjoy the city and student life in the area?”, “Do you have a good sense of community at your uni?”, “Have you found it easy to make friends at uni?”, “What is your living situation, approximately how much does it cost per month?”, “Do you feel it is financially affordable?”, “How is the teaching quality and academic experience at your uni? Do you feel prepared to handle the rigour?”, “Did you know many other people with similar backgrounds going to your university?”

who overlap in *at least one* of the following characteristics: gender, region of hometown, ethnicity.³⁰ Participants in T1 and T2 receive their mentor assignments at the end of the in-person workshop after completing the midline survey. Mentors reach out to students over email and schedule time for a call. We recruited 245 volunteer mentors who either currently attend or previously attended a UK university for their undergraduate degree.³¹ Table OA8 presents summary statistics for our mentor pool; the vast majority of our mentors are current university students under the age of 25.

Outcomes.— The baseline, midline, and endline surveys collect the universities a student wishes to apply to, as well as their beliefs about various academic and social aspects of universities, allowing us to document belief evolution and changes in application portfolios over the course of the experiment. Specifically, for each of Oxford University and one of the video universities (randomly selected), the surveys ask students to report the percentage chance that they would (a) get an offer from, (b) make friends and fit in at, or (c) graduate successfully from each university. Student demographic characteristics and academic history (e.g., teacher-predicted grades and coursework) are collected in the baseline and midline surveys. We collect realized application outcomes directly from schools.

B. Socioeconomic Gaps in Baseline Beliefs about Admissions, Academic Success, and Social Fit

We begin by exploring socioeconomic gaps in beliefs about Oxford University as a benchmark for elite universities in our experimental sample.³² Panel A in Figure 6 shows that low-SES students, defined here as either first-generation university attendees or low-income students, have pessimistic beliefs about their probability of fitting in at Oxford compared to their high-SES counterparts with similar grades (we control for A-level tariff points in the regressions). On average, low-SES students have a 9 percentage point ($p = 0.001$) lower perceived probability of fitting in and making friends at Oxford than other students. However, we do not see similar patterns for the probability of receiving an

30. Participants listed up to three universities from which they would like to receive a mentor. Mentor assignment was subject to capacity: one from one of the universities that they requested, and one from an academically matched university that they did *not* request, where we use the same procedure to academically match participating students and universities as we do for the video treatment. Appendix OA7.A provides additional details on our mentor assignment algorithm.

31. Mentors were recruited from a variety of sources, including Sutton Trust, directly from several universities, DiscoverEconomics, as well as from STEM Ambassadors, a STEM-focused volunteering platform that we partnered with for recruitment.

32. We ask about Oxford to have a benchmark for students' beliefs about a particular elite university that all students are asked about; none of our participating schools are in or near Oxford, so no schools would have a particular local connection to this university.

offer or graduating successfully from Oxford.

Our baseline surveys also ask students to report what they believe are “typical” A-level grades achieved by students who attend Oxford University. This allows us to compare students’ perceptions of academic requirements against ground truth measures of admitted cohorts. Panel B in Figure 6 shows that low-SES students are well-calibrated about grade requirements at Oxford. If anything, low-SES students are slightly overly optimistic, with around 30% estimating grades that are below the 25th percentile.

Because we do not have a measure of ground truth for social fit, socioeconomic gaps in beliefs about the chances of fitting in may reflect *correct* beliefs about negative social experiences for low-SES who attend elite universities. While this is certainly possible and in line with qualitative evidence from Ivy League universities in the US (Jack 2020), our results suggest that students are *overly* pessimistic in the absence of exposure. First, in our experimental sample, beliefs about social fit become more optimistic in response to our treatments as discussed in Section VI.E. Second, breakthrough effects estimated in Section V.B persist in subsequent years. If a breakthrough student had a negative social experience, we would expect the effect to decay in subsequent years. Our preferred interpretation of the results is that low-SES students are *overly* pessimistic about their social fit, but we cannot test this directly.³³

These patterns are consistent with the notion that lack of exposure in the sense of persistently observing *non-attendance* to an elite university can generate pessimism as outlined in our framework in Section IV. Moreover, this lack of exposure primarily affects beliefs about aspects of universities that are demographic-group specific and highly university specific, such as social fit. We do not find any evidence that less exposed students are pessimistic or otherwise less informed about more general aspects of elite universities that are likely to be common across demographic groups and transferable across universities.

C. Results: Shifts in University Applications

Our primary outcomes are indicators for whether a student applied to (i) either of their two assigned video universities, (ii) either of their assigned mentor universities, and (iii) any of these assigned “treatment” universities. We assign latent video and mentor universities to all participants to ensure that outcomes are defined regardless of treatment status. We measure applications using records that we collect directly from schools, supplemented

33. Beyond our own results, an external survey (Neves, Stephenson, and Armstrong 2026) found that 31% of Russell Group university attendees felt that university had been better than expected while only 11% said it had been worse, and students who said it was better than expected frequently mentioned social factors in free-text explanations of this response.

with students’ endline reports of their submitted applications where school records are not yet available.³⁴ Our preferred specifications report estimates among the 567 students (89% of the sample) who did not explicitly indicate that they do not wish to participate in the mentorship treatment; results are similar, but slightly noisier, when estimated on the full sample (Appendix Figures OA2 and OA3).³⁵

Intent-to-Treat.—We run the following intent-to-treat (ITT) specification:

$$(8) \quad Y_i = \alpha + \beta T_i + \varepsilon_i$$

where Y_i denotes the application outcome of interest for student i and T_i is an indicator for being randomized into either treatment arm (pooling T1 and T2).³⁶ Since T_i is randomly assigned, β identifies the causal effect of *assignment* to videos and mentors regardless of whether the student took up the treatment. When we estimate effects separately for the two mentor arms, we replace T_i with separate indicators for assignment to T1 (demographically dissimilar mentors) and T2 (demographically matched mentors).

Panel (a) of Figure 7 presents the ITT estimates. Treatment assignment increases the probability of applying to at least one assigned treatment university by 14 percentage points ($p = 0.004$). Turning to the individual components, treated students are 10 percentage points ($p = 0.008$) more likely to apply to one of their two assigned video universities and 11 percentage points ($p = 0.032$) more likely to apply to one of the mentor universities. The result that both videos and mentors are about equally as effective in inducing applications suggests that the breakthroughs discussed in Section V.A are not driven by students wishing to attend a university at which they already have a friend. While the mentor treatment could plausibly generate a friendship, watching a video clip without interacting with the speaker does not. This indicates that it is *new information* rather than a new friendship that generates shifts in university applications.

Assignment to a demographically matched mentor (T2) increases applications to the

34. Of the 473 students in our sample with endline outcomes, 395 are sourced from school records and 78 are supplemented with endline survey results. Our results are very similar when we exclude the 78 students.

35. Specifically, we exclude students who answered “Definitely not” or “Probably not” when asked at baseline whether they would like to be matched with a mentor. Estimates in the full sample ($N = 640$) are positive but attenuated, consistent with limited scope for effects among students with no interest in engaging with the program.

36. We pre-registered relevant pooled comparisons between these arms, as well as the comparisons of individual treatment arms. Following the discussion in Muralidharan, Romero, and Wüthrich (2025), pooled comparisons should be interpreted as a weighted average of the effects of one treatment averaging over assignment to other treatments – so the pooled comparison of (T1, T2) vs. C can be interpreted as the effects of mentor access pooling across whether or not the student was assigned a demographically similar mentor.

mentor universities by 21 percentage points ($p = 0.001$), whereas assignment to a demographically dissimilar mentor (T1) has no detectable effect, with a point estimate of 0.4 percentage points ($p = 0.952$). Our breakthrough results (Section V.A) could not distinguish between demographic *homophily in friendship* (i.e., students are more likely to interact with the breakthrough student when demographically similar) and *differential responsiveness* to information as an underlying mechanism for the observed heterogeneity by demographic overlap. Our experimental results reveal that students are indeed more responsive to information when it comes from a demographically similar mentor. Similarly, Appendix Table OA4 shows that treatment effects of videos on applications are concentrated among demographically matched³⁷ video speakers, although variation in video demographic match was not formally randomized.

Local Average Treatment Effects.—We instrument take-up with random assignment and estimate the local average treatment effect using two-stage least squares (2SLS):

$$(9) \quad D_i = \gamma_0 + \gamma_1 T_i + u_i$$

$$(10) \quad Y_i = \delta_0 + \delta \widehat{D}_i + v_i$$

where D_i denotes take-up of the treatment corresponding to the outcome (watching the assigned videos for the video-university outcome, any interaction with a mentor for the mentor-university outcome, and either form of engagement for the pooled outcome).³⁸ The coefficient δ identifies the local average treatment effect (LATE) for compliers (Imbens and Angrist 1994); with one-sided noncompliance, compliers are simply treated students who take up the treatment, so the LATE coincides with the average effect of treatment on the treated (TOT). For the arm-specific mentor estimates, we restrict the sample to the control group and the relevant arm and instrument take-up with that arm’s assignment indicator.

Panel (b) of Figure 7 presents the LATE estimates. Students who take up either treatment are 21-22 percentage points more likely to apply to a treatment university. Interacting with a demographically matched mentor raises the probability of applying to a mentor university by 49 percentage points ($p = 0.001$) among the treated, while the corresponding estimate for demographically dissimilar mentors is null. Take-up was higher in the dissimilar arm (42%) than in the similar arm (35%), suggesting that the null effect of dissimilar mentors

37. Since all treated participants saw both a male and a female speaker in their two videos, we define demographic match for videos based on just ethnicity and region of the UK.

38. We designate students who spend at least 100 seconds on each video as having “watched” the videos. Results are robust to longer horizons such as 200 seconds.

is not driven by differences in take-up.

D. Results: Heterogeneity by SES

Our framework predicts that treatment effects should be concentrated among low-SES students: they are the students who lack social exposure to elite universities and, as documented in Section VI.B, the students who hold pessimistic beliefs about their social fit at such universities. The video and mentor treatments provide precisely the university-specific social exposure that low-SES students may lack in their existing networks. Figure 8 presents ITT and TOT estimates separately by socioeconomic status. We find that treatment effects are concentrated among low-SES students across both video and mentor treatments. Low-SES point estimates are two to four times larger than high-SES ones across outcomes. In the control group, low-SES students apply to their assigned video universities at a rate of 11%, compared with 17% among high-SES students; the treatment effect for low-SES students is large enough to close this baseline SES gap in applications.

E. Mechanisms: Shifts in Beliefs about Social Fit

The socioeconomic gaps in social beliefs about Oxford at baseline along with large treatment effects among low-SES students suggest that changes in application behavior are likely mediated by rightward shifts in beliefs about the social environment at elite universities.³⁹ We now directly examine treatment effects on beliefs about admissions chances, social fit, and graduation probability separately by SES.

Figure 9 illustrates the full distributions of belief updates between the baseline and midline survey for low-SES students. We see a distinct rightward shift in the distribution of belief updates about fitting in and making friends for students who received the video treatment relative to those who didn't, shown in panel (a), with no effect on the probability of receiving an offer or graduating successfully (panels (b) and (c)). We reject equality in distributions between treatment and control for social belief updates ($p = 0.024$), but not for beliefs about offer likelihood ($p = 0.732$) or graduation probability ($p = 0.386$). We find no effects among high-SES students in any of the three beliefs (Figure 10). A placebo analysis finds no statistically significant shifts in any of the three beliefs about Oxford University among low-SES students (Figure OA1).⁴⁰ As a robustness check, Table

39. If lack of exposure resulted in uncertainty, university application choices may still shift in response to acquisition of new information among risk averse students.

40. None of the videos discussed Oxford, so university-specific belief updating should have no effect on beliefs about Oxford. However, similarly elite universities (including the University of Cambridge) were included, so the fact that we find no updating about Oxford indicates the university-specific nature of belief

OA5 reports regression estimates with positive updates, negative updates, no updates, and belief gaps as outcomes. We find no statistically significant treatment effects on any outcome among high-SES students. Among low-SES students, we find positive treatment effects on positive updates and negative treatment effects on negative updates. As with application effects, treatment effects on beliefs are concentrated among students who saw a demographically matched video speaker.

F. Mechanisms: Topics Discussed with Mentors

In addition to these outcomes collected from student surveys, we conduct surveys of mentors who have contacted students, and ask them to provide information about their discussions with each of their mentees.⁴¹

Table 2 reports the topics that students discuss with mentors, as reported by mentors in a survey.⁴² The two most commonly discussed topics are student life at the mentor’s university (70% of conversations) and life at university and fitting in (68% of conversations). Thus in these organic conversations students are most likely to want to discuss aspects of social fit, substantially more than advice about choosing applications (53%), careers after university (36%), or how to succeed academically at university (30%). Panel (b) reports what mentors perceived as the importance of different factors to their mentee’s university application decisions.⁴³ The factor with the highest average importance reported was whether students would fit in and enjoy their time at university, with a mean importance of 3.8, and in a majority of conversations (56%) mentors reported a weakly higher importance score for this factor than any other factor. Prospects of academic success on the course, careers, prospects of getting an offer and course content were all reported as less important.

This is consistent with the results from Section VI.B illustrating that video exposure affects social beliefs more than offer or graduation beliefs; both sets of results indicate that the social environment at university is the primary topic that students seek to learn about from interactions with past students.

shifts, consistent with our previous findings with breakthrough students in Section V.A.

41. As discussed in section VI.A, we did not prescribe topics for students to talk about with mentors. The topics that students discuss with their mentors thus reflect what students are most interested in learning from past university attendees and choose to discuss with them. They are incentivized in the sense that students need to actually ask mentors about a topic to learn relevant information. This avoids potential demand effects that would arise if we directly ask students what they are considering in their university applications, as students may think they are ‘supposed’ to decide based on factors like course content and teaching quality that are usually reported in university guides, and respond with these answers.

42. We ask mentors to select the topics from a multiple-choice list that they discussed with each of their mentees; the topics selected are reported in panel (a).

43. For each of the factors listed in panel (b), mentors reported importance on a 5-point scale from ‘Not at all important’ to ‘extremely important’.

VII. Conclusion

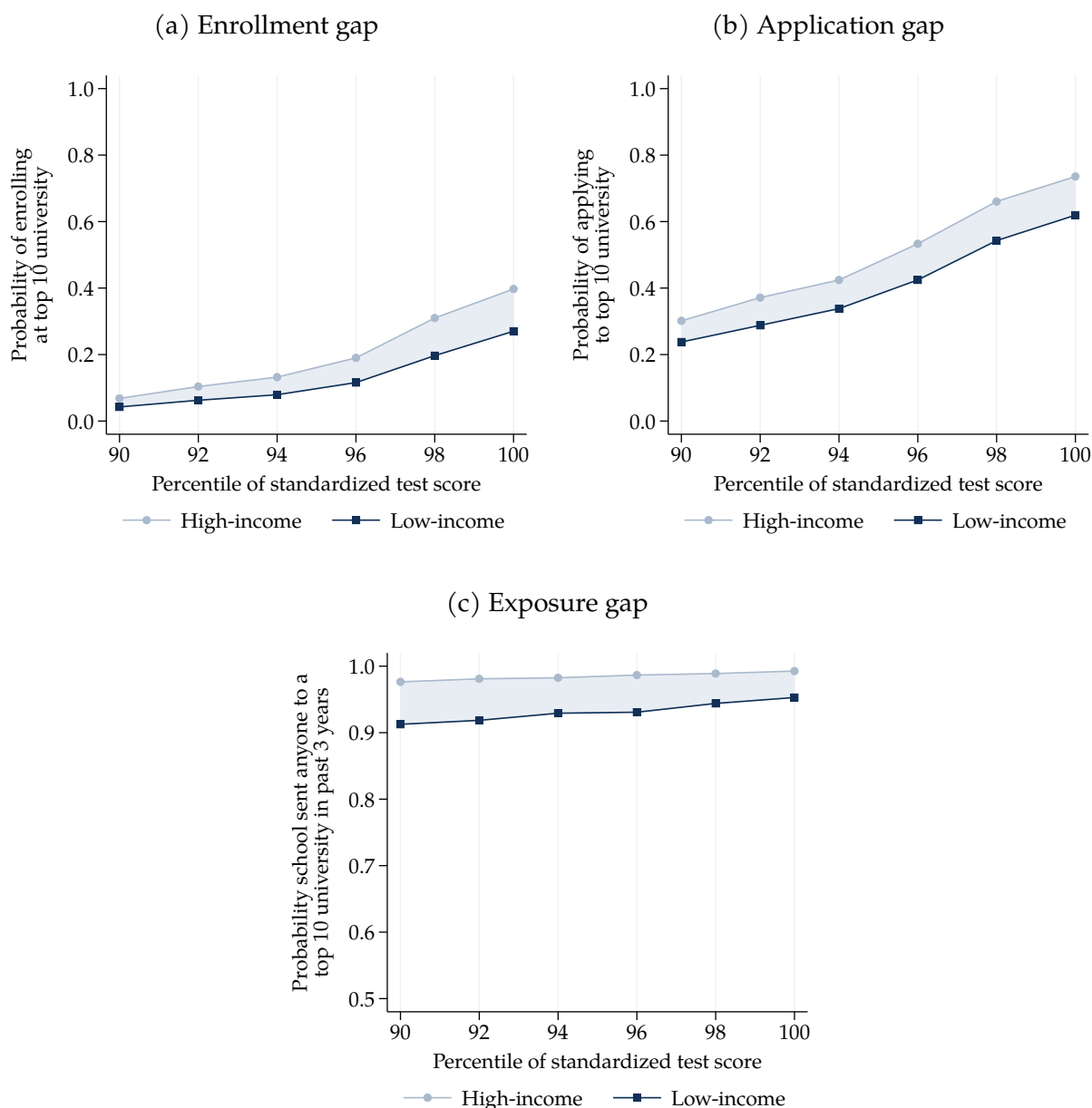
Socioeconomically disadvantaged students enroll at elite universities at lower rates compared to high-SES students with similar academic backgrounds. Widespread efforts to address these disparities typically involve increasing transparency on admission requirements or directly lowering admission thresholds for low-SES students. But much of what students care about when making application choices, such as whether *someone like them would fit in*, can be learned only through direct contact with past attendees. Students at socioeconomically disadvantaged schools are less likely to interact with schoolmates who go on to attend prestigious universities. Our paper illustrates how this lack of exposure can deter subsequent cohorts of students from applying to elite universities, dynamically reinforcing this gap. Our back-of-envelope calculation attributes 15–30% of the socioeconomic gap in elite university applications to differences in exposure.

A student who *breaks through* to an elite university, however, can break this cycle by conveying tacit information that is both demographic group-specific and university-specific. Our surveys with prospective university applicants indicate that low-SES students are pessimistic about their chances of fitting in at elite universities, but are neither miscalibrated on the grades required for admission nor pessimistic about their chances of admission or graduation. Our experiment, which randomizes exposure to elite universities outside of participants’ existing social networks, shifts beliefs about social fit, but has no effect on admissions or graduation beliefs. Consistent with our findings in the administrative data, experimentally induced exposure increased applications to treatment universities with particularly strong effects when participants and mentors were demographically similar.

Low-SES students in our experiment are *more responsive* to elite exposure than high-SES students. Yet the presence of more high-SES classmates can generate negative relative-rank effects that offset increases in university aspirations among low-SES students (Cattan, Salvanes, and Tominey 2025). By offering social connectedness without class-rank effects, our experiment highlights external mentorship and social exposure as a promising avenue for addressing inequality of opportunity across socioeconomically stratified school networks. From a policy perspective, our results also suggest that lowering admissions requirements for disadvantaged students (e.g., affirmative action in the US and contextual offers in the UK) may be insufficient to close socioeconomic gaps at elite universities. At the same time, policies that require universities to admit high-achieving students from a wider range of schools, such as the Top 10 Percent Rule in Texas, may have positive spillover effects on the applications from future cohorts. Quantifying the implications of these dynamic spillovers for optimal university admissions policy is an important avenue for future research.

VIII. Exhibits

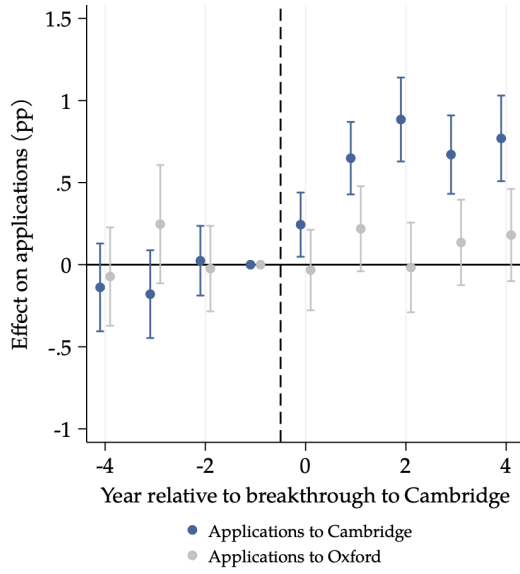
Figure 1: Socioeconomic gaps in enrollment, applications, and exposure at elite universities



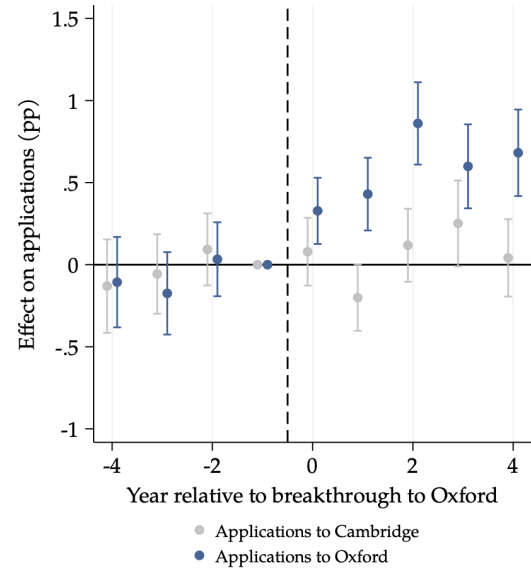
Notes: Panels (a) and (b) plot the probability of enrolling at (panel (a)) and applying to (panel (b)) a top 10 university, separately by parental income, across the distribution of standardized test scores (GCSE percentile). Low-income students are students whose home postcode is in the bottom 40% of neighbourhoods as ranked by the Index of Multiple Deprivation (IMD); high-income students are students in the top two quintiles of IMD. Panel (c) plots the corresponding gap in exposure to top 10 universities, defined the probability of attending a school where at least one student attending a top 10 university in the preceding three years.

Figure 2: Effects of exposure to breakthroughs on university outcomes

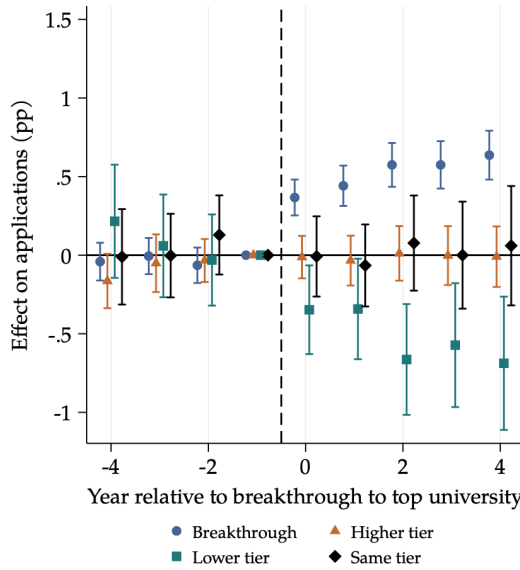
(a) Applications, breakthrough to Cambridge



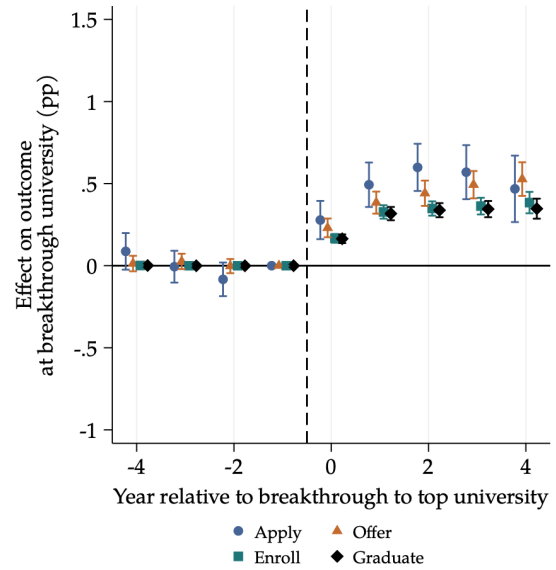
(b) Applications, breakthrough to Oxford



(c) Application portfolio outcomes



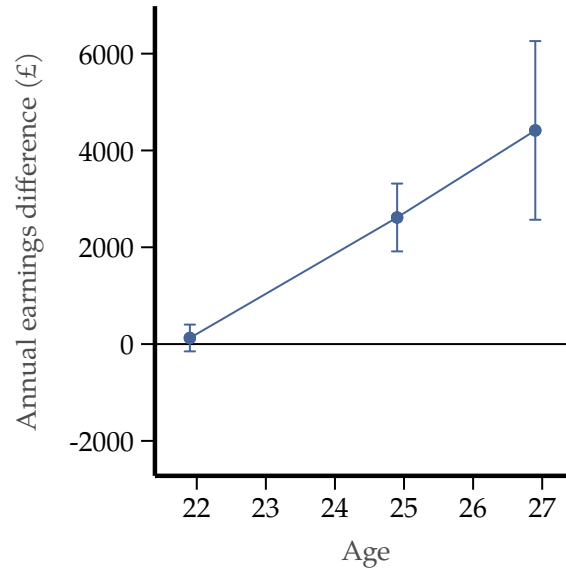
(d) Offer, enrollment and graduation outcomes



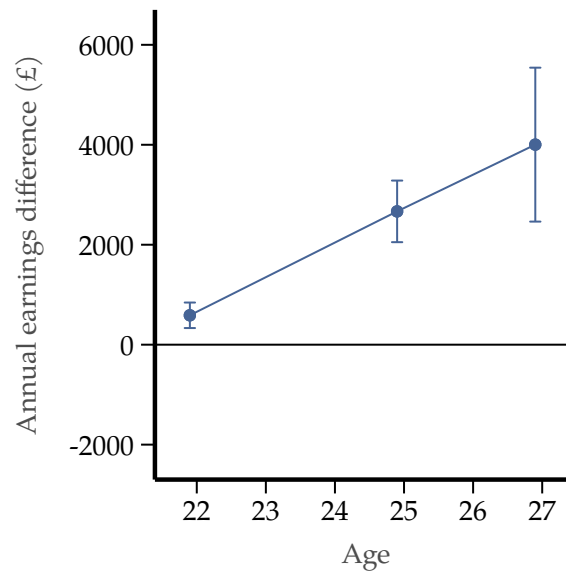
Notes: Panels (a) and (b) estimate equation 5 for breakthroughs to Cambridge and to Oxford respectively, where the outcome is applications to that university; these regressions include school and year fixed effects. Panels (c) and (d) estimate equation 6, pooling across top 10 universities (top 5 in c), and include school-university and year-university fixed effects. Panel (c) reports four mutually exclusive and exhaustive application outcomes based on portfolios including: the breakthrough university; a university in the same tier; a university in a higher tier; and only universities in a lower tier. Panel (d) reports indicators for applying to, receiving an offer from, enrolling at, and graduating from the breakthrough university; cohorts after 2016 are excluded from panel (d) as graduation is not observed for these cohorts. Universities are ranked by the mean A-level grades of their enrolled students. We estimate breakthrough effects using the Sun and Abraham estimator and multiply coefficients and SEs by 100 so they can be interpreted in percentage point terms. 95% confidence intervals reported based on standard errors clustered at the school level.

Figure 3: Labor market outcomes for students induced into elite universities

(a) Earnings effects relative to matched control schools

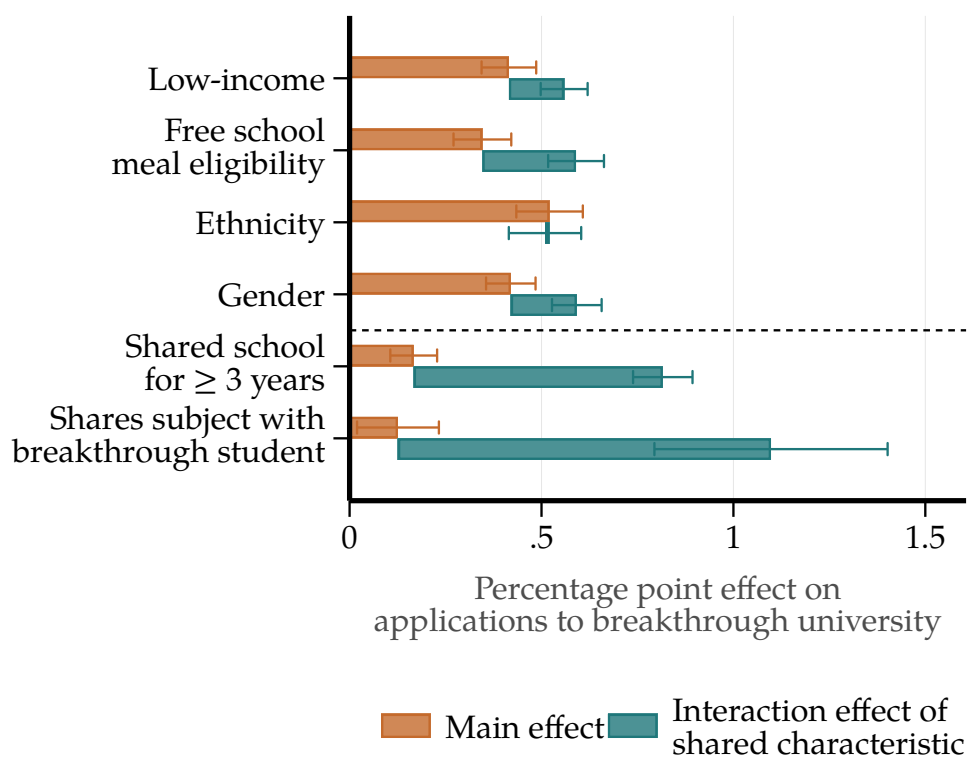


(b) Earnings effects relative to matched control cohorts



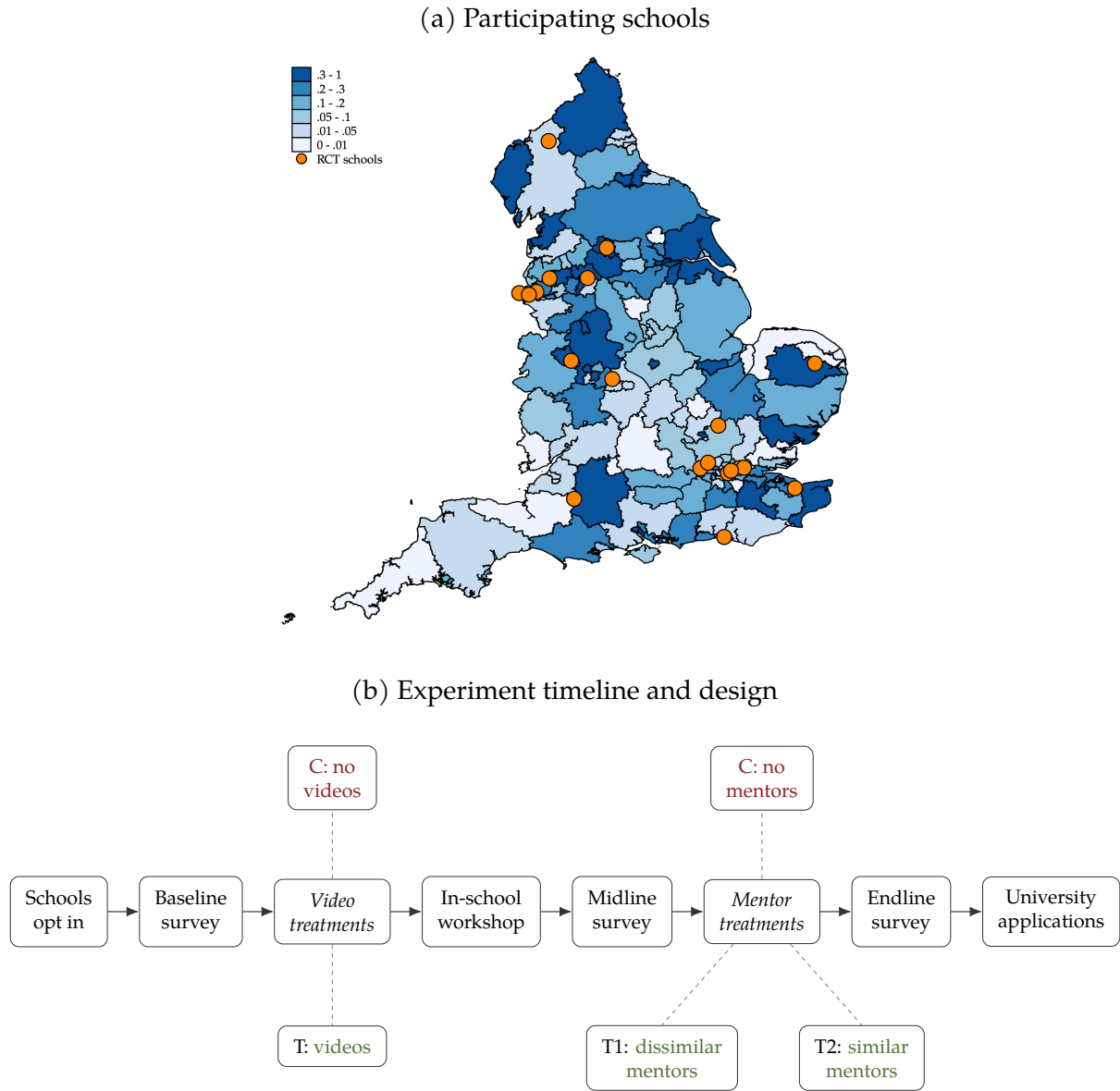
Notes: 95% confidence intervals throughout, based on standard errors clustered at the school level. Panels A and B report regressions of earnings at the specified age on an indicator for applying to the breakthrough university following a breakthrough, controlling for matched pair effects. Matched control students are drawn from a control set consisting of students applying for the same major in the same year at control high schools in panel (a), and of students at the same school prior to treatment in panel (b). Within the control set for each student, we also exactly match on the student's ventile in the sample GCSE grade distribution and an indicator for whether the student is taking 3 or more A-levels, and then select one nearest neighbour by Mahalanobis distance, matching on gender, neighbourhood income decile, and GCSE grades.

Figure 4: Heterogeneity in the effect of breakthroughs on applications, by similarity between the student and the breakthrough student



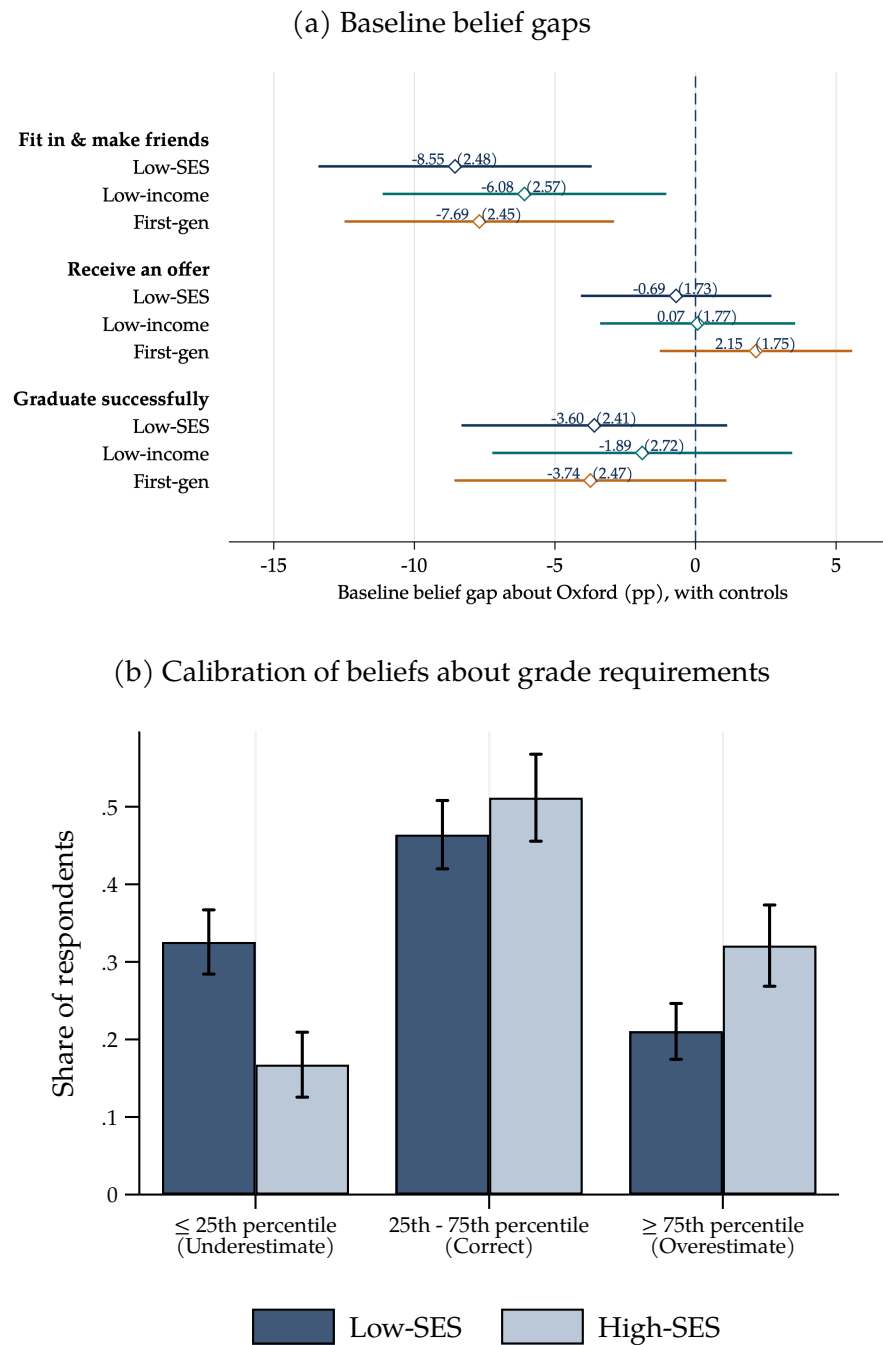
Notes: Coefficients from difference-in-difference regressions estimating the effect of breakthroughs on applications to the breakthrough university, pooled across breakthroughs to top 10 universities (with the exception of the shares subject regression, for which we pool across the top 7). For each characteristic listed on the vertical axis, we interact the Treated $\times$ Post indicator with an indicator for whether the student shares that characteristic with the breakthrough student at their school. The blue bars report the main Treated $\times$ Post effect for students who do not share the characteristic, and the red bars the additional effect for students who do. ‘Shared school for ≥ 3 years’ denotes students who attended the same age-16 school as the breakthrough student (by construction, as students are compared to a breakthrough student from the school they attended at age 18, these students share a school for at least 3 years). Coefficients are multiplied by 100 so they can be interpreted in percentage point terms. Regressions include school-by-breakthrough-university and year-by-breakthrough-university fixed effects. 95% confidence intervals reported based on standard errors clustered at the school level.

Figure 5: Overview of Experimental Design and Sample



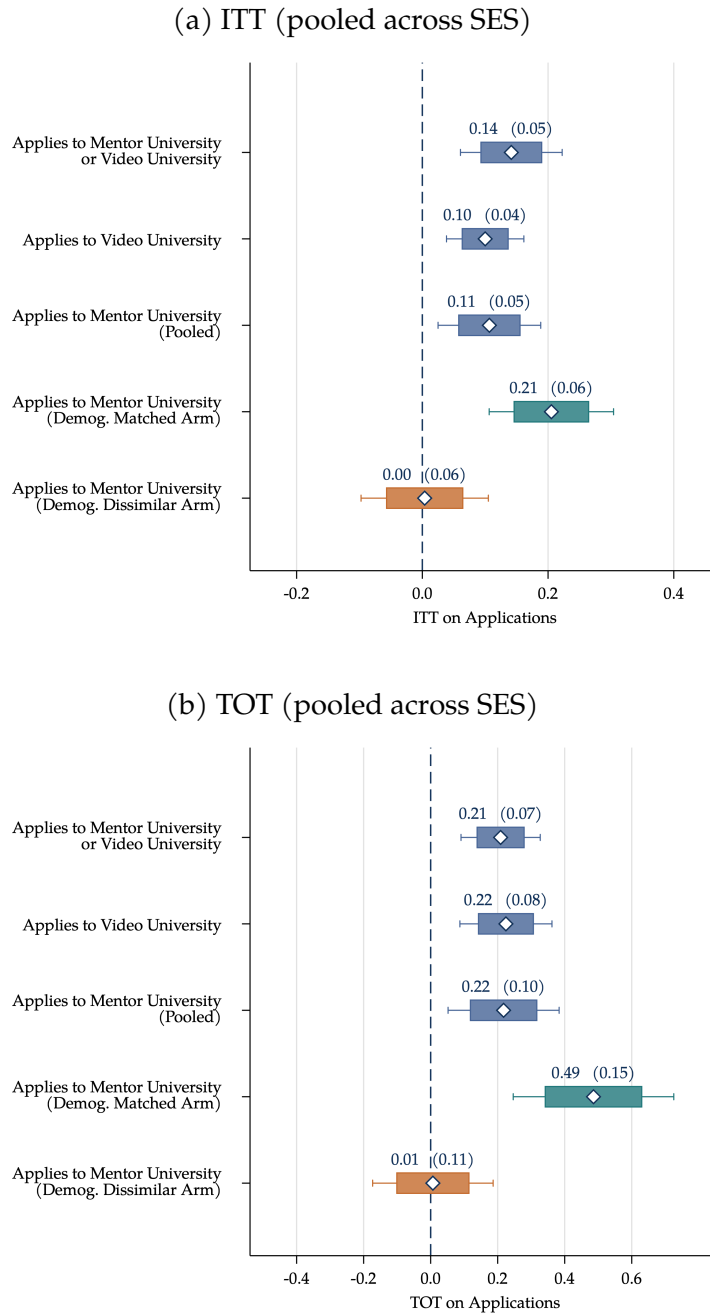
Notes: Panel (a) maps the locations of schools participating in our RCT within England; each orange dot represents a participating school. The underlying map plots the share of students in each ITL 3 region of England who attended high schools that sent no one to a top 10 university in the preceding three years, as of 2018. One participating school in Wales is omitted from the map. Panel (b) outlines the stages of the RCT and the treatment arms, as implemented in the Spring 2025 wave.

Figure 6: Baseline beliefs about elite universities



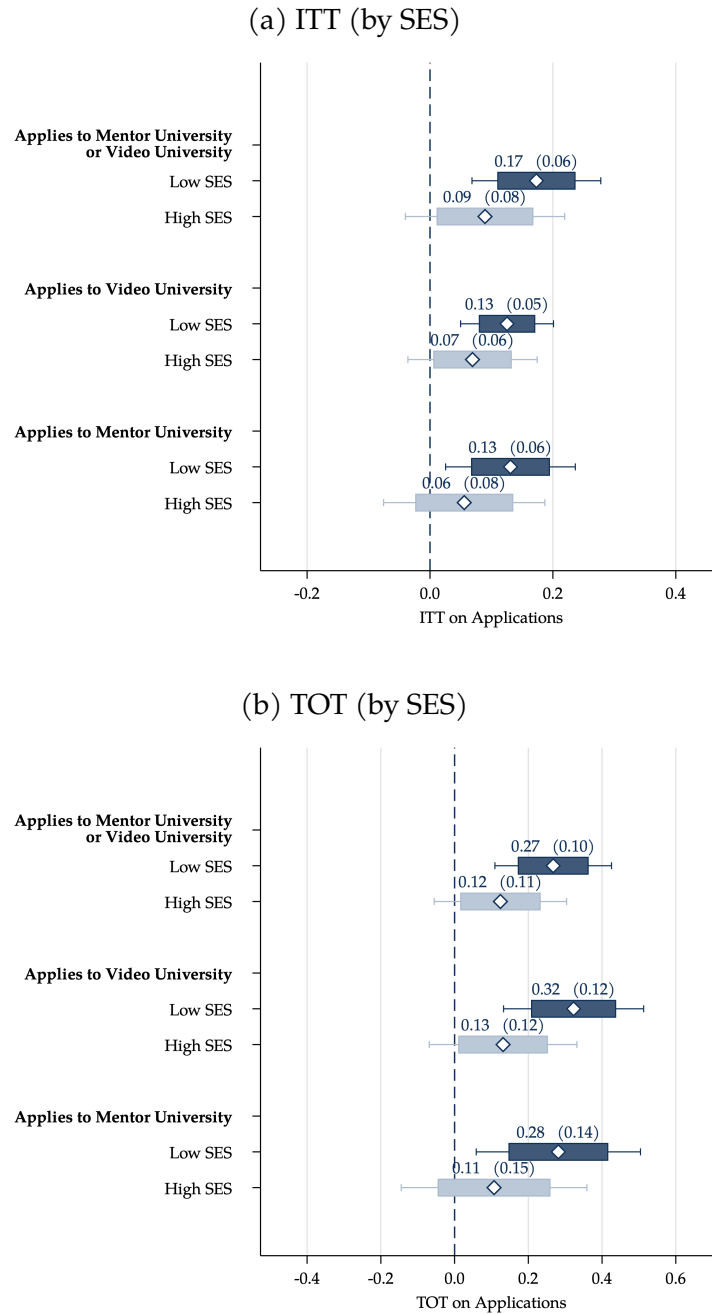
Notes: Sample restricted to RCT participants who completed both baseline and midline surveys. Beliefs are subjective probabilities, between 0 and 100, of 'receiving an offer from', 'fitting in and making friends at', or 'graduating successfully from' Oxford University. Low-SES students are students who are *either* low-income or first generation. *Panel (a)*: coefficients from regressions of each belief on the specified student characteristic and a control for the student's predicted A-level grades, reported as the average percentage point difference in probability relative to other students; 90% confidence intervals. *Panel (b)*: the share of respondents whose estimate of typical grades at Oxford falls below the 25th percentile, between the 25th and 75th percentile, or above the 75th percentile of the actual grade distribution of enrolled students, separately for low- and high-income students; 90% confidence intervals.

Figure 7: Treatment effects on university applications



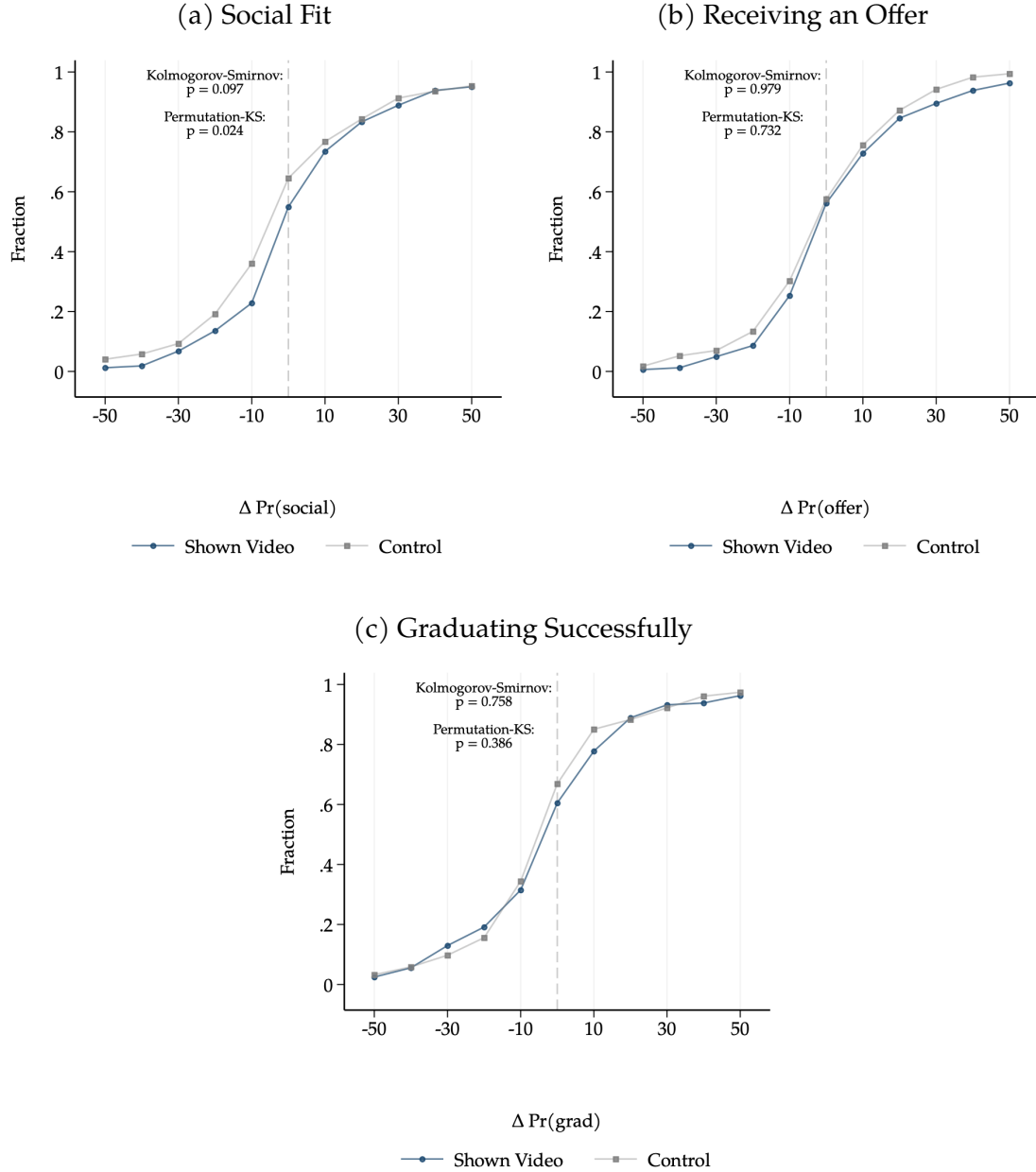
Notes: Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. We regress the specified application outcome on either the pooled treatment arm assignment T (in the first three rows of each panel) or the demographic match arms T1 and T2 (the demog. dissimilar and demog. matched arms respectively), in each case relative to the control group C. In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for takeup with the treatment arm dummies and identify the treatment-on-treated effects. The sample is restricted to students who responded ‘Maybe’, ‘Probably’, or ‘Definitely’ when asked whether they would be interested in a mentor.

Figure 8: Treatment effects on university applications, by SES



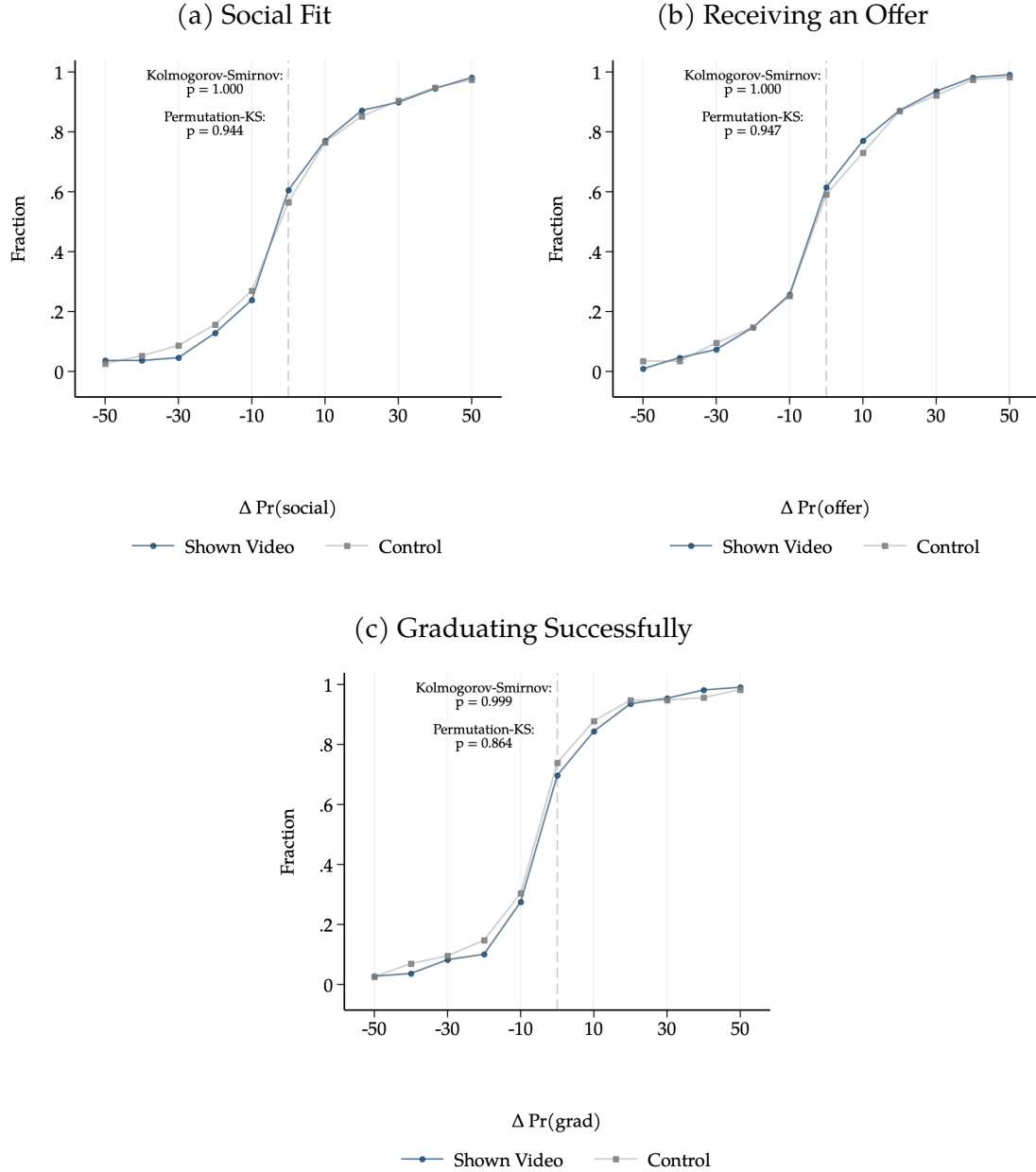
Notes: Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. Low-SES students are those who either have a home in the bottom 40% of neighborhood deprivation or who have no parents who have attended university; all others are high-SES. We regress the specified application outcome on either the pooled treatment arm assignment T , with the sample restricted to the specified SES group. In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for take-up with the treatment arms and identify the treatment-on-treated effects. The sample is restricted to students who responded ‘Maybe’, ‘Probably’, or ‘Definitely’ when asked whether they would be interested in a mentor.

Figure 9: Treatment effects on beliefs about video university, low-SES



Notes: Sample restricted to students who completed both baseline and midline surveys. Each panel plots the cumulative distribution of the percentage point change between the baseline and midline survey in the reported probability of the specified outcome at the student's assigned video university, separately for students who were and were not assigned to the video treatment arms. We pool across arms T1 and T2. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of the two distributions. Elicited beliefs were restricted to be multiples of 10, so distributions are discrete.

Figure 10: Treatment effects on beliefs about video university, high-SES



Notes: Sample restricted to students who completed both baseline and midline surveys. Each panel plots the cumulative distribution of the percentage point change between the baseline and midline survey in the reported probability of the specified outcome at the student's assigned video university, separately for students who were and were not assigned to the video treatment arms. We pool across arms T1 and T2. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of the two distributions. Elicited beliefs were restricted to be multiples of 10, so distributions are discrete.

Table 1: Summary statistics

				Experimental sample	
	(1)	(2)	(3)	(4)	(5)
	Age-18 school leavers	University applicants	Event study sample	Full sample	Endline observed
<i>Panel A. Demographic characteristics</i>					
Female	51.2	56.0	57.3	42.2	43.8
Low-income	37.2	28.2	26.3	35.7	37.3
Free school meal eligible	21.6	12.5	11.6	—	—
Parents attended university	—	56.0 [†]	57.7 [†]	54.5	53.5
White	80.0	80.6	81.1	56.7	53.7
Black	4.6	4.8	4.5	6.9	7.6
Asian	9.2	10.2	10.2	24.8	28.3
Other	6.3	4.4	4.2	11.6	10.4
<i>Panel B. School region</i>					
Northern England	28.2	28.7	28.9	42.8	37.6
The Midlands	19.6	19.3	19.2	3.0	1.5
Southern England	38.0	36.9	36.8	33.1	36.2
London	14.1	15.1	15.1	20.8	24.5
<i>Panel C. Academic characteristics</i>					
Taking ≥ 3 A-levels	35.2	66.6	68.9	85.0	85.6
Achieved tariff points (med.)	104	112	112	—	—
Predicted tariff points	—	—	—	126.4	125.1
<i>N</i>	7,164,386	2,920,445	2,290,950	640	473

Notes: All entries are percentages unless otherwise noted. The event study sample is students at schools that either experience a breakthrough or are a control for one. The RCT sample is students completing both the baseline and midline survey. Low-income means a home postcode in the bottom 40% of the Index of Multiple Deprivation. A-level tariff points convert letter grades to a 0–56 scale, summed over a student’s top 3 grades (maximum 168). Column (5) reports the control group mean and column (6) the coefficient on an indicator for assignment to a mentor treatment arm, both in percentage points, with heteroskedasticity-robust standard errors in parentheses.

[†] Parental university attendance in LEO is observed only for students who themselves attend university.

Table 2: Mentor conversation topic reports

(a) Topics Discussed in Mentor-Participant Conversations

	Conversations where topic was discussed (%)
Student Life at Mentor's University	70
Life at University and Fitting In	68
Mentor's Course (university-major)	66
Application Advice (e.g., Personal Statement)	64
Choice of Course (university-major)	53
Post-degree Career	36
Academic Success at University	30
<i>N</i>	103

(b) Mentor-reported importance of choice factors for students

	Importance (1–5 scale)		Z-score	Most important (%) [†]
	Mean	SE		
Whether they would fit in and enjoy their time	3.80	0.10	0.18	56.7
Whether they would succeed academically	3.65	0.11	0.03	35.1
Jobs that the course could help them to get	3.59	0.11	-0.03	38.1
Whether they could get an offer from the course	3.58	0.11	-0.04	43.3
Course content and teaching quality	3.46	0.10	-0.16	29.9
<i>N</i>	103	103	103	103

Notes: Descriptive statistics on topics discussed in calls and emails with mentors, based on the post-mentorship survey. In panel (a), mentors were asked a multiple choice question to select all topics they had discussed with each of their mentees, from the topics listed in the table; we tabulate the share of conversations for which the specified topic was selected. In panel (b), mentors were asked how important each of the listed factors was to the student they met with as the student decided which course to apply to, on a qualitative scale from “Not at all important” to “Extremely important”. We code these responses onto a 1–5 scale (where 5 is “extremely important”) and report the mean response, standard error, Z-score (based on the pooled mean and standard deviation of importance scores across factors), and the share of students for which each factor was (possibly jointly) most important according to their mentor.

[†] Ties were allowed when ranking importance, so shares do not sum to 1.

References

- Altmejd, Adam, Andrés Barrios-Fernández, Marin Drlje, Joshua Goodman, Michael Hurwitz, Dejan Kovac, Christine Mulhern, Christopher Neilson, and Jonathan Smith. 2021. "O Brother, Where Start Thou? Sibling Spillovers on College and Major Choice in Four Countries*." *The Quarterly Journal of Economics* 136, no. 3 (August 1, 2021): 1831–1886.
- Andrews, Rodney J., Scott A. Imberman, and Michael F. Lovenheim. 2020. "Recruiting and supporting low-income, high-achieving students at flagship universities." *Economics of Education Review* 74 (February): 101923.
- Angrist, Joshua, David Autor, and Amanda Pallais. 2022. "Marginal Effects of Merit Aid for Low-Income Students." *The Quarterly Journal of Economics* 137, no. 2 (April 8, 2022): 1039–1090.
- Avdeev, Stanislav, Nadine Ketel, Hessel Oosterbeek, and Bas van der Klaauw. 2024. "Spillovers in fields of study: Siblings, cousins, and neighbors." *Journal of Public Economics* 238 (October 1, 2024): 105193.
- Banerjee, Abhijit V. 1992. "A Simple Model of Herd Behavior." *The Quarterly Journal of Economics* 107, no. 3 (August 1, 1992): 797–817.
- Barr, Andrew, and Benjamin Castleman. 2025. "Increasing Degree Attainment among Low-Income Students: The Role of Intensive Advising and College Quality." *American Economic Review* 115, no. 11 (November): 4075–4103.
- Barrios-Fernández, Andrés. 2022. "Neighbors' Effects on University Enrollment." *American Economic Journal: Applied Economics* 14, no. 3 (July): 30–60.
- Barrios-Fernández, Andrés, Christopher Neilson, and Seth Zimmerman. 2026. "Elite Universities and the Intergenerational Transmission of Human and Social Capital." *American Economic Review* 116, no. 6 (June 1, 2026): 2120–2165.
- Bechichi, Nagui, and Gustave Kenedi. 2024. "Older Schoolmate Spillovers on Higher Education Choices." https://gustavekenedi.github.io/assets/pdf/bechichi_kenedi_older_schoolmate_jmp.pdf.
- Bhopal, Kalwant, and Martin Myers. 2023. "Elite universities aren't hotbeds of 'wokery': our research shows they're rife with racism and classism." *The Guardian* (January 30, 2023).
- Black, Sandra E., Kalena E. Cortes, and Jane Arnold Lincove. 2015. "Academic Undermatching of High-Achieving Minority Students: Evidence from Race-Neutral and Holistic Admissions Policies." *American Economic Review* 105, no. 5 (May): 604–610.
- Britton, Jack, Laura van der Erve, Chris Belfield, Anna Vignoles, Matt Dickson, Yu Zhu, Ian Walker, Lorraine Dearden, Luke Sibietta, and Franz Buscha. 2022. "How much does degree choice matter?" *Labour Economics* 79 (December 1, 2022).

- Burtonshaw, Sally, Richard Eyre, Jonathan Simons, and Pete Whitehead. 2024. *A review of collaborative support for improving equality of opportunity in access to higher education*. February.
- Campbell, Stuart, Lindsey Macmillan, Richard Murphy, and Gill Wyness. 2022. "Matching in the Dark? Inequalities in Student to Degree Match." *Journal of Labor Economics* 40, no. 4 (October 1, 2022): 807–850.
- Cattan, Sarah, Kjell G. Salvanes, and Emma Tominey. 2025. "First-Generation Elite: The Role of School Social Networks." *American Economic Review* 115, no. 12 (December): 4369–4403.
- Chetty, Raj, David J Deming, and John N Friedman. 2026. "Diversifying Society's Leaders? The Determinants and Causal Effects of Admission to Highly Selective Private Colleges." *The Quarterly Journal of Economics* 141, no. 1 (February 1, 2026): 51–145.
- Chetty, Raj, John N Friedman, Emmanuel Saez, Nicholas Turner, and Danny Yagan. 2020. "Income Segregation and Intergenerational Mobility Across Colleges in the United States*." *The Quarterly Journal of Economics* 135, no. 3 (August 1, 2020): 1567–1633.
- Chetty, Raj, Matthew O. Jackson, Theresa Kuchler, Johannes Stroebe, Nathaniel Hendren, Robert B. Fluegge, Sara Gong, et al. 2022. "Social capital I: measurement and associations with economic mobility." *Nature* 608, no. 7921 (August): 108–121.
- Cohodes, Sarah R., Helen Ho, and Silvia C. Robles. 2022. *STEM Summer Programs for Underrepresented Youth Increase STEM Degrees*. NBER Working Paper 30227. National Bureau of Economic Research, July.
- Conley, Timothy G, and Christopher R Udry. 2010. "Learning about a New Technology: Pineapple in Ghana." *American Economic Review* 100, no. 1 (March 1, 2010): 35–69.
- Dale, Stacy B., and Alan B. Krueger. 2002. "Estimating the Payoff to Attending a More Selective College: An Application of Selection on Observables and Unobservables." *The Quarterly Journal of Economics* 117, no. 4 (November 1, 2002): 1491–1527.
- . 2014. "Estimating the Effects of College Characteristics over the Career Using Administrative Earnings Data." *The Journal of Human Resources* 49 (2): 323–358.
- Dillon, Eleanor Wiske, and Jeffrey Andrew Smith. 2020. "The Consequences of Academic Match between Students and Colleges." *Journal of Human Resources* 55 (3): 767–808.
- Dizon-Ross, Rebecca. 2019. "Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments." *American Economic Review* 109, no. 8 (August 1, 2019): 2728–2765.
- Dynarski, Susan, C. J. Libassi, Katherine Michelsmore, and Stephanie Owen. 2021. "Closing the Gap: The Effect of Reducing Complexity and Uncertainty in College Pricing on the Choices of Low-Income Students." *American Economic Review* 111, no. 6 (June): 1721–1756.

- Eyster, Erik, and Matthew Rabin. 2010. "Naïve Herding in Rich-Information Settings." *American Economic Journal: Microeconomics* 2, no. 4 (November): 221–243.
- Hoxby, Caroline M., and Christopher Avery. 2012. *The Missing "One-Offs": The Hidden Supply of High-Achieving, Low Income Students*. NBER Working Paper 18586. National Bureau of Economic Research, December.
- Hoxby, Caroline M., and Sarah Turner. 2013. *Expanding College Opportunities for High-Achieving, Low Income Students*. SIEPR Discussion Paper 12-014. Stanford Institute for Economic Policy Research.
- . 2015. "What High-Achieving Low-Income Students Know about College." *American Economic Review* 105, no. 5 (May): 514–517.
- Hull, Peter. 2025. "'One Weird Trick' to Characterize Effective Populations in Design-Based Specifications." September. <https://www.dropbox.com/scl/fi/lgba6bh15jm72i3rxft1/EffectivePop.pdf?rlkey=u4wg8539uuecp404mcr9rfrly&raw=1>.
- Imbens, Guido W., and Joshua D. Angrist. 1994. "Identification and Estimation of Local Average Treatment Effects." *Econometrica* 62, no. 2 (March): 467.
- Jack, Anthony Abraham. 2020. *The privileged poor: how elite colleges are failing disadvantaged students*. Cambridge, Massachusetts: Harvard University Press.
- Larroucau, Tomás, Ignacio A. Rios, Anaïs Fabre, and Christopher Neilson. 2025. *College Application Mistakes and the Design of Information Policies at Scale*. Working Paper 34164. National Bureau of Economic Research, August.
- Malmendier, Ulrike, Ryan Oprea, and Laura Veldkamp. 2026. "Information Resonance." https://business.columbia.edu/sites/default/files-efs/citation_file_upload/Resonance_MOV_jul2026.pdf.
- Michelman, Valerie, Joseph Price, and Seth D Zimmerman. 2022. "Old Boys' Clubs and Upward Mobility Among the Educational Elite." *The Quarterly Journal of Economics* 137, no. 2 (April 8, 2022): 845–909.
- Mountjoy, Jack, and Brent R. Hickman. 2021. *The Returns to College(s): Relative Value-Added and Match Effects in Higher Education*. NBER Working Paper 29276. National Bureau of Economic Research, September.
- Mulhern, Christine. 2023. "Beyond Teachers: Estimating Individual School Counselors' Effects on Educational Attainment." *American Economic Review* 113, no. 11 (November 1, 2023): 2846–2893.
- Muralidharan, Karthik, Mauricio Romero, and Kaspar Wüthrich. 2025. "Factorial Designs, Model Selection, and (Incorrect) Inference in Randomized Experiments." *The Review of Economics and Statistics* 107, no. 3 (May 21, 2025): 589–604.
- Murphy, Richard, and Gill Wyness. 2020. "Minority report: the impact of predicted grades on university admissions of disadvantaged groups." *Education Economics* 28, no. 4 (July 3, 2020): 333–350.

- Neves, Jonathan, Rose Stephenson, and Charlotte Armstrong. 2026. *Student Academic Experience Survey 2026*. Advance HE.
- Office for National Statistics. 2023. *Longitudinal Education Outcomes SRS Iteration 2 Standard Extract - England*. Accessed May 19, 2024. <https://ons.metadata.works/browser/dataset?id=369>.
- Pallais, Amanda. 2015. "Small Differences That Matter: Mistakes in Applying to College." *Journal of Labor Economics* 33, no. 2 (April): 493–520.
- Sanders, Michael, Simon Burgess, Raj Chande, Catherine Dilnot, Eliza Kozman, and Lindsey Macmillan. 2018. "Role models, mentoring and university applications - evidence from a crossover randomised controlled trial in the United Kingdom." *Widening Participation and Lifelong Learning* 20, no. 4 (November 15, 2018): 57–80.
- Sanders, Michael, Raj Chande, and Eliza Selley. 2017. *Encouraging People into University*. Behavioral Insights Team.
- Sun, Liyang, and Sarah Abraham. 2021. "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects." *Journal of Econometrics*, Themed Issue: Treatment Effect 1, 225, no. 2 (December 1, 2021): 175–199.
- Sutton Trust, The. 2025. *Elitist Britain 2025*. September 18, 2025.
- Tadjfar, Nagisa, and Kartik Vira. 2025. *Prediction or Prejudice? Standardized testing and university access*.
- Van Der Erve, Laura, Elaine Drayton, and Jack Britton. 2021. *Which university degrees are best for intergenerational mobility?* The IFS, November 24, 2021.
- Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. *Stacked Difference-in-Differences*. NBER Working Paper 32054. National Bureau of Economic Research, January.
- Wyness, Gill. 2023. "Mismatch in Higher Education." In *Oxford Research Encyclopedia of Economics and Finance*. June 21, 2023.
- Zimmerman, Seth D. 2014. "The Returns to College Admission for Academically Marginal Students." *Journal of Labor Economics* 32, no. 4 (October): 711–754.

Online Appendix

Friends in Higher Places: Social Fit and University Choice

OA1. Definitions of Constructed Variables

Region: We generally ‘region’ to refer to the nine ITL 1 regions of England, unless otherwise specified. These are the largest statistical subdivisions of England: the North East, the North West, Yorkshire and the Humber, the West Midlands, the East Midlands, East of England, London, the South East, and the South West. For some analysis, we aggregate the regions into the North (including the North East, the North West, and Yorkshire and the Humber), the Midlands (West and East Midlands), the South (including the South East, the South West, and the East of England), and London.

Neighbourhood income: Our main measure of socioeconomic background is students’ decile of the Index of Multiple Deprivation, which is a composite measure of deprivation constructed by the UK government based on incomes, unemployment, education, health, housing, and environment. This measure is defined based on the Lower-Level Super Output Area (LSOA) of a student’s home residence; LSOAs are neighbourhoods with an average population of 1500, roughly equivalent to a US Census Block Group. As shorthand, we refer to students in the bottom 4 (most deprived) deciles of IMD as ‘low-income’ or ‘from low-income neighbourhoods’ throughout.

GCSE grade percentile: As described in section II.A, all English students complete GCSEs in Maths, English, and Science, as well as some optional subjects. For our primary measure of ability, we focus on the core subjects since these are not affected by selection into who takes the subject. Students receive letter grades (A* – G) in each GCSE subject they complete until 2017, after which the grading switched to a 9–1 numerical scale. The mapping between letter grades and number grades was not one-to-one. To create a comparable measure across time, we therefore convert grades in each core subject into a percentile within each year based on the distribution of grades in that subject across all students who complete GCSEs. We then take the mean of these percentiles for each student across their core subjects. In some cases, we use the subject-specific percentiles.

A-Level tariff points: We convert letter A-Level grades to a numerical score using the UCAS tariff points scheme: A* = 56, A = 48, B = 40, C = 32, D = 24, E = 16. We then take the total of these tariff points across the student’s subjects, restricting to the top 3 subjects for students who take more than 3 A-levels, for a score out of 168 (A*A*A*).

Distance to university: We observe the LSOA (see the neighbourhood income definition above) that each student lived in in each year of school, which provides a reasonably

precise location measure; LSOAs have a median area of 0.5km^2 and a mean area of 4.3km^2 . For universities, we directly observe the university's region but do not observe any more precise location. As a proxy for the university's location, we consider all students who attend that university and report that they live with their parents during term time (which generally means that they live near enough to the campus to commute regularly to the university, particularly since our sample period is pre-COVID so that remote lectures were less common). We then take the midpoint of these student locations, based on students' last observed LSOA while at school, and infer that this is the approximate location of the university. This is not an exact measure, but acts as a reasonable proxy; spot-checks produce roughly correct university locations.

Earnings: Our primary measure of earnings is the total earnings in pounds received in a tax year from an individual's primary employer, conditioned on receiving positive earnings. We exclude earnings from self-employment because these are only included for tax years after 2013, meaning that it would be impossible to consistently include self-employment earnings for the whole sample. We restrict to individuals with positive earnings recorded because the data does not generally let us distinguish between individuals with 0 earnings in a year, individuals who have no earnings from employment but positive self-employment earnings, and individuals who have positive earnings but did not work in the UK in the given tax year or do not appear in the tax data for some other reason. Where we report earnings age profiles including zero earners, we assume that anyone who is observed in the educational data in an appropriate cohort, but is not observed in the tax data in the year corresponding to that age, had earnings of 0. We do not observe hours worked, so annual earnings may reflect part-time work or employment spells lasting less than a full year, which is why mean earnings can be well below the annual equivalent of full-time minimum wage. Earnings are deflated to 2018 levels using CPIH (Consumer Prices Index including owner occupiers' housing costs). Conversions into US dollars are at the 2018 PPP exchange rate for household final consumption expenditures as reported by the OECD 'Annual Purchasing Power Parities and exchange rates' dataset, which is $\text{£}0.7811 = \$1.0000$.

OA2. Additional Institutional Details

Each of the four nations of the UK – England, Scotland, Wales, and Northern Ireland – has a somewhat different educational system. Our administrative data is drawn from English students, and all but one of the schools taking part in the RCT are in England (one school is in Wales), so we focus on institutional details for England below.

Students in both England and Wales are required to remain in education until age 18. At age 16, students complete General Certificate of Secondary Education (GCSE) qualifications in specific subjects. Students are required to complete GCSEs in English, Mathematics, and Science, and can take additional optional GCSEs depending on their preferences and the GCSEs offered by their school; the median student completes 8 GCSEs. Full-time education at ages 15–16 is focused on preparing students for GCSEs. After completing GCSEs, students must remain in education, but need not stay in full-time education. Academically oriented students typically remain in full-time education to complete GCE Advanced Level (A-Level) qualifications in specific subjects, but alternative routes include vocational qualifications (most commonly Business and Technology Education Council—BTEC—certificates), full-time apprenticeships, or part-time employment alongside part-time training. A-Level students pick a small number of subjects—typically 3–4—to study, again based on both their preferences and the subjects offered by their school. It is reasonably common to change schools after completing GCSEs to attend a dedicated ‘sixth form college’ or ‘further education college’ that offers only 16–18 education, but many schools also offer 11–18 education. Courses receive applications from students without observing the other universities that students apply to. Admissions decisions are generally made at the course level, rather than holistically by the university.

After students receive offers, they choose up to two offers to accept, one as a ‘firm choice’ (a student’s top choice that they plan to attend if they meet any offer conditions), and one as an ‘insurance choice’ (a course that will accept them if they fail to meet the conditions for their firm choice but achieve that course’s conditions; students thus generally select an insurance choice with more lenient conditions than the firm choice). Students then complete exams and coursework required for their A-Levels or other qualifications, and receive grades later in the year. Depending on the grades they may attend their firm choice or their insurance choice, or if they fail to meet the conditions for either offer they can enter the ‘Clearing’ process – an after-market ‘scramble’ facilitated by UCAS in which universities list remaining open course slots and students can put in new direct applications to these courses given their realised (rather than predicted) grades. If they choose not to enter this process or fail to find an available course as part of it, they are free to reapply in the

following year or not to attend university.

University tuition for domestic students is capped by the government. As almost all universities charge tuition at the cap, this means that there is no variation in tuition costs for students between universities. The government provides universal income-contingent loans covering all tuition costs to all students: under the current policy regime, students pay no tuition upfront and repay 9% of their annual income in excess of £25,000 after graduating. Interest rates are linked to inflation. Any debt remaining unpaid after 40 years is cancelled. Thresholds, repayment rates, and write-off periods have changed over time, but the essential income-contingent structure of loans has been in place since reforms to tuition fees in 2012. The government also provides maintenance loans to cover living expenses: the amount offered depends on parental income and whether students live with their parents, and there is a supplement for students living in London. These are paid back in the same way as tuition loans.

OA3. Supplemental Exhibits: Robustness Checks and Alternative Specifications

Table OA1: Application event studies for top 10 universities with ability controls

	(1) Apply	(2) Apply	(3) Apply	(4) Apply	(5) Apply	(6) Apply
$1\{t = -4\} \times 1\{\text{Treated}\}$	0.000708 (0.000378)	0.000831* (0.000376)	0.000545 (0.000398)	-0.000914 (0.000717)	-0.000391 (0.000711)	-0.000656 (0.000740)
$1\{t = -3\} \times 1\{\text{Treated}\}$	-0.000348 (0.000353)	-0.000344 (0.000358)	-0.000344 (0.000379)	-0.000903 (0.000700)	-0.000712 (0.000708)	-0.000921 (0.000739)
$1\{t = -2\} \times 1\{\text{Treated}\}$	-0.0000626 (0.000365)	-0.0000712 (0.000365)	-0.000253 (0.000383)	-0.000752 (0.000702)	-0.000844 (0.000696)	-0.00115 (0.000731)
$1\{t = -1\} \times 1\{\text{Treated}\}$	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
$1\{t = 0\} \times 1\{\text{Treated}\}$	0.00337*** (0.000377)	0.00284*** (0.000371)	0.00304*** (0.000392)	0.00361*** (0.000710)	0.00290*** (0.000700)	0.00277*** (0.000726)
$1\{t = 1\} \times 1\{\text{Treated}\}$	0.00489*** (0.000442)	0.00440*** (0.000427)	0.00431*** (0.000437)	0.00591*** (0.000783)	0.00504*** (0.000791)	0.00501*** (0.000807)
$1\{t = 2\} \times 1\{\text{Treated}\}$	0.00590*** (0.000473)	0.00554*** (0.000458)	0.00539*** (0.000465)	0.00661*** (0.000800)	0.00566*** (0.000819)	0.00567*** (0.000847)
$1\{t = 3\} \times 1\{\text{Treated}\}$	0.00505*** (0.000482)	0.00480*** (0.000474)	0.00479*** (0.000489)	0.00494*** (0.000913)	0.00375*** (0.000900)	0.00364*** (0.000929)
$1\{t = 4\} \times 1\{\text{Treated}\}$	0.00543*** (0.000523)	0.00541*** (0.000516)	0.00545*** (0.000526)	0.00510*** (0.000804)	0.00399*** (0.000812)	0.00363*** (0.000842)
<i>N</i>	8,133,835	8,080,870	7,284,345	1,969,650	1,962,180	1,812,640
Sample mean	0.0180	0.0178	0.0191	0.0126	0.0125	0.0131
Pre-treatment mean	0.0184	0.0184	0.0196	0.0108	0.0108	0.0114
GCSE controls	N	Y	Y	N	Y	Y
A-level controls	N	N	Y	N	N	Y
School-level matching	N	N	N	Y	Y	Y

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Notes: Data: LEO. Coefficients from event studies of the effects of breakthroughs on applications to the breakthrough university using different sets of controls, pooled across top 10 universities. All regressions include school-by-breakthrough-university and year-by-breakthrough-university fixed effects. Column (2) includes GCSE percentile by breakthrough university effects; column (3) adds controls for the number of A-levels and the number of facilitating (more academic) A-levels completed; column (4) has no ability controls, but restricts to schools that are matched on 2007–2009 observables and includes match pair-by-year-by-breakthrough-university fixed effects; column (5) adds GCSE percentile by breakthrough university effects to the matched specification; column (6) adds controls for the number of A-levels and the number of facilitating (more academic) A-levels completed to this specification. 95% confidence intervals reported based on standard errors clustered at the school level.

Table OA2: Application event studies for top 10 universities by distance to the breakthrough university

	(1)	(2)	(3)	(4)	(5)	(6)
	Region		Median dist.		Dist. (miles)	
	Different	Same	Below	Above	≤ 30	> 30
$1\{t = -4\} \times 1\{\text{Treated}\}$	-0.000353 (0.000754)	-0.00369 (0.00354)	0.000630 (0.00118)	-0.000818 (0.000930)	-0.000740 (0.00189)	-0.000390 (0.000794)
$1\{t = -3\} \times 1\{\text{Treated}\}$	-0.000622 (0.000738)	-0.00306 (0.00394)	0.0000477 (0.00113)	0.0000154 (0.00107)	0.000449 (0.00201)	-0.000479 (0.000840)
$1\{t = -2\} \times 1\{\text{Treated}\}$	-0.000901 (0.000730)	-0.0000121 (0.00474)	0.00147 (0.00126)	-0.00168 (0.00101)	0.000176 (0.00242)	-0.00110 (0.000814)
$1\{t = -1\} \times 1\{\text{Treated}\}$	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
$1\{t = 0\} \times 1\{\text{Treated}\}$	0.00276*** (0.000741)	0.00626 (0.00405)	0.00459*** (0.00123)	0.00229* (0.00104)	0.00283 (0.00239)	0.00251** (0.000802)
$1\{t = 1\} \times 1\{\text{Treated}\}$	0.00496*** (0.000821)	0.00265 (0.00410)	0.00495*** (0.00130)	0.00520*** (0.00116)	0.00167 (0.00234)	0.00538*** (0.000948)
$1\{t = 2\} \times 1\{\text{Treated}\}$	0.00506*** (0.000856)	0.0153*** (0.00450)	0.00789*** (0.00135)	0.00297* (0.00124)	0.00790** (0.00251)	0.00390*** (0.000980)
$1\{t = 3\} \times 1\{\text{Treated}\}$	0.00331*** (0.000932)	0.00901 (0.00523)	0.00478*** (0.00142)	0.00167 (0.00135)	0.00290 (0.00284)	0.00306** (0.00108)
$1\{t = 4\} \times 1\{\text{Treated}\}$	0.00315*** (0.000828)	0.0116* (0.00569)	0.00461*** (0.00135)	0.00346** (0.00132)	0.00521 (0.00273)	0.00381*** (0.000983)
<i>N</i>	1,703,905	107,585	881,675	857,295	346,370	1,392,910
Sample mean	0.0125	0.0237	0.0148	0.0103	0.0164	0.0116
Pre-treat mean	0.0108	0.0203	0.0125	0.00883	0.0137	0.00998

Standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Notes: Data: LEO. Coefficients from event studies of the effects of breakthroughs on applications to the breakthrough university for students with different distances to the breakthrough university, pooled across top 10 universities. All regressions include school-by-breakthrough-university and year-by-breakthrough-university fixed effects. Column (1) includes only students in a different (ITL 1) region from the breakthrough university, while column (2) includes only students in the same region as the breakthrough university. Column (3) restricts to students below the median distance from the breakthrough university (which is calculated separately for each breakthrough university), and column (4) to students above the median distance. Column (5) restricts to students within 30 miles of the breakthrough university, and column (6) to students located more than 30 miles from the breakthrough university. 95% confidence intervals reported based on standard errors clustered at the school level.

Table OA3: Mean characteristics for compliers (students who apply in response to a breakthrough) and broader samples

Variable	(1) Complier mean	(2) Mean for enrollees at breakthrough universities	(3) Event study sample mean	(4) Event study treated schools mean
<i>Panel A: Breakthroughs to universities ranked 1–10</i>				
Female	0.460	0.482	0.576	0.571
White	0.707	0.803	0.809	0.81
Low-income	0.297	0.147	0.331	0.304
FSM eligible	0.188	0.057	0.147	0.135
Northern England	0.203	0.181	0.311	0.309
The Midlands	0.205	0.139	0.212	0.197
Southern England	0.318	0.472	0.332	0.337
GCSE percentile	88.4	92.3	68.4	70.8
≥ 3 A-levels	0.921	0.954	0.620	0.688

Notes: Comparison of mean characteristics for compliers – treating breakthroughs as an instrument for applications to the breakthrough university – with the population of all students who enroll at the breakthrough university (in column 2), for all students in the analysis sample for event studies (in column 3) and for all students at treated schools in the analysis sample (in column 4). Characteristics for compliers are estimated using the method described in Hull (2025); other characteristics are raw means within the specified sample. ‘FSM eligible’ denotes eligibility for Free School Meals. ‘Low-income’ denotes students in the poorest two quintiles of our neighbourhood deprivation measure. GCSE percentiles are average percentiles of age-16 test scores estimated separately by year. Taking ≥ 3 A-levels is the standard qualification for academic-track students.

Table OA4: Demographic Match Heterogeneity in Video Outcomes

	Applied			Positive belief update		
	Pooled (1)	Low-SES (2)	High-SES (3)	Pooled (4)	Low-SES (5)	High-SES (6)
Treated	-0.027 (0.075)	0.004 (0.077)	-0.111 (0.197)	-0.033 (0.056)	0.005 (0.070)	-0.117 (0.096)
Demog. Match	-0.066 (0.055)	-0.043 (0.057)	-0.185 (0.144)	-0.010 (0.058)	-0.001 (0.076)	-0.084 (0.094)
Treated × Demog. Match	0.166* (0.086)	0.169* (0.095)	0.209 (0.208)	0.149* (0.083)	0.193* (0.108)	0.146 (0.133)
Control mean	0.130	0.098	0.174	0.390	0.355	0.435
N	423	254	164	565	334	224

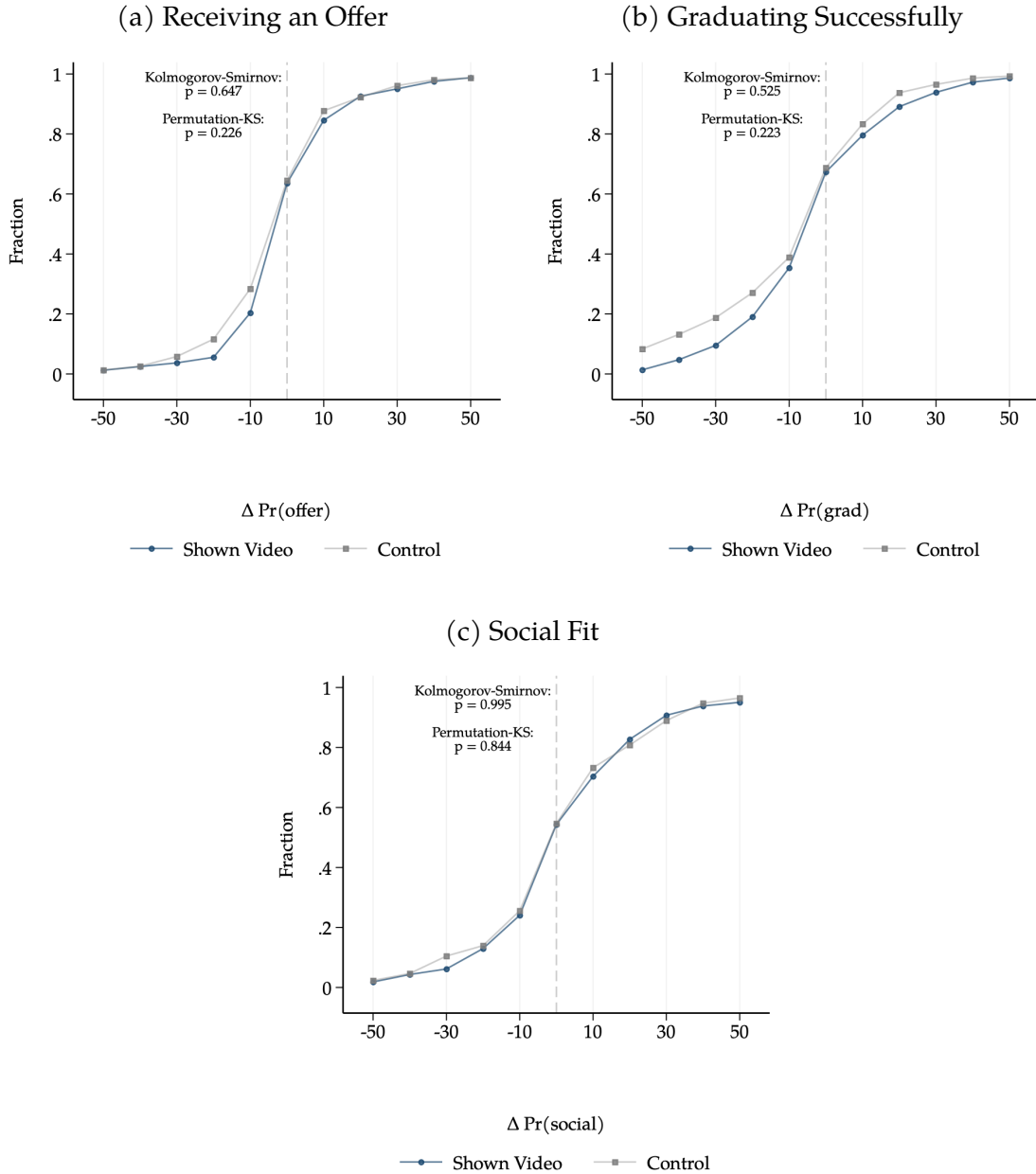
Notes: Data: RCT. Reports heterogeneity in video treatment effects on applying to one of the video universities (in columns 1–3) or the probability of updating positively about the university (in columns 4–6) by whether the video speakers was demographically matched with the student. Demographic match is defined as sharing *either* ethnicity, region, or both with the student. Low-SES students are those who are either in the poorest two quintiles of neighbourhood deprivation or did not attend university; other students are high-SES. The sample in columns 1–3 is restricted to students for whom we have final applications data from their school or their endline survey response.

Table OA5: Belief updating about video university, by SES

	Positive Update		Negative Update		No Update		Δ Belief	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low-SES	High-SES	Low-SES	High-SES	Low-SES	High-SES	Low-SES	High-SES
<i>Panel A. Social fit</i>								
Video treatment	0.104*	-0.038	-0.151***	-0.027	0.047	0.065	5.334*	-0.776
	(0.055)	(0.067)	(0.050)	(0.059)	(0.052)	(0.064)	(2.922)	(3.271)
Control mean	0.36	0.43	0.36	0.27	0.28	0.30	2.12	4.09
N	329	224	329	224	329	224	329	224
<i>Panel B. Receive an offer</i>								
Video treatment	0.008	-0.025	-0.047	0.011	0.039	0.014	4.235*	-0.387
	(0.056)	(0.067)	(0.049)	(0.060)	(0.050)	(0.064)	(2.394)	(2.974)
Control mean	0.43	0.41	0.29	0.25	0.28	0.34	2.94	3.65
N	329	224	329	224	329	224	329	224
<i>Panel C. Graduate successfully</i>								
Video treatment	0.066	0.038	-0.036	-0.025	-0.030	-0.013	1.732	1.470
	(0.055)	(0.061)	(0.054)	(0.061)	(0.053)	(0.067)	(2.881)	(2.780)
Control mean	0.33	0.26	0.34	0.30	0.33	0.43	0.26	-1.39
N	312	224	312	224	312	224	312	224

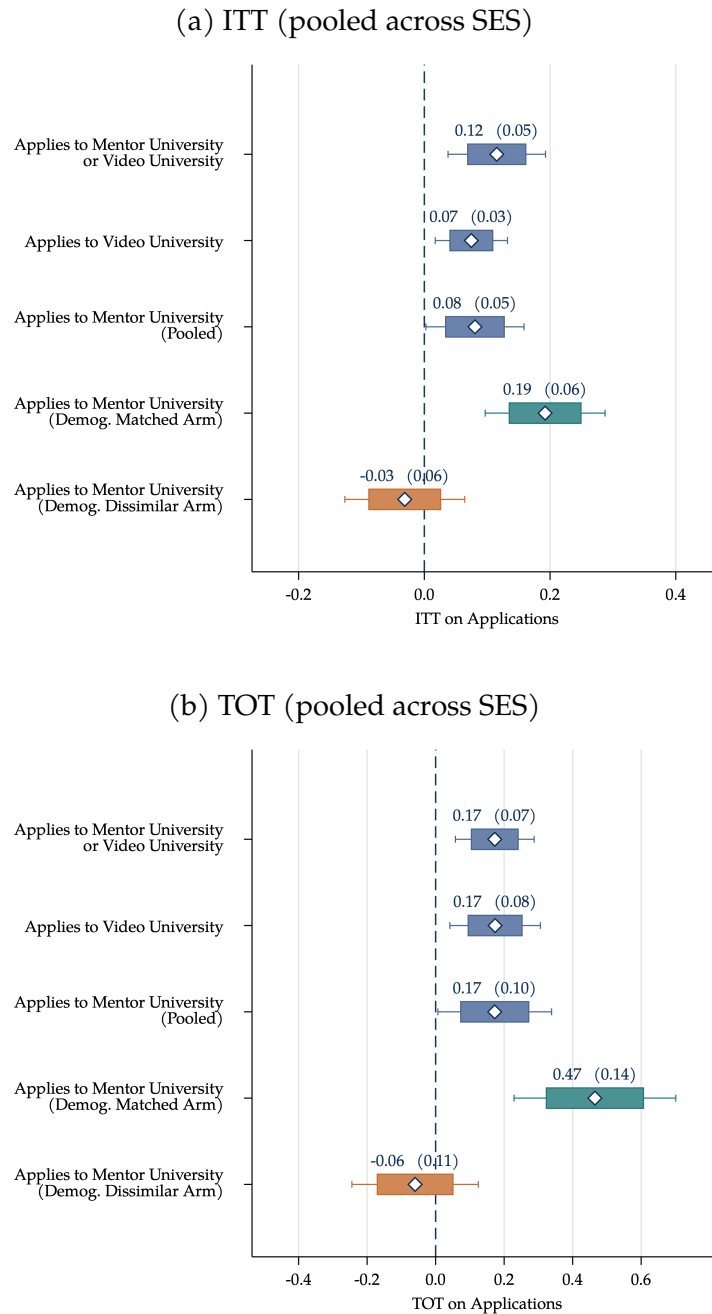
Notes: Data: RCT. Table reports the effect of watching a video on the probability of updating beliefs positively, updating negatively, and not changing beliefs, and on the mean change in beliefs, separately for low-SES and high-SES students. All beliefs relate to the university that was randomly selected as the belief elicitation university from the two universities discussed in a student's assigned videos. Belief questions ask how likely students are to 'fit in and make friends, if you attended', 'receive an offer from', and 'graduate successfully from' the specified university; elicitation restricts beliefs to percentages that are multiples of 10. Updates are changes in beliefs between the baseline survey, taken just before videos, and the midline survey, taken a few days after watching the videos. Low-SES students are those who either have a home address in the bottom two quintiles of neighbourhood income or whose parents did not attend university.

Figure OA1: Placebo treatment effects on university beliefs (Oxford), low-SES



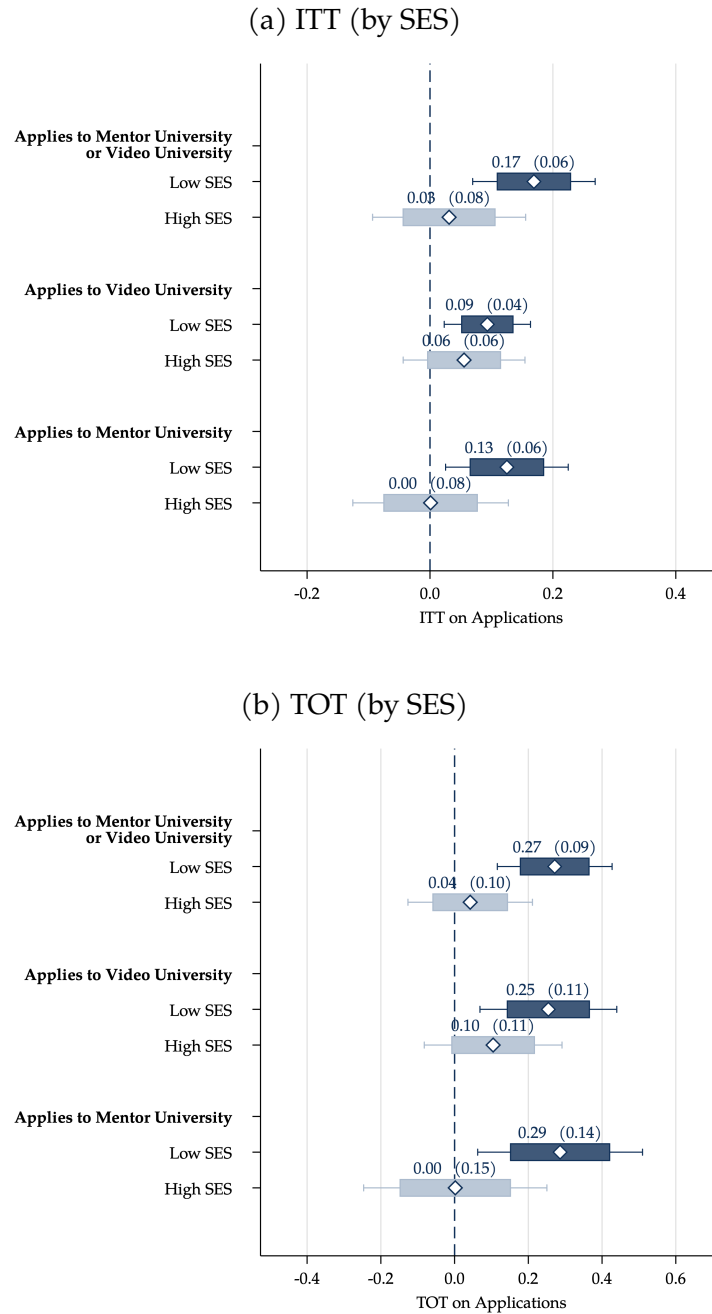
Notes: Data: RCT. Sample restricted to students who completed both baseline and midline surveys. Each panel plots the cumulative distribution of the percentage point change between the baseline and midline survey in the reported probability of the specified outcome at the student's assigned video university, separately for students who were and were not assigned to the video treatment arms. We pool across arms T1a, T1b, T2a and T2b. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of the two distributions. Elicited beliefs were restricted to be multiples of 10, so distributions are discrete; updates are truncated to between -40 and $+40$. Beliefs about social fit shift markedly in response to the video treatment, while beliefs about offer and graduation probabilities do not.

Figure OA2: Treatment effects on university applications



Notes: Robustness check for Figure 7: here we include students who said they were ‘definitely not’ or ‘probably not’ interested in receiving a mentor in the sample, while they are excluded from the sample in Figure 7. Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. We regress the specified application outcome on either the pooled treatment arm assignment T (in the first three rows of each panel) or the demographic match arms T1 and T2 (the demog. dissimilar and demog. matched arms respectively), in each case relative to the control group C. In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for takeup with the treatment arm dummies and identify the treatment-on-treated effects.

Figure OA3: Treatment effects on university applications, by SES



Notes: Robustness check for Figure 8: here we include students who said they were ‘definitely not’ or ‘probably not’ interested in receiving a mentor in the sample, while they are excluded from the sample in Figure 8. Treatment effects of videos and mentors on applying to the universities discussed. ‘Mentor university’ is defined as one of the 1–2 universities that a student’s assigned mentors attended, and ‘video university’ is defined as one of the 2 universities that the speakers in a student’s assigned videos attended. The outcome in each case is a binary indicator for applying to at least one of the specified universities. Low-SES students are those who either have a home in the bottom 40% of neighborhood deprivation or who have no parents who have attended university; all others are high-SES. We regress the specified application outcome on either the pooled treatment arm assignment T_i , with the sample restricted to the specified SES group. In panel (a), we report intent-to-treat estimates, and in panel (b) we instrument for takeup with the treatment arms and identify the treatment-on-treated effects. The sample is restricted to students who responded ‘Maybe’, ‘Probably’, or ‘Definitely’ when asked whether they would be interested in a mentor.

OA4. Effects of equalizing application rates on socioeconomic enrollment gaps

Adapting a decomposition described by Chetty, Deming, and Friedman (2026) to our context, we observe that the probability of enrollment at a given university through the main UCAS application scheme can be decomposed as:

$$\begin{aligned} P(\text{enroll}) &= P(\text{apply}) \\ (OA1) \quad &\times P(\text{receive offer} \mid \text{apply}) \\ &\times P(\text{accept offer} \mid \text{receive offer}) \\ &\times P(\text{enroll} \mid \text{accept offer}) \end{aligned}$$

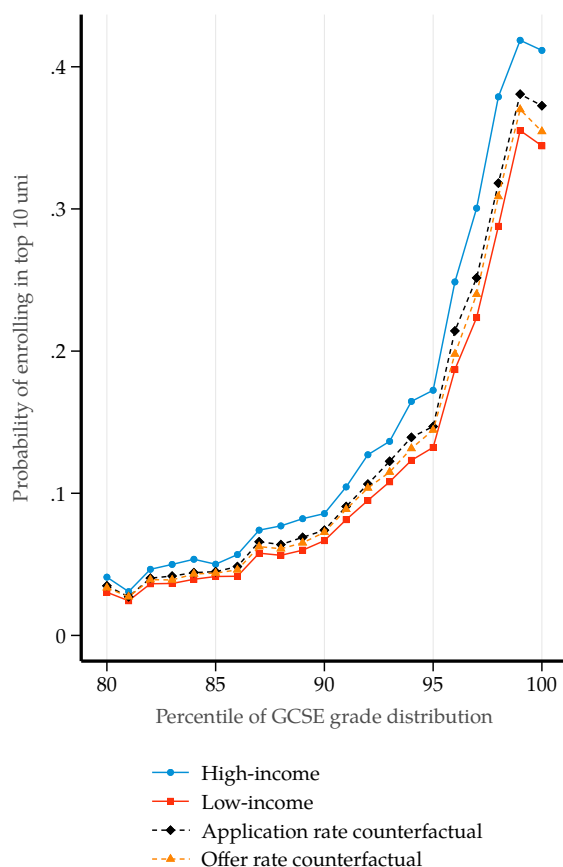
Focusing on students who enroll at university in the main scheme, we can thus decompose overall enrollment gaps (i.e., $P(\text{enroll} \mid \text{high income}) - P(\text{enroll} \mid \text{low income})$) into components explained by differences in application rates, offer rates, offer acceptance rates, and conditional enrollment rates. Taking logs of (OA1) produces an additive decomposition in terms of log points; we can also predict the enrollment rate for low-income students if they applied at the same rate as high-income students by taking

$$\begin{aligned} &P(\text{enroll} \mid \text{low income, high income application rate}) \\ &= P(\text{apply} \mid \text{high income}) \\ &\quad \times P(\text{receive offer} \mid \text{apply, low income}) \\ &\quad \times P(\text{accept offer} \mid \text{receive offer, low income}) \\ &\quad \times P(\text{enroll} \mid \text{accept offer, low income}) \end{aligned}$$

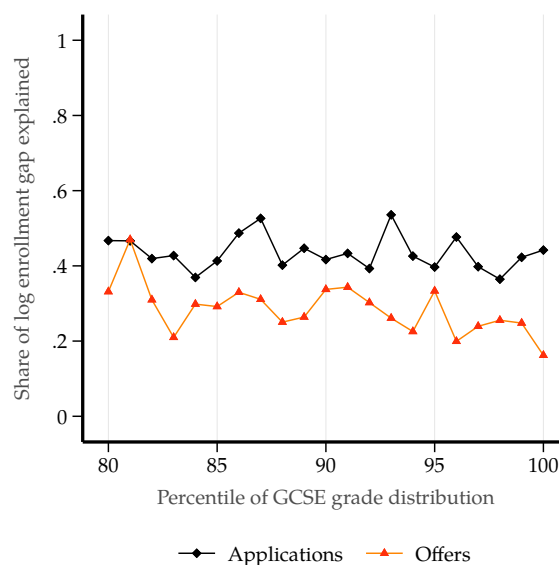
Estimating this decomposition reveals that disparities in applications explain a large fraction of overall disparities in enrollment. We estimate it conditional on students' ability, as measured by their percentile in the national GCSE distribution, focusing on the probability of enrollment in one of the top 10 universities. In the 90th–100th percentile of the GCSE distribution, where application and enrollment rates are highest, differences in application rates explain the largest share of differences in enrollment rates out of the four components of the decomposition; the share of enrollment differences explained by application differences averages around 40–50% and is consistently higher than that explained by offer differences. This is a higher share than found by Chetty, Deming, and Friedman in the context of Ivy-Plus enrollment gaps in the US, where only 30% of the differences in the excess enrollment of the top 1% could be attributed to application gaps, compared with 57% that could be explained by admissions.

Figure OA4: Decomposition of gaps in attendance at top 10 universities

(a) Actual and counterfactual enrollment rates



(b) Share of enrollment gap explained



Notes: Data: LEO. In panel (a), solid lines plot the probability of enrolling at one of the top 10 universities conditional on percentile in the national GCSE grade distribution (truncating at 80 since applications to top universities are very low below this rate), and conditional on enrolling through the main UCAS scheme. Application rate counterfactual calculated mechanically by multiplying application rate for high income students by offer rate, offer acceptance rate, and conditional enrollment rate for low-income students. Offer rate counterfactual similarly calculated by multiplying offer rate for high income students by application rate, offer acceptance rate, and conditional enrollment rate for low-income students. In panel (b), we take logs of the four components of the enrollment rate to get an additive decomposition, and plot the share attributable to applications and offers (i.e. the log difference in application rate and offer rate as a fraction of the log difference in enrollment rate).

OA5. Effects of equalizing exposure rates on socioeconomic application disparities

We combine estimates of differences in exposure to students attending top universities between high-income and low-income students with our estimates of the treatment effects of breakthroughs. Taking the latter as the causal effect of inducing exposure to a university on applications, we can now conduct a simple back-of-the-envelope exercise to quantify the effect that equalising exposure across income groups would have on application rates. Specifically, let the exposure rates for low-income and high-income students to university u – specifically, the probability of low or high-income students attending a school where at least one student has attended that university in the last three years – be $e_u^l, e_u^h \in [0, 1]$ respectively, and let the treatment effect of exposure at university u , as estimated from the difference-in-difference coefficient in the breakthrough event studies, be Δ_u . Then we can predict

$$(OA2) \quad Pr(\text{apply}_u \mid l, e_u^h) = Pr(\text{apply}_u \mid l, e_u^l) + \Delta_u(e_u^h - e_u^l)$$

We pool GCSE grades into ventiles (since we are not powered to estimate treatment effects conditional on exact percentiles), and then estimate the difference-in-differences regression separately within each ventile of GCSE grades for university u to get a grade-dependent treatment effect, $\Delta_u(g)$. We then combine this with exposure rates similarly calculated by GCSE ventile, $e_u^h(g), e_u^l(g)$, to get

$$(OA3) \quad Pr(\text{apply}_u \mid g, l, e_u^h) = Pr(\text{apply}_u \mid g, l, e_u^l) + \Delta_u(g)(e_u^h(g) - e_u^l(g))$$

Finally, we sum these effects over each of the top 10 universities to get

$$(OA4) \quad Pr(\text{apply top 10} \mid g, l, e^h) = Pr(\text{apply top 10} \mid g, l, e^l) + \sum_{u \in \{1, \dots, 10\}} \Delta_u(g)(e_u^h(g) - e_u^l(g))$$

making use of the empirical result that breakthroughs to university u do not affect applications to any similarly ranked university u' to simplify the calculation.

OA6. Earnings effects of graduating from selective universities

From an economic perspective, the primary reason to care about access to elite universities is that the university a student attends is consequential for their labor market outcomes. Since our administrative data link university applications and attendance to earnings, test scores, and demographics for a very large sample of students, we can precisely estimate earnings for graduates of individual universities across the distribution of university selectivity. We use this to illustrate the potential returns to students going to a higher-ranked university. Throughout this analysis, as well as in the rest of the paper, we rank universities by the mean A-level grades of students who are enrolled at each university.

We first plot the age profile of earnings for graduates from different universities from 18–30, finding that the earnings of graduates overtake those of non-attendees or dropouts by around age 23. Our primary outcome in this analysis is annual earnings from an individual’s primary employer, conditional on receiving positive earnings in the tax year.⁴⁴ Figure OA5a illustrates how mean annual earnings change over time for students with different university outcomes. University attendees earn less than non-attendees until age 22, but start to earn substantially more than non-attendees after this age. We also clearly see that graduating from a higher-ranked university is associated with higher earnings, and that the gaps between higher- and lower-ranked universities are larger than those between lower-ranked universities and non-attendees. By age 30, non-attendees at university have annual earnings around £20,000, graduates from an average university (below the top 30) have earnings around £26,000, and graduates from a top 10 university have earnings around £44,000. We also see from this figure that the relative earnings differentials have largely stabilised by age 27, suggesting that earnings at this age are a reasonable proxy for future earnings differences. This motivates our focus on earnings at age 27 as our primary measure of earnings in the remainder of this paper.⁴⁵ The whole distribution of earnings at age 27 for top-10 graduates lies to the right of that for graduates of lower-ranked universities, not just the mean.

We next attempt to adjust for selection, now looking university-by-university at earnings effects. For student i who graduates from university u and is aged 27 in year t , we estimate

44. See appendix OA1 for an explanation of the precise earnings outcome used. Earnings age profiles that instead assign 0 earnings to individuals who do not appear in the tax data show the same pattern.

45. More precisely, since tax years and academic years are misaligned – tax years start in April while academic years start in September – we use earnings in the tax year starting 9 years after the student completed their high school education, meaning students born between April and August would be 26 at the start of the tax year while all other students would be 27.

the regression

$$(OA5) \quad Y_{itu} = \alpha + \beta X_i + \theta_u + \delta_t + \varepsilon_{itu}$$

where X_i is a vector of individual-level observables, θ_u is a university fixed effect, and δ_t a year fixed effect. The omitted university dummy is the modal university in the sample.

We plot the fixed effect for each university θ_u after applying empirical Bayes shrinkage to point estimates of the university effects. Specifically, let s_u be the estimated standard error of the coefficient θ_u and $\mu_\theta, \sigma_\theta^2$ be the mean and variance of the estimated θ_u coefficients across universities; then

$$(OA6) \quad \theta_u^* = \frac{\sigma_\theta^2}{\sigma_\theta^2 + s_u^2} \theta_u + \frac{s_u^2}{\sigma_\theta^2 + s_u^2} \mu_\theta$$

$$(OA7) \quad s_u^* = \sqrt{\frac{\sigma_\theta^2}{\sigma_\theta^2 + s_u^2}} s_u$$

While we cannot identify universities by name,⁴⁶ we plot earnings against each university's rank by the mean of the top 3 A-level grades, converted to UCAS tariff points (see Appendix OA1), achieved by students who enroll at the university. The modal (omitted) university is ranked 56, close to the median.

Figure OA5b plots the results of this analysis. Specification 1 includes no controls in X_i and thus reflects only the raw differences in earnings across universities after applying empirical Bayes shrinkage. Specification 2 controls for observables: demographics, test scores, and major. It also introduces fixed effects for the exact portfolio of universities that the student receives an offer from (a subset of those that they apply to). Offer portfolios proxy for unobserved components of ability. If we could assume that university choice conditional on an offer portfolio is independent of ability, as argued in Dale and Krueger (2002, 2014) and Mountjoy and Hickman (2021), these estimates could be interpreted as causal effects on earnings. However, this is a strong assumption in this context⁴⁷, so we instead interpret offer portfolios as another observable that can help us absorb more variation in ability.

Figure OA5c plots the coefficients for each university from just this latter specification,

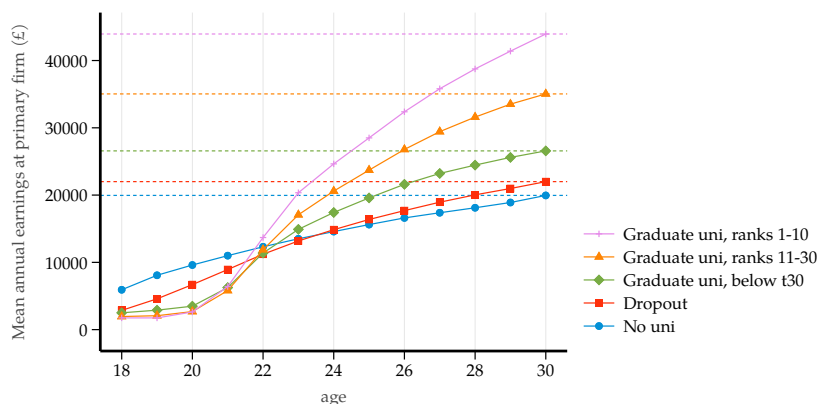
46. See Britton et al. (2022) for analysis of the earnings returns to different universities in the UK that is able to name universities. Their analysis is conducted without data on applications and offers, so they are not able to include offer set controls in their specifications, but they find a similar pattern of high returns for the most selective universities and a flat selectivity-earnings gradient beyond these elite universities.

47. In particular, because offers are typically conditional on future grades and more selective universities have stricter conditions, students may select into accepting offers from more selective universities based on residual private information about their own ability.

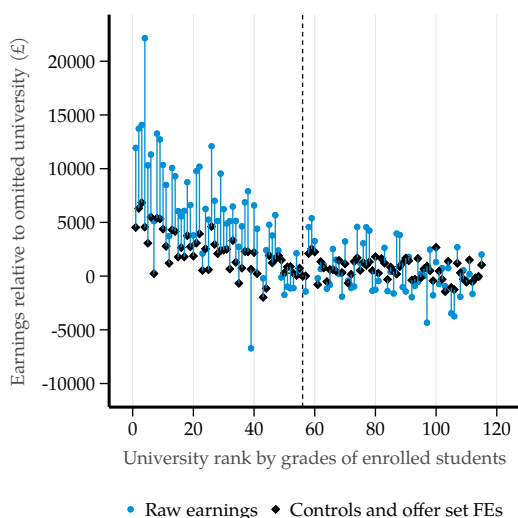
to better illustrate the range of estimated earnings effects, and adds confidence intervals to the estimates. Points in black have estimates that are significantly different from the mean university fixed effect (relative to the omitted university) after shrinkage. We see that most of the top 10–20 universities have distinctly higher returns than the median university – the average coefficient for the top 10 universities, relative to the omitted university, is £4,614. This is much lower than the descriptive gaps in earnings, but still a substantial increase, corresponding to 24% of mean earnings for all individuals at age 27 and 18% of mean earnings for all university graduates. Outside top institutions, there is a flat gradient between selectivity and earnings effects, and most coefficients are not significantly different from the mean effect of £1,220. Access to the most selective universities is thus particularly important for students’ earning potential, and socioeconomic disparities in students’ access to these universities are likely to perpetuate income inequality.

Figure OA5: Earnings returns to universities at age 27 across the distribution of university academic selectivity

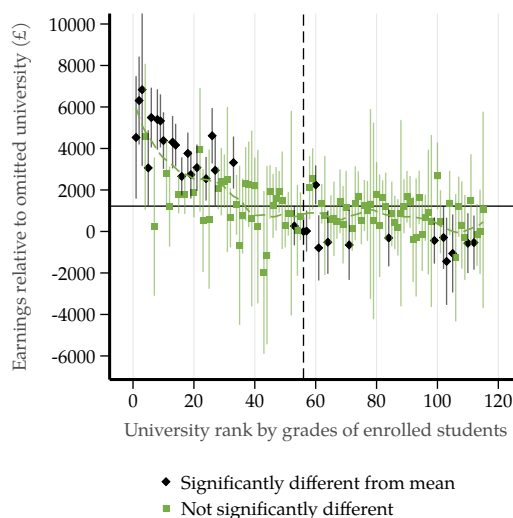
(a) Earnings trajectories by age



(b) Unconditional and conditional earnings effects



(c) Conditional earnings effects



Notes: Estimates of the earnings return to different universities. Data: LEO. All earnings are in GBP and adjusted to inflation in 2018. Panel (a) plots mean earnings conditional on age and university outcome, and on positive earnings. We restrict to students starting a degree at age 18 for dropouts and university graduates; dropouts are those who do not earn a degree within 8 years of completing high school, and graduates are those who earn a degree within 4 years of completing high school. Students completing a degree in 5–8 years are excluded. Panel (b) plots unconditional earnings gaps (in blue) and conditional earnings gaps (in green) between graduates of different universities. The unconditional gap records the coefficient on each university in a regression of earnings at age 27 on university fixed effects, omitting the university at rank 56, after applying empirical Bayes shrinkage to university effect estimates. Universities are ranked on the X-axis by the mean A-level tariff points of their enrolled students (see Appendix OA1). The conditional specification adds controls for gender, ethnicity, neighbourhood income decile, GCSE grades, A-level grades, major, and offer set to the regression. Panel (c) simply plots the conditional estimates shown in panel (b), rescaling the Y-axis and reporting 95% confidence intervals (based on standard errors after applying empirical Bayes shrinkage clustered at the high school level). Estimates that are significantly different from the mean at the 5% level are highlighted in black.

OA7. Supplemental Experimental Details

A. Details of Treatments

Active control workshop.—We designed the workshop to provide key information about the application process. Specifically, our materials discussed how universities make admissions decisions; statistics on grades and qualifications at different universities; statistics on earnings and students’ perspectives on different universities; the application process and timeline; ‘degree apprenticeships’ and other hybrid courses including work components; student loans, cost of living and student finance; advice on students’ personal statements; suggestions about where to find more information; and a description of the mentorship and visit components of the programme. The statistics provided were the most novel component of the workshop, as other information largely mirrored easily accessible information about the application process. Appendix Figures [OA51–OA58](#) show the statistics presented. We inform students about how universities make admission decisions—including information about how they may have more lenient offer conditions for low-income students—and provide statistics about the qualifications and grades required for admission at different universities, resulting earnings, and statistics from student surveys about their sense of belonging on their course, regret about course choice, and overall satisfaction.

Conceptually, this active control workshop ensured that all participating students had a baseline level of information about the university application process *in general*, allowing our treatments to then shock beliefs and preferences about *specific universities*. It helped to calibrate students’ prior beliefs about how ability and social type might affect academic and non-academic payoffs at university by providing statistics about these outcomes, reducing variation in these priors across students. The workshops also served several logistical functions. Having a component of the programme available to all students rather than just the treatment group was important for school recruitment, as schools would have been less enthusiastic about a programme in which a large fraction of their students would get no benefit. Having a highly salient in-school component of the programme—students were taken out of their lessons to attend the workshop—substantially raised the salience of the programme and engagement with the other treatments, particularly as we instructed workshop presenters to describe and promote the mentorship and visit treatments during the workshop (see Appendix Figure [OA67](#)).

Videos.—Students in the relevant treatment groups were shown two videos towards the end of the baseline survey. Appendix Figure [OA20](#) illustrates how the videos were displayed to students in the survey. Speakers were asked to discuss: the school and city the student attended; their A-Levels / other qualifications; how they made their decisions

about which universities to apply to; where they got offers from and ultimately attended; student life in their area at university; fitting in and making friends; cost of living and expenses; and teaching quality on their course. Videos provide exposure to a particular university in a manner that is more detailed and personal than generic online information, but more scalable than direct conversations online or in-person. The design of the video replicates what a student might learn in a conversation about university, but without the interactivity and potential for follow-ups of a full conversation. As illustrated in Appendix Figure OA19, we introduced these videos as students describing their application process and university experience in a way that might be helpful for students; we did not imply that we were encouraging students to apply to these particular universities.

We selected videos to show students relating to universities that were aspirational but realistic given their predicted grades. Earlier in the baseline survey, we ask students to enter their predicted grades, and converted them into UCAS tariff points, a metric provided by UCAS to compare grades across different qualifications (see Appendix OA1). We then grouped universities into four tiers based on the distribution of students' grades, as recorded in the public statistics on the discoveruni.gov.uk website. Specifically, universities were assigned to one of four tiers based on the 25th percentile of UCAS tariff points of enrolled students at the university. Appendix Table provides the cutoffs for each tier and their A-level letter grade equivalents, as well as some examples of universities in each tier. We then assigned *students* to tiers based on their predicted tariff points, using the 25th percentile groups described in Appendix Table as cutoffs. We assigned all students to at least tier 3 even if their predicted grades fell below the tier 3 cutoff, on the grounds that lower-ranked universities would have the potential to discourage ambitious applications, and only assigned students to tier 1 if they were taking A-levels and had predicted grades exceeding the cutoff, on the grounds that the most selective courses generally do not accept vocational alternatives to A-levels.

Mentors.—We recruited a set of current university students and recent graduates to act as volunteer mentors for students in the programme. The majority of these mentors signed up through STEM Ambassadors, a STEM-focused volunteering platform. Some were recruited through other channels, such as direct outreach via university partners or via AtkinsRealis, an engineering organisation that supported the programme. Table OA8 describes the characteristics of the mentors taking part in the programme. The majority are current undergraduate students, but a substantial minority were older. The sample of mentors is disproportionately female, whereas our student sample is disproportionately male, but the mentor sample has similar levels of economic disadvantage compared to the RCT sample. As in our RCT sample, there is a much lower share of white students in the

sample of mentors compared with the broader student population.

Students assigned to the mentorship treatment were connected with 1–2 mentors from this pool. We sent students and mentors an email to connect them, and encouraged them to get in touch to arrange a call to talk about university applications and life at university. We suggested that mentors and mentees could discuss “[mentor’s] course, life at [mentor’s university], uni life in general, and the application process”, but did not provide a script for mentors or prescribe topics, as we wanted to allow for organic conversations and to treat the topics that students chose to discuss as an outcome of interest. Mentors were asked to have at least one 15-minute call with their students, to answer any further questions over email, and to arrange follow-up calls if the student was interested and the mentor was available. Students were also able to ask questions of their mentors over email if they preferred not to have a call.

During the midline survey, and before students were informed whether they would be connected with a mentor, students were asked (a) whether they would like a mentor, and (b) to name three universities that they would be interested in receiving a mentor from. We connected each student in the relevant treatment arm with a mentor from one of these three universities, subject to availability of a mentor in our mentor pool.⁴⁸ We also identified a second mentor from a less familiar university to connect students with, based on their university tier and the subject that they intended to apply for. In both cases, students in treatment arm T1 were matched with mentors with whom they did *not* share a gender, ethnicity, or home region of the UK, while students in arm T2 matched with mentors with whom they shared at least one of these characteristics. We describe the matching algorithm in full in Appendix Section [OA7.B](#).

B. Algorithm for mentor assignment

Students are first assigned to a treatment arm T1 or T2. We assign a latent treatment arm for students in the control group, so that these students can be notionally assigned mentors according to the same procedure as treated students.

We identify a tier-based and / or a preference-based mentor for each student—we target assigning each treated student one mentor of each type, but will assign only one mentor if there are no eligible mentors with remaining capacity. For preference-based mentors, we take the set of mentors from one of the three universities that the student requested. We then restrict to mentors with the appropriate demographics (different gender, ethnicity and

48. Students who responded ‘definitely not’ to the question of whether they wanted a mentor were not connected with a mentor to avoid wasting student and mentor time.

region for students in T1; same gender, ethnicity or region for students in T2). Ethnicities are categorised into white, black, Asian, and other; regions are the 9 ITL 1 statistical regions of England. We then select the mentor with the highest capacity to take on new mentors remaining, and assign this mentor to the student. The mentor's capacity is decremented by 1 after being assigned, starting from the number of students that they initially told us they could take on.

For tier-based mentors, we similarly restrict on demographics based on the treatment arm. We exclude any mentors from universities that the student requested a mentor from, or that they reported a parent or sibling as having attended, so that the tier-based mentor is unfamiliar. We allow for overlap with the video treatment university. We then restrict to universities in the same tier that the student was assigned to. Next, we try to match on students' majors. We start by looking for a match on the exact major; if there are no available mentors within the exact major, we then use a more aggregated definition of major, and if there are still none then we take all mentors within the remaining universities. We then assign this mentor as the tier-based mentor; if there are multiple available mentors suitable to be matched, we take the mentor with the highest remaining capacity.

Table OA6: Balance table

	Sample means					Differences in means	
	All (1)	C (2)	T (3)	T1 (4)	T2 (5)	T-C (6)	T2-T1 (7)
<i>Panel A: demographics</i>							
Female	0.422 (0.020)	0.455 (0.027)	0.386 (0.028)	0.396 (0.040)	0.375 (0.039)	-0.069* (0.039)	-0.021 (0.056)
Low-SES	0.602 (0.019)	0.593 (0.027)	0.613 (0.028)	0.599 (0.040)	0.627 (0.040)	0.020 (0.039)	0.028 (0.056)
White	0.567 (0.020)	0.566 (0.027)	0.569 (0.028)	0.539 (0.040)	0.599 (0.040)	0.003 (0.039)	0.060 (0.057)
Black	0.069 (0.010)	0.096 (0.016)	0.039 (0.011)	0.039 (0.016)	0.039 (0.016)	-0.057*** (0.020)	0.001 (0.022)
Asian	0.248 (0.017)	0.222 (0.023)	0.278 (0.026)	0.292 (0.037)	0.263 (0.036)	0.056 (0.034)	-0.029 (0.051)
<i>Panel B: UK region</i>							
N. England	0.428 (0.020)	0.413 (0.027)	0.444 (0.028)	0.461 (0.040)	0.428 (0.040)	0.031 (0.039)	-0.033 (0.057)
The Midlands	0.030 (0.007)	0.030 (0.009)	0.029 (0.010)	0.039 (0.016)	0.020 (0.011)	-0.001 (0.013)	-0.019 (0.019)
S. England	0.331 (0.019)	0.329 (0.026)	0.333 (0.027)	0.299 (0.037)	0.368 (0.039)	0.004 (0.037)	0.070 (0.054)
London	0.208 (0.016)	0.225 (0.023)	0.190 (0.022)	0.195 (0.032)	0.184 (0.032)	-0.035 (0.032)	-0.011 (0.045)
<i>Panel C: Academic characteristics</i>							
Taking ≥ 3 A-levels	0.850 (0.014)	0.832 (0.020)	0.869 (0.019)	0.838 (0.030)	0.901 (0.024)	0.037 (0.028)	0.064* (0.038)
Predicted tariff points	126.400 (2.066)	125.078 (2.937)	127.843 (2.901)	124.416 (4.091)	131.316 (4.110)	2.765 (4.128)	6.900 (5.799)
<i>Panel D: Baseline beliefs and mentorship interest</i>							
Baseline social belief (Oxford)	50.734 (1.234)	50.359 (1.717)	51.144 (1.778)	51.234 (2.468)	51.053 (2.569)	0.785 (2.472)	-0.181 (3.562)
Baseline offer belief (Oxford)	25.062 (0.893)	25.719 (1.241)	24.346 (1.288)	23.766 (1.772)	24.934 (1.876)	-1.372 (1.789)	1.168 (2.581)
Baseline graduation belief (Oxford)	61.290 (1.290)	63.376 (1.804)	59.150 (1.839)	61.364 (2.685)	56.908 (2.507)	-4.225 (2.576)	-4.456 (3.673)
Interested in mentor	0.886 (0.013)	0.880 (0.018)	0.892 (0.018)	0.883 (0.026)	0.901 (0.024)	0.012 (0.025)	0.018 (0.036)
N	640	334	306	154	152		

Notes: Balance table reporting means, differences in means, and standard errors for the specified characteristics across different treatment arms in the RCT. Column 1 reports characteristics for the full sample; columns 2 and 3 report characteristics for the control arm C and the pooled treatment arm T (pooling over arms T1 and T2); columns 4 and 5 report characteristics within each of the sub-treatment arms T1 and T2; and columns 6 and 7 report the difference between arms T and C, and between arms T2 and T2. Low-SES students are those who either have a home address in the bottom two quintiles of neighbourhood income or who do not have parents who attended university. Heteroskedasticity-robust standard errors reported. For differences, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA7: University tiers for videos

Tier	UCAS tariff cutoff (25th percentile)	Letter grade equivalent	Restricted to A-level students?	Universities
1	144	AAA	Y	Cambridge, Imperial, LSE, Glasgow
2	128	ABB	N	Bath, Warwick, Durham, Leeds, Edinburgh
3	0	—	N	Loughborough, Newcastle, QMUL, Queen's Belfast, Nottingham, Sheffield, Southampton, York

Notes: Table indicating the university tiers used in the RCT. We report the UCAS tariff point cutoff for the tier (based on the 25th percentile of tariff points for attendance), the equivalent in terms of A-level letter grades, whether we restrict students assigned to this tier to be those taking A-levels (this is the case for tier 1), and some examples of universities in each tier.

Table OA8: Mentor Pool Summary Statistics

Characteristic	Percentage of mentors
<i>Demographics</i>	
Female	62.8
Current university student	59.0
First-gen uni attendee	42.2
Low-income	28.9
<i>Age</i>	
18–21	48.2
22–25	28.2
26+	23.7
<i>Ethnicity</i>	
White	60.0
Black	7.8
Asian	22.5
Mixed / other	9.8
<i>N</i>	245

Notes: Characteristics of participating mentors, as reported by mentors in our recruitment survey.

OA8. Survey Questions

A. *Baseline survey*

Figure OA6: Student name and school

Select your school

Enter your first name

Enter your surname

Figure OA7: Student academic year

Which school year are you currently in?

☐ Year 13

☐ Year 12

Figure OA8: Student qualifications

What qualifications are you studying for?

A-levels	☐
BTECs	☐
T-levels	☐

Figure OA9: Student predicted grades

What are your **predicted grades** in your A-level subjects?

If you are doing fewer than 4, you can select N/A for some subjects. If you don't have predicted grades yet, enter your best guess for each of your subjects.

	A-Level Predicted Grade
Subject 1	A* ▼
Subject 2	A ▼
Subject 3	B ▼
Subject 4	N/A ▼

Figure OA10: Student major choice

Which of these best represents the subject that you will apply to university for?

Agriculture, Food and Related Studies

Allied Health

Architecture, Building and Planning

Biosciences

Business and Management

Celtic Studies

Chemistry

Combined and General Studies

Computing

Creative Arts and Design

Economics

Education and Teaching

Engineering

English Studies

General, Applied and Forensic Sciences

Geography, Earth and Environmental Studies

Health and Social Care

Figure OA11: University choice

If you had to apply to five universities today, which five universities would you pick for Mathematical Sciences courses?

Choice 1:
University of Cambridge

Choice 2:
The London School of Economics and Political Science (LSE)

Choice 3:
The University of Warwick

Choice 4:
University College London (UCL)

Choice 5:
University of York

Figure OA12: University top choice

Which one of these would be your top choice to attend?

University of Cambridge ☐

The London School of Economics and Political Science (LSE) ☐

The University of Warwick ☐

University College London (UCL) ☐

University of York ☐

Figure OA13: University choice motivation (open-text)

In a few sentences, can you explain why you picked these 5 courses? What about your top choice?

Figure OA14: Admission beliefs

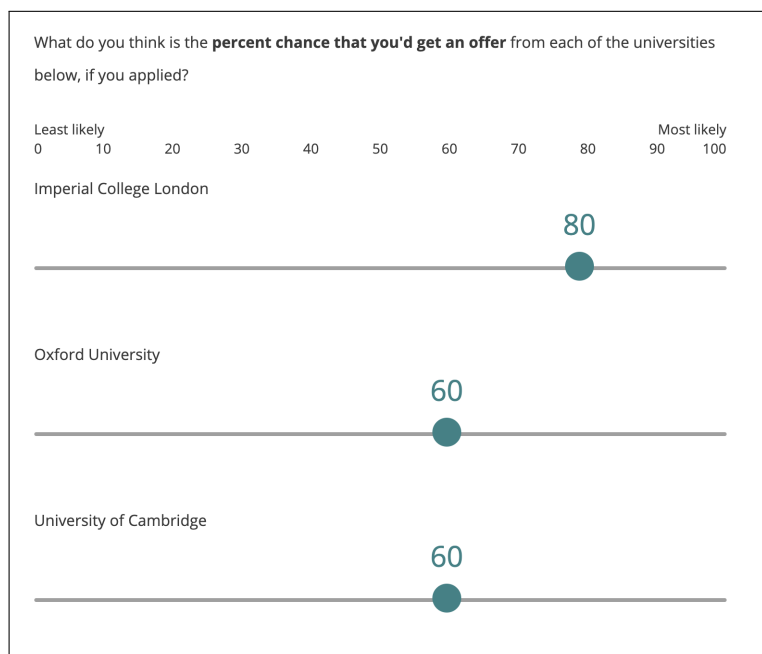


Figure OA15: Social fit beliefs

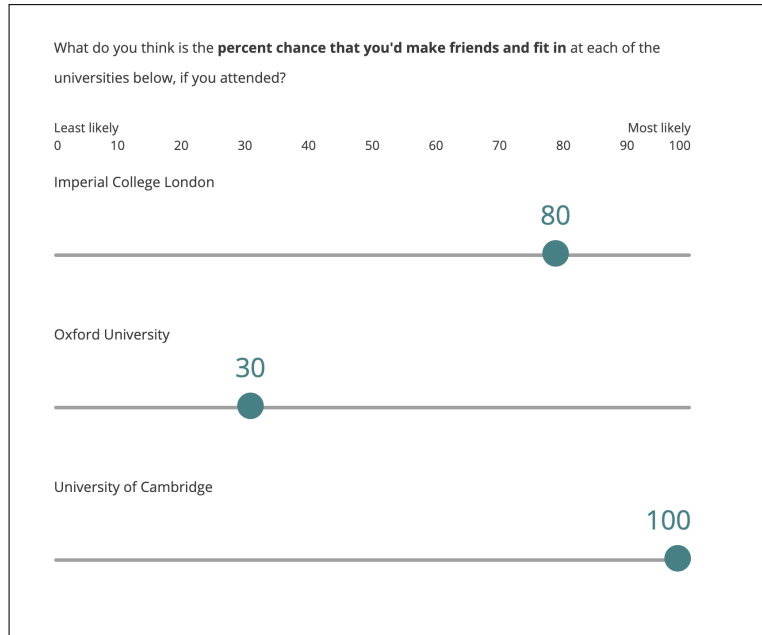


Figure OA16: Graduation beliefs

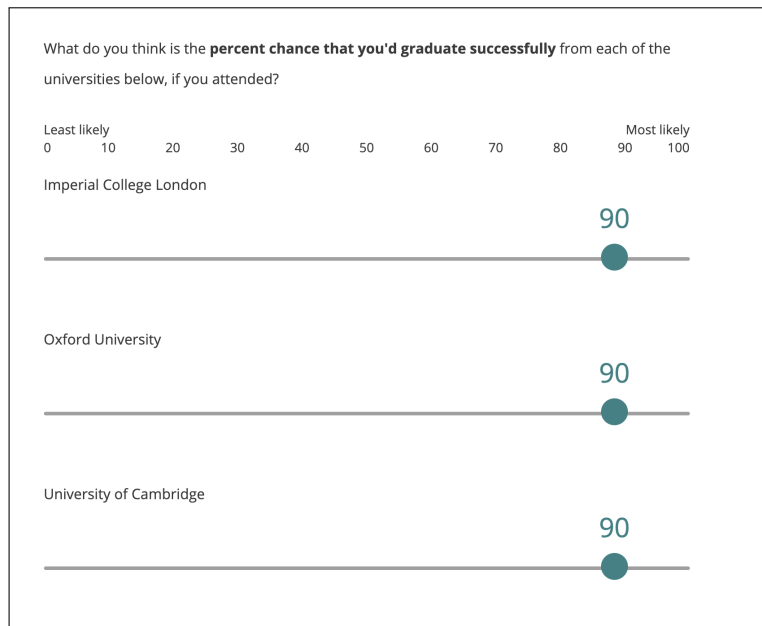


Figure OA17: Typical grades beliefs

What do you think are **typical A-level grades (e.g. ABB)** for students at each of the universities below in your subject?

Imperial College London

Oxford University

University of Cambridge

Figure OA18: Beliefs about typical university at students' school

Which university do you think **students at** [REDACTED] **most commonly attended** in the past few years?

The University of Liverpool ▼

Figure OA19: Introduction to videos

Next, you will watch two videos from current university students describing their application process and university experience. **Please watch both videos carefully.** We can tell if you watch the videos in full or not, so please try to watch both of them to the end. We hope that these videos will be helpful for your university application process!

Figure OA20: Video page

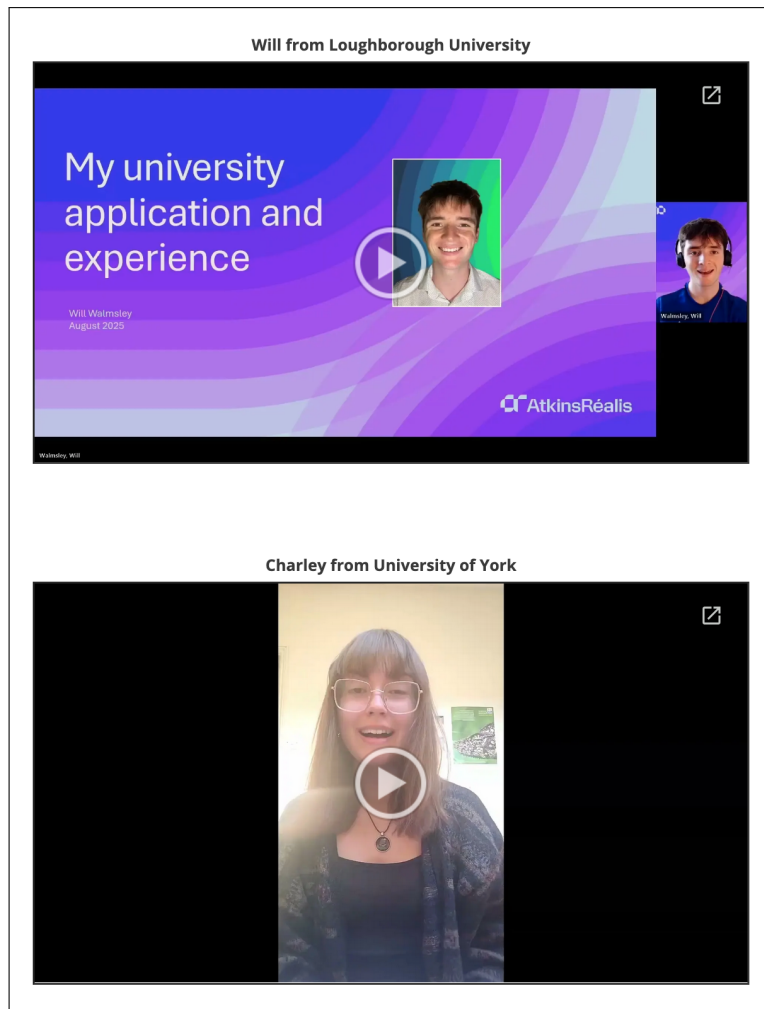


Figure OA21: Video text responses

Optional: tell us something here about what you took away from the videos. For example, were there some things that surprised you? What did you find most helpful? Least helpful? Are there any follow up questions you would like to ask the speakers?

Figure OA22: Re-elicitation of admission beliefs after video

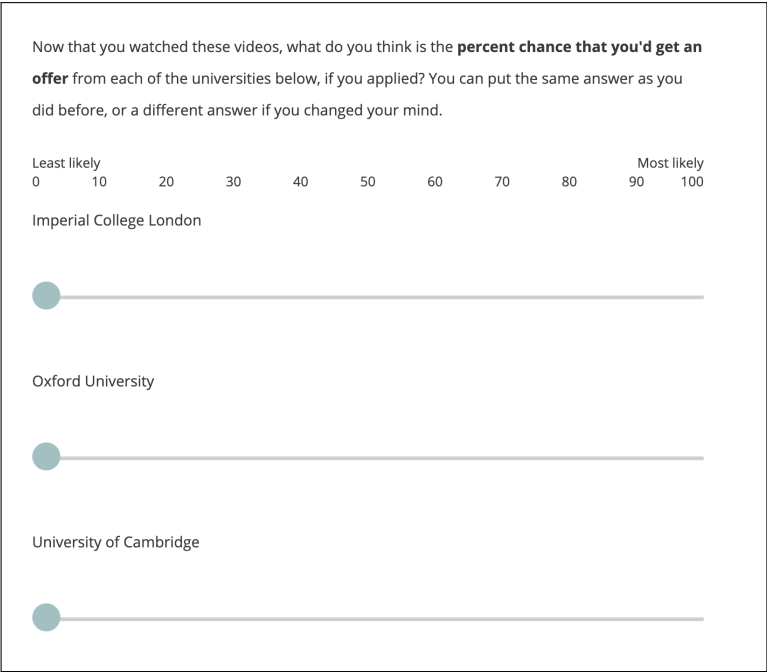


Figure OA23: Re-elicitation of social beliefs after video

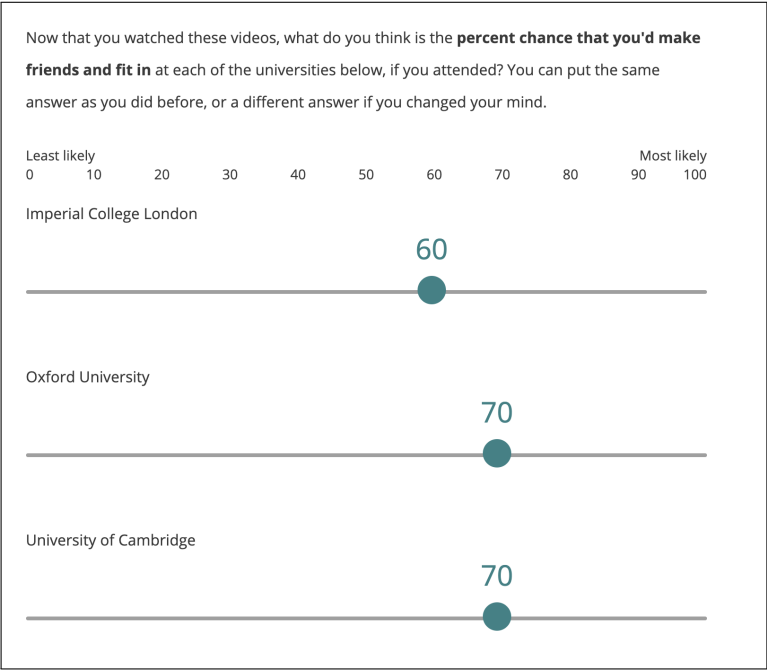


Figure OA24: Re-elicitation of graduation beliefs after video

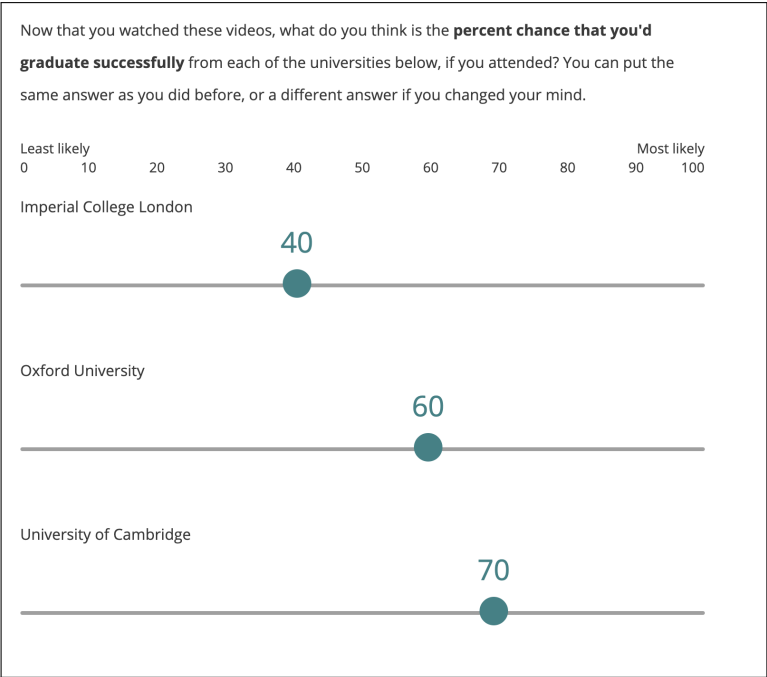


Figure OA25: End of survey message

Thanks for taking the time to answer our survey! If you have any questions, concerns, or comments please email either ntadjfar@mit.edu or kvira@mit.edu.

We hope you enjoy the workshop!

B. Midline survey

Figure OA26: Student name and school

Thank you for participating in this research! In this survey, we'll ask you questions about yourself and how you are thinking about the university application process as of today. We expect this survey to take less than 10 minutes.

You may have seen similar questions in a previous survey. Feel free to answer the same or differently. We're curious about whether or not your thought process has evolved, so there are no right answers.

Your answers to this survey are **confidential** and will only be accessible by MIT researchers Nagisa Tadjfar and Kartik Vira. **Your friends and teachers will not be able to see your responses.**

Select your school and name below.

School

Full name

Figure OA27: Student major choice

Which of these subjects are you planning on applying to university for? If you are considering applying for multiple subjects or joint courses, select the subject that you're most confident that you want to apply for. If none of these are exactly what you plan to apply for, pick the best fit from the list below.

Figure OA28: University choice

You indicated Economics as your main subject. If you had to apply to five universities today, which five universities would you pick for Economics courses?

Choice 1:

Choice 2:

Choice 3:

Choice 4:

Choice 5:

Figure OA29: University top choice

If you were guaranteed to get an offer from all five courses and meet the offer conditions, which one would be your top choice to attend?

University of Cambridge	☐
The London School of Economics and Political Science (LSE)	☐
University of York	☐
University College London (UCL)	☐
The University of Warwick	☐

Figure OA30: Family university attendees

Which of these applies to you? Select as many boxes as apply.

I have a parent or guardian who went to university	☒
I have an older sibling who went to or is currently in university	☐
I have another close relative who went to or is currently in university	☐
I have an older friend who went to or is currently in university	☐
None of the above	☐

Figure OA31: Universities attended by family members

You indicated above that you have a parent or guardian who went to university. If they went to university in the UK, please select the university(s) that they attended.

You indicated above that you have an older sibling who went to or is currently in university. If they went to university in the UK, please select the university(s) that they attended.

Figure OA32: Names of friends at school

Please list the full names of 3 friends you talk to regularly at your school.

	First name	Surname
Friend 1		
Friend 2		
Friend 3		

Figure OA33: A-level subjects and predicted grades

What are the subjects that you're taking and your predicted grades in each one? Enter up to 4 subjects below.

	Qualification (e.g. A-level)	Subject title (e.g. History)	Predicted grade
Subject 1			
Subject 2			
Subject 3			
Subject 4			

Figure OA34: Ethnicity

What best describes your ethnic origin?

☐ White

☐ Black/African/Caribbean

☐ Asian (Indian, Pakistani, Bangladeshi, Chinese, any other Asian background)

☐ Mixed two or more ethnic groups

☐ Other (Arab or any others)

☐ Prefer not to say

Figure OA35: Gender

What is your gender?

☐ Male

☐ Female

☐ Non-binary

☐ Prefer not to say

Figure OA36: Student home postcode

What is your home postcode?

Figure OA37: Interest in mentorship

After this workshop, many of you will be matched with current university student mentors who can answer any university-related questions you might have. Please let us know about your preferences for mentors in the questions below.

Would you be interested in being given a mentor?

Yes, definitely ☐

Yes, probably ☐

Maybe ☐

Probably not ☐

Definitely not ☐

Figure OA38: Mentor preferences

What would be your **top** choice of university to get connected to a current student from?

What would be your **second** choice of university to get connected to a current student from?

What would be your **third** choice of university to get connected to a current student from?

Figure OA39: Mentor preference explanation

In a few sentences, can you explain how you picked these 3 universities to receive mentors?

Figure OA40: Number of universities visited previously

Have you visited any universities in-person (e.g. for Open Days)?

Yes, I visited multiple universities

☐

Yes, I visited one university

☐

No, but I plan on visiting universities before I apply

☐

No, I have not visited any universities and do not plan on visiting any

☐

Figure OA41: Obstacles to university visits

Have any of the factors below stopped you from visiting a university that you are interested in?

I already know which university I want to go to

☐

I don't think visiting the university is helpful

☐

It would cost too much money

☐

It would take too long to travel there and back

☐

I didn't know it was an option / hadn't thought about it

☐

Other

☐

None of the above

☐

Figure OA42: Names of universities visited

Which universities have you visited?

Figure OA43: Request for university to visit

As part of this programme, **many of you will receive up to £75 in travel funding to visit a university** of your choice in person. Which university would you like to visit using these funds?
You will only be reimbursed to visit this particular university, so please choose carefully.

Figure OA44: Admission beliefs

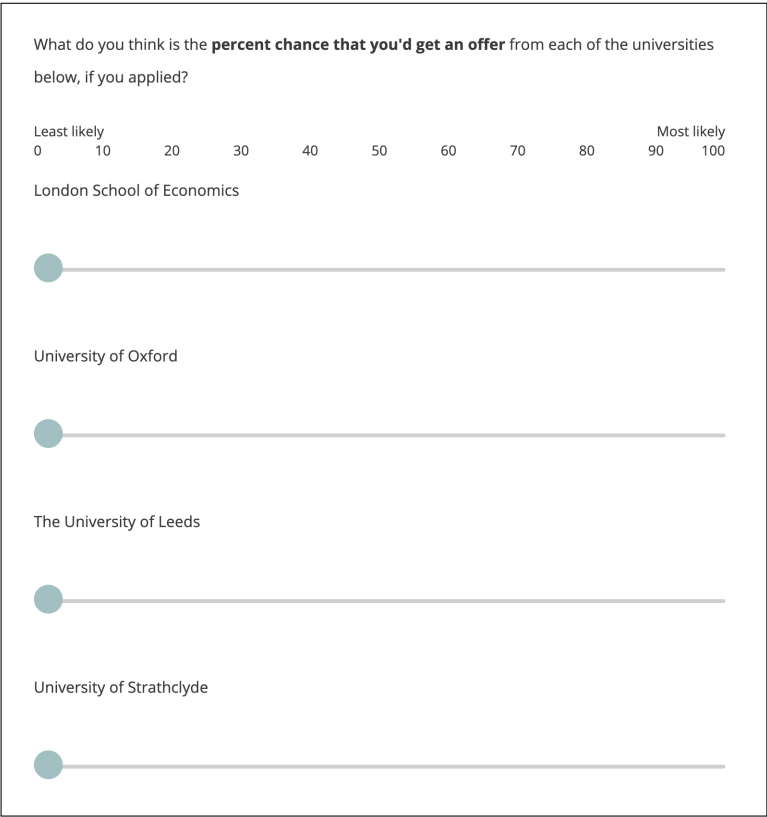


Figure OA45: Social fit beliefs



Figure OA46: Graduation beliefs



Figure OA47: Typical grades beliefs

What do you think are **typical A-level grades (e.g. ABB)** for students at each of the universities below in your subject?

London School of Economics

University of Oxford

The University of Leeds

University of Strathclyde

Figure OA48: Ending screen

Congratulations! You are eligible to be matched with up to two mentors after this workshop.

Kartik Vira (kvira@mit.edu) or Nagisa Tadjfar (ntadjfar@mit.edu) from the MIT research team will reach out to you to put you in touch with your mentor(s). We'll contact you over email and over text using the details that you provided, so watch out for messages from us.

Mentors have generously volunteered their time to help you with the university process, and have a lot of useful information that should help you feel more comfortable with the application process and life at uni, so please make sure to reply to your mentors once we put you in touch with them!

Thanks for participating in this workshop - we hope you found it useful and that it'll help with your university decisions. Good luck!

OA9. Experimental materials

We present a selection of the key slides from the workshop, excluding some transition slides and slides that were less relevant to university application decisions.

Figure OA49: Presenter introduction

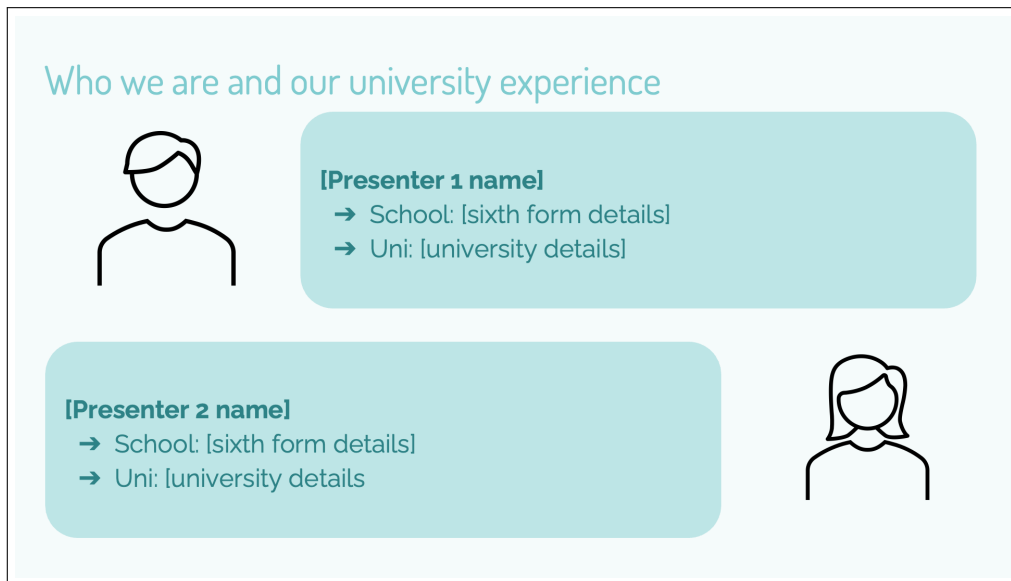


Figure OA50: Application timeline

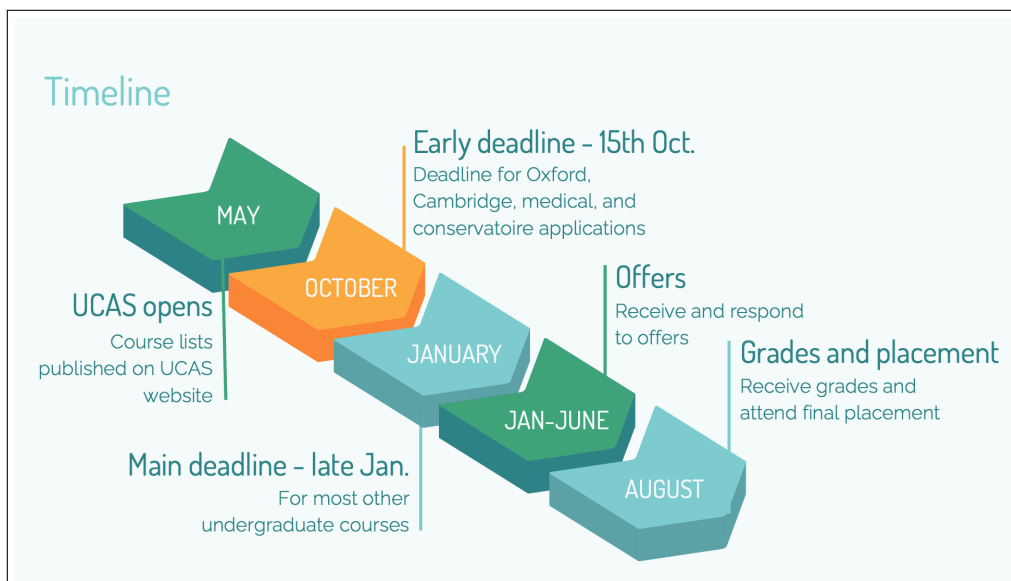


Figure OA51: How universities make admissions decisions

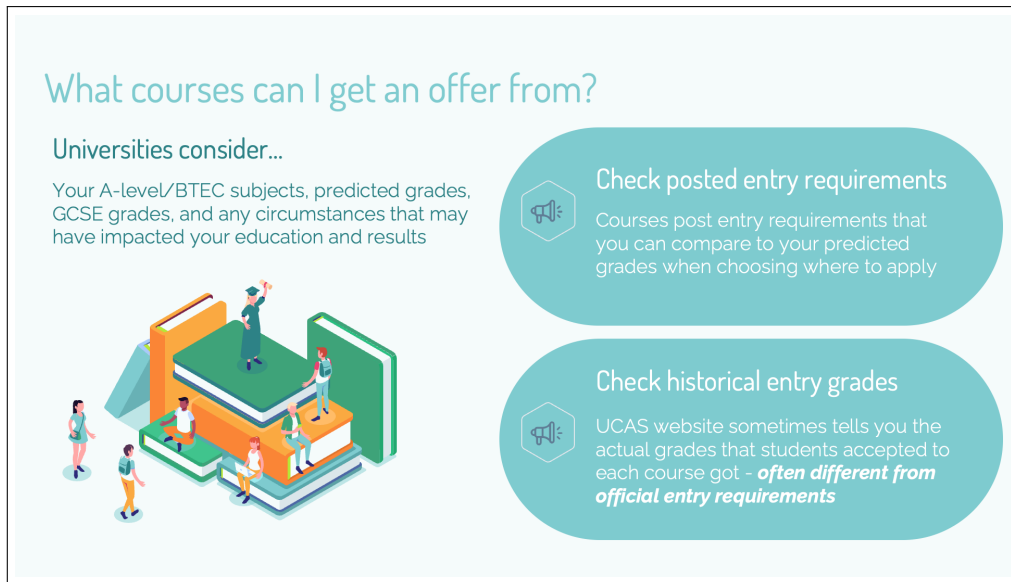


Figure OA52: Contextual admissions



Figure OA53: Russell Group student qualifications

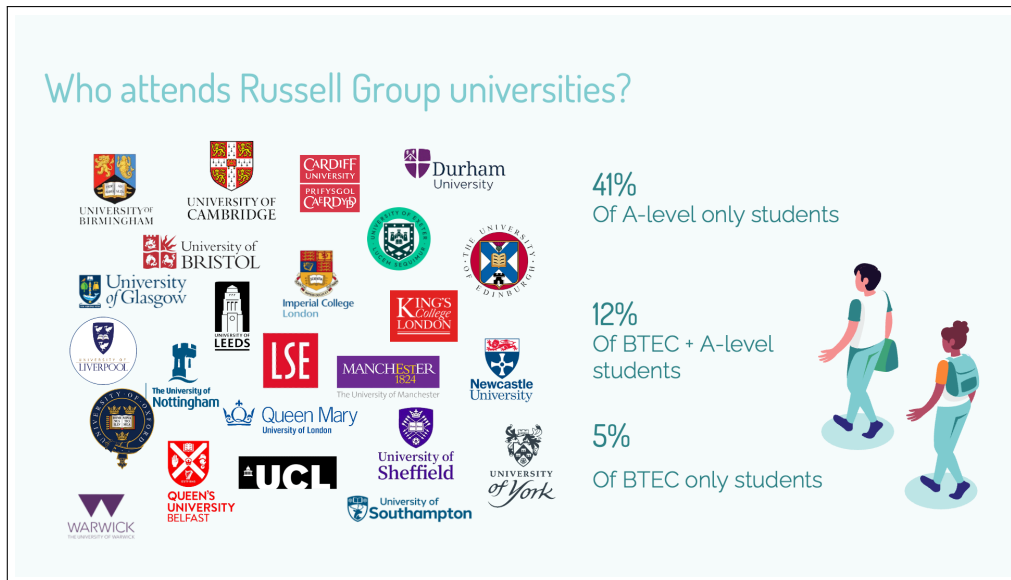


Figure OA54: Russell Group student grades

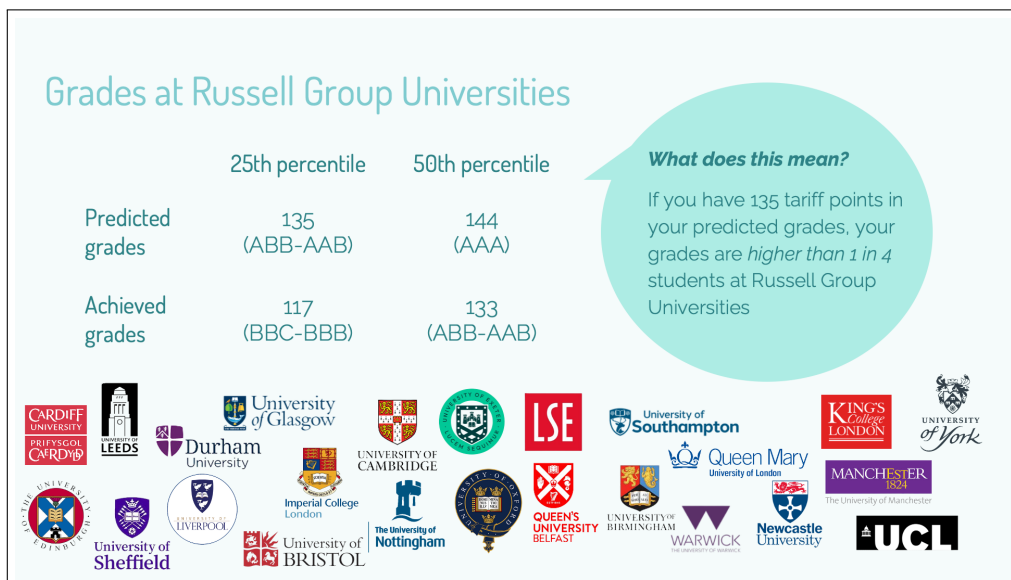


Figure OA55: Required grades at different universities

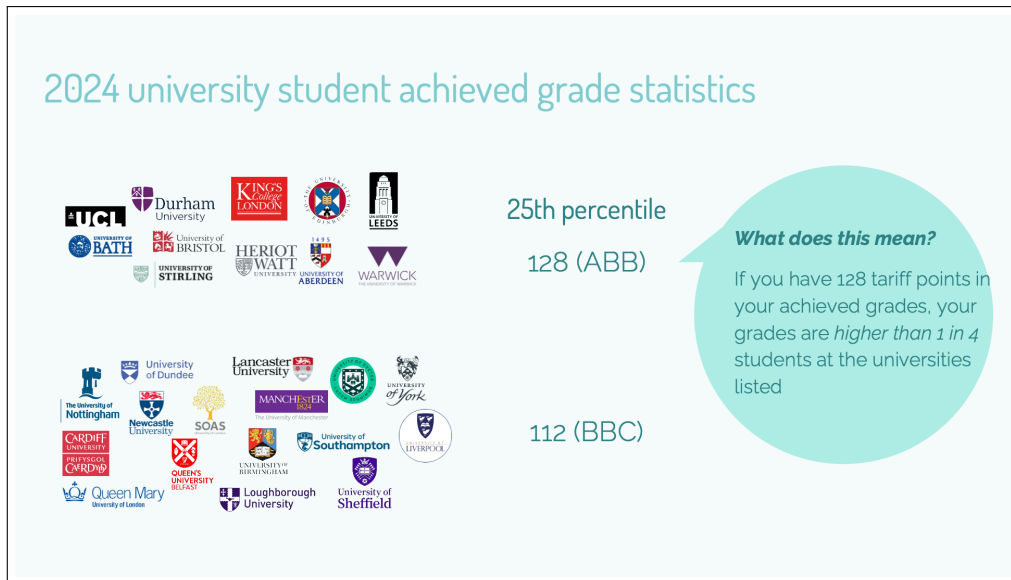


Figure OA56: Earnings at different universities



Figure OA57: Student regret statistics

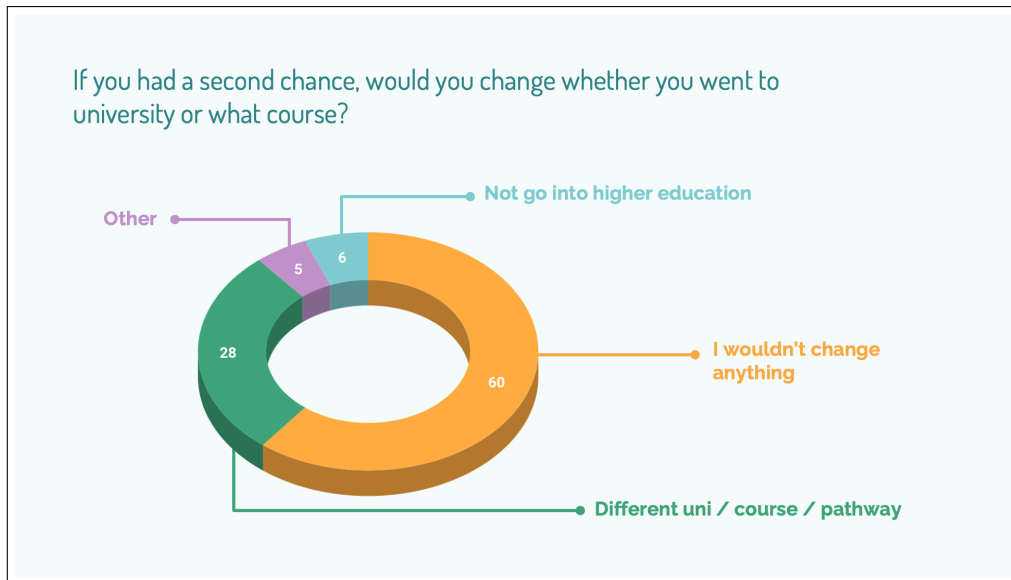


Figure OA58: Student belonging and course statistics

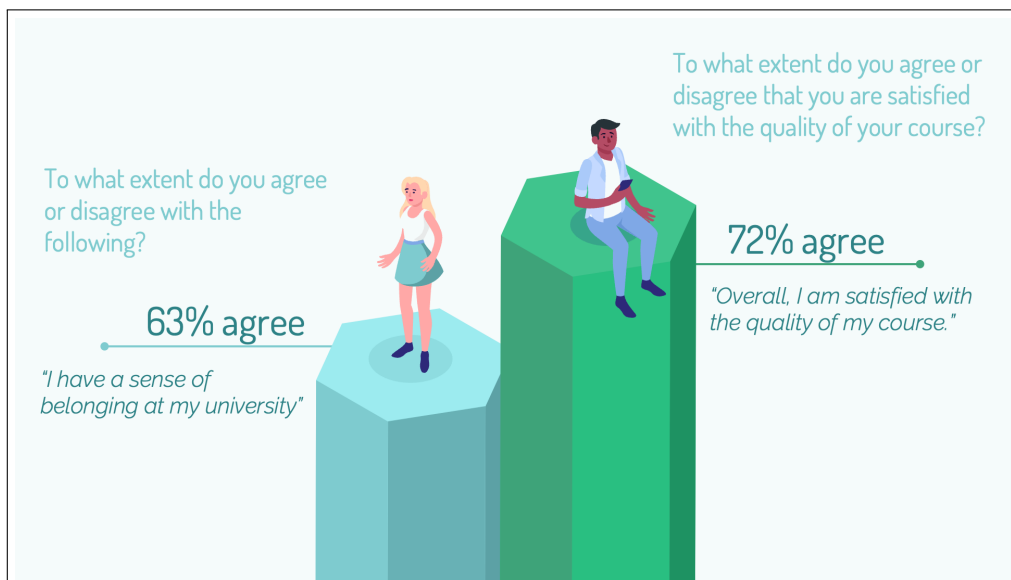


Figure OA59: Responding to offers

Responding to your offers

You may still need to meet offer conditions for your insurance, so **choose a course with lower offer conditions** than your firm!



Firm

- where you'll go if you meet the conditions in your offer
- choose this as long as you have *at least one offer*

Insurance

- can consider you if you *miss your firm conditions* and your firm doesn't accept you
- choose this if your firm choice gave you a conditional offer

Figure OA60: Tuition and cost of living

Fees and cost of living

Tuition fees

£9,535 / year at almost all UK universities - fully covered by student loans

Part-time work

55% of students did some part-time work during term time in 2023



Maintenance loans

Maintenance loans can support cost of living - up to £14,000 / year depending on where you live at uni and your parents' income

University support

Universities have hardship funds and might also provide discounts for food, travel etc. as well as affordable housing options. Be sure to ask universities *explicitly* about these as they may not be advertised!

Figure OA61: Personal statement format

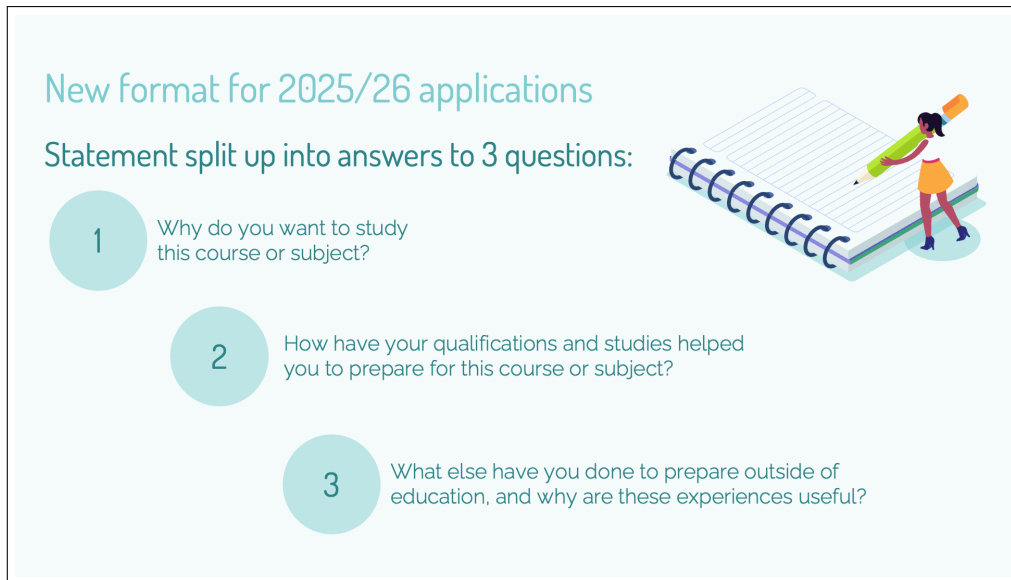


Figure OA62: Personal statement advice – what not to do

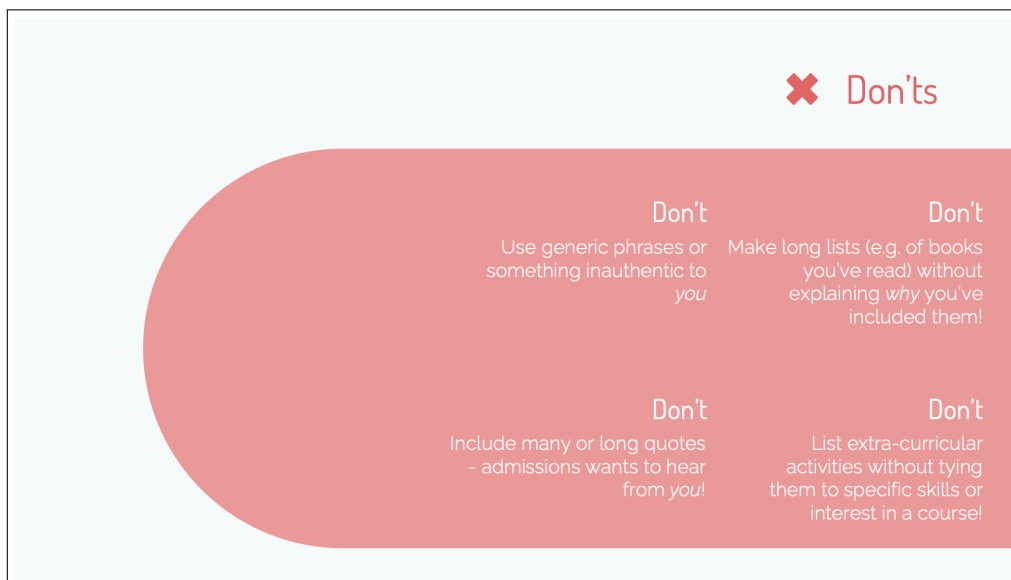


Figure OA63: Personal statement advice – what to do

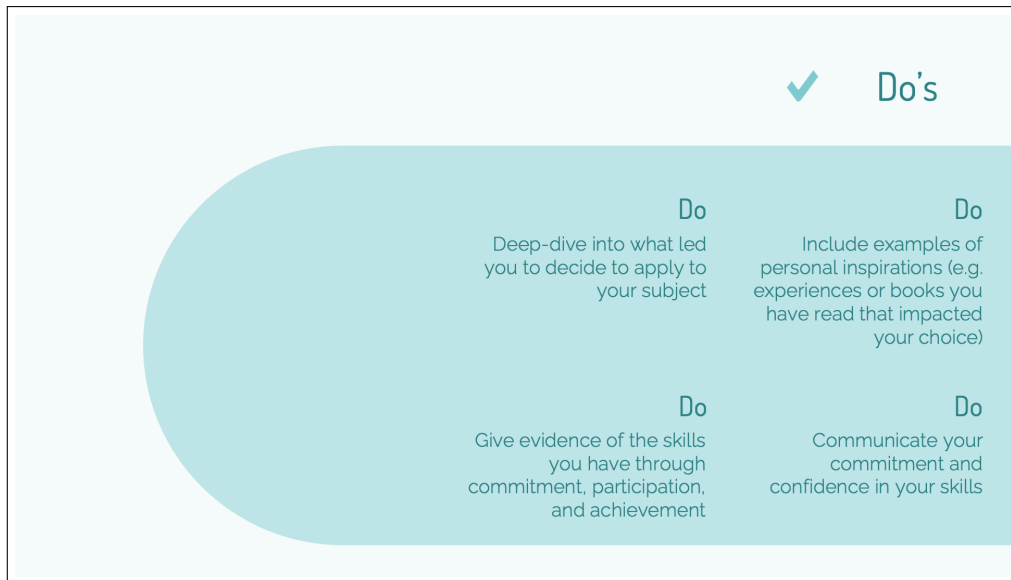


Figure OA64: Online resources

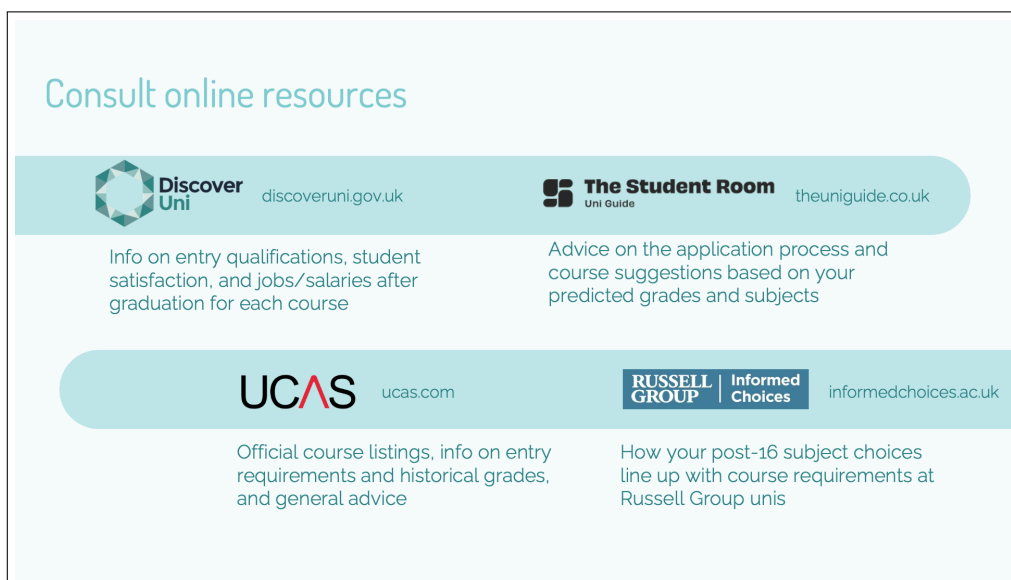


Figure OA65: Other sources of information

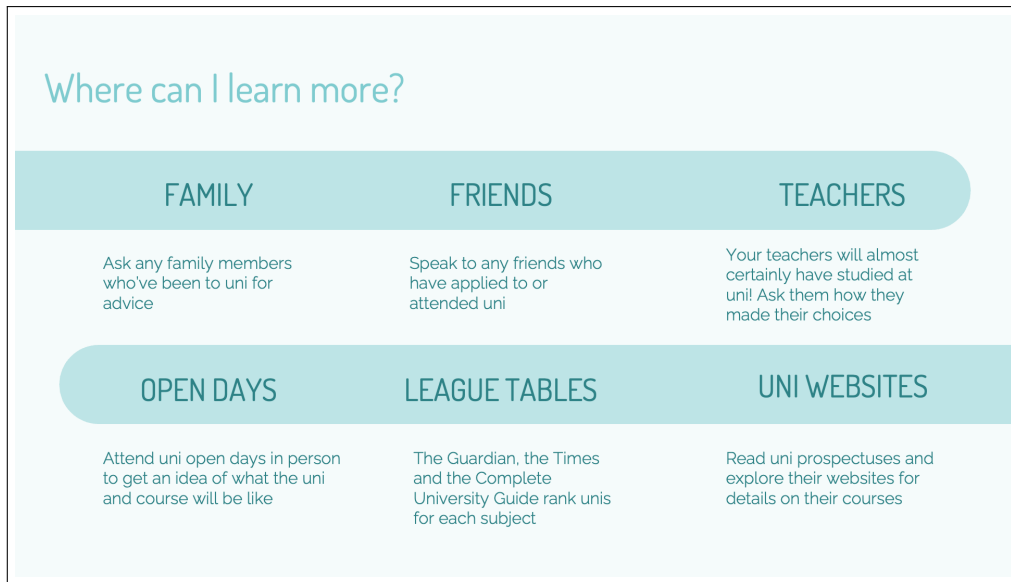


Figure OA66: University visits



Figure OA67: Description of mentorship / visit treatments

As part of this study, many of you will have the chance to...



Receive 2 mentors who are current uni students

- ◆ You can use these conversations to **learn more about unis you are curious about** (but might not know much about)
- ◆ Get **advice / tips on your application**, personal statements, etc.

Subsidised visits to universities

Receive financial support of **up to £75** to visit a university of your choice in person



Figure OA68: Survey QR code

Survey #2: tell us about your plans and preferences for mentors!



